

GD25LF256F

DATASHEET

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1 FEATURES

- ◆ 256M-bit Serial NOR Flash
 - 32M-Byte
 - 256 Bytes per programmable page
- ◆ Standard, Dual, Quad SPI, QPI, DTR
 - Standard SPI: SCLK, CS#, SI, SO, WP#, RESET#
 - Dual SPI: SCLK, CS#, IO0, IO1, WP#, RESET#
 - Quad SPI: SCLK, CS#, IO0, IO1, IO2, IO3, RESET#
 - QPI: SCLK, CS#, IO0, IO1, IO2, IO3, RESET#
 - SPI DTR (Double Transfer Rate) Read
 - 3 or 4-Byte Address Mode
- ◆ High Speed Clock Frequency
 - 166MHz for fast read
 - Dual I/O Data transfer up to 332Mbits/s
 - Quad I/O Data transfer up to 664Mbits/s
 - QPI Mode Data transfer up to 664Mbits/s
 - DTR Quad I/O Data transfer up to 104MBytes/s
- ◆ Allows XIP (eXecute in Place) Operation
 - High speed Read reduce overall XIP instruction fetch time
 - Continuous Read with Wrap further reduce data latency to fill up SoC cache
- ◆ Software/Hardware Write Protection
 - Write protect all/portion of memory via software
 - Enable/Disable protection with WP# Pin
 - Individual Block Protection
- ◆ Data Integrity Check
 - On-chip ECC (1-bit correction/2-bit detection every 8-Byte)⁽¹⁾
 - CRC detects accidental changes to raw data
- ◆ Safety features
 - Safe boot detects POR failure, and provides retry options
- ◆ Fast Program/Erase Speed
 - Page Program time: 0.25ms typical
 - Sector Erase time: 30ms typical
 - Block Erase time: 0.12/0.15s typical
 - Chip Erase time: 75s typical
- ◆ Flexible Architecture
 - Sector of 4K-Byte
 - Block of 32/64K-Byte
 - Erase/Program Suspend/Resume
- ◆ Low Power Consumption
 - 15μA typical stand-by current
 - 2μA typical power-down current
- ◆ Advanced Security Features
 - 128-bit Unique ID
 - 3x4K-Byte Security Registers With OTP Lock
- ◆ Single Power Supply Voltage
 - Full voltage range: 1.65~2.0V
- ◆ Endurance and Data Retention
 - Minimum 100,000 Program/Erase Cycles
 - 20-year data retention typical
- ◆ Package Information
 - WSON8 (6x5mm)
 - WSON8 (8x6mm)
 - TFBGA-24ball (5x5 Ball Array)
 - SOP16 300mil

Note:

1. When ECC is enabled, it is required to program minimum one or multiple aligned 8-Byte granularities. Every aligned 8-Byte granularity should only be programmed once before Erase to ensure correct ECC operations.

2 GENERAL DESCRIPTIONS

The GD25LF256F (256M-bit) Serial NOR Flash supports the standard Serial Peripheral Interface (SPI), and supports the Dual and DTR Dual SPI, and supports the Quad and DTR Quad SPI: Serial Clock, Chip Select, Serial Data I/O0 (SI), I/O1 (SO), I/O2, I/O3, DQS, ECS# and RESET#. The Dual I/O data is transferred with speed of 332Mbit/s, and the Quad I/O data is transferred with speed of 664Mbit/s. The DTR Dual I/O data is transferred with speed of 416Mbits/s, and the DTR Quad I/O data is transferred with speed of 832Mbits/s.

CONNECTION DIAGRAM AND PIN DESCRIPTION

Figure 1 Connection Diagram for WSON8 package

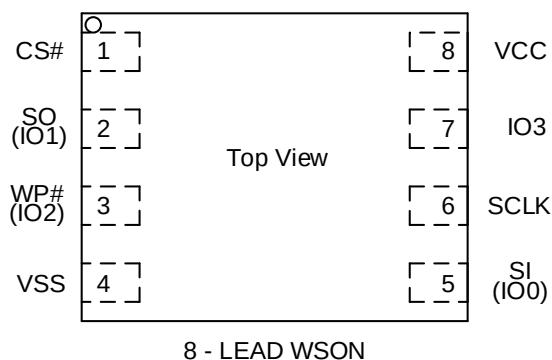


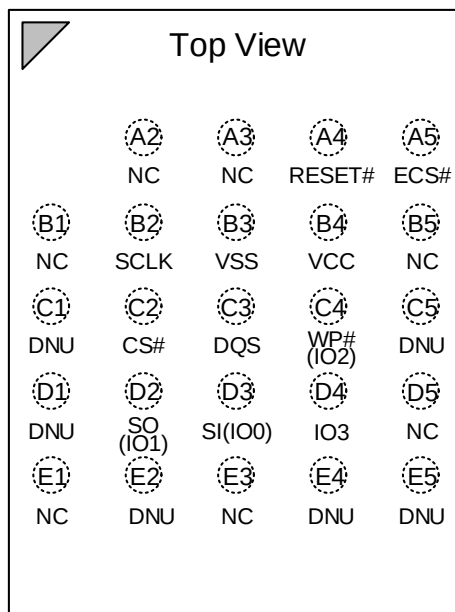
Table 1 Pin Description for WSON8 package

Pin No.	Pin Name	I/O	Description
1	CS#	I	Chip Select Input
2	SO (IO1)	I/O	Data Output (Data Input Output 1)
3	WP# (IO2)	I/O	Write Protect Input (Data Input Output 2)
4	VSS		Ground
5	SI (IO0)	I/O	Data Input (Data Input Output 0)
6	SCLK	I	Serial Clock Input
7	IO3	I/O	Data Input Output 3
8	VCC		Power Supply

Note:

1. CS# must be driven high if chip is not selected. Please don't leave CS# floating any time after power is on.

Figure 2 Connection Diagram for TFBGA24 5x5 ball array package



24-BALL TFBGA (5x5 ball array)

Table 2 Ball Description for TFBGA24 5x5 ball array package

Pin No.	Pin Name	I/O	Description
A4	RESET#	I	Reset Input
A5	ECS#	O	ECC Correction Signal
B2	SCLK	I	Serial Clock Input
B3	VSS		Ground
B4	VCC		Power Supply
C2	CS#	I	Chip Select Input
C3	DQS	O	Data Strobe Signal Output
C4	WP# (IO2)	I/O	Write Protect Input (Data Input Output 2)
C1/D1/C5/ E2/E4/E5	DNU		Do Not Use (It may connect to internal signal inside)
D2	SO (IO1)	I/O	Data Output (Data Input Output 1)
D3	SI (IO0)	I/O	Data Input (Data Input Output 0)
D4	IO3	I/O	Data Input Output 3

Notes:

1. CS# must be driven high if chip is not selected. Please don't leave CS# floating any time after power is on.
2. The DNU ball must be floating. It may connect to internal signal inside.
3. The NC ball is not connected to any internal signal. It is OK to connect it to the system ground (GND) or leave it floating.
4. The RESET# pin is a dedicated hardware reset pin regardless of device settings or operation states. If the hardware reset function is not used, it is recommended to connect it to VCC in the system but leaving it floating is OK.

Figure 3 Connection Diagram for SOP16 package

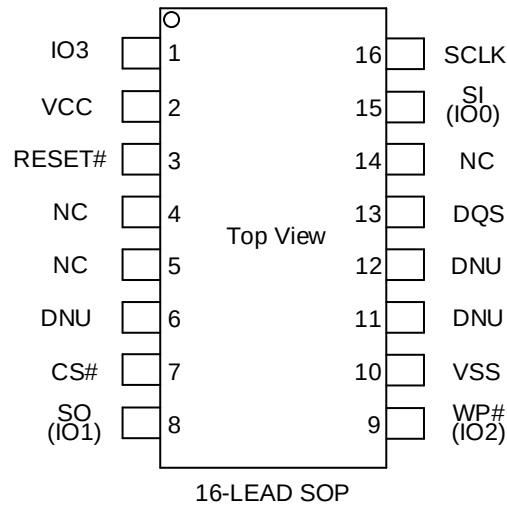


Table 3. Pin Description for SOP16 Package

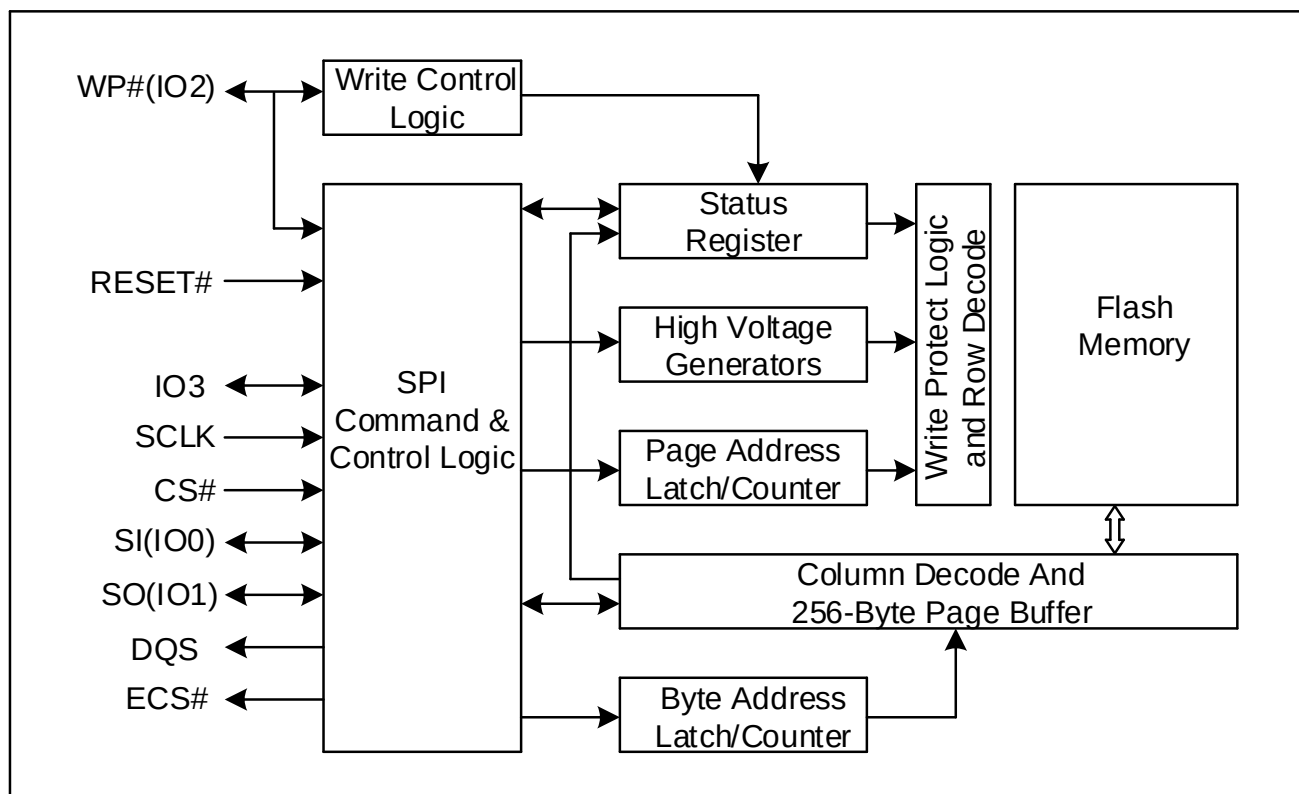
Pin No.	Pin Name	I/O	Description
1	IO3	I/O	Data Input Output 3
2	VCC		Power Supply
3	RESET#	I	Reset Input
7	CS#	I	Chip Select Input
8	SO (IO1)	I/O	Data Output (Data Input Output 1)
9	WP# (IO2)	I/O	Write Protect Input (Data Input Output 2)
10	VSS		Ground
6/11/12	DNU		Do Not Use (It may connect to internal signal inside)
13	DQS	O	Data Strobe Signal Output
15	SI (IO0)	I/O	Data Input (Data Input Output 0)
16	SCLK	I	Serial Clock Input

Notes:

1. CS# must be driven high if chip is not selected. Please don't leave CS# floating any time after power is on.
2. The DNU pin must be floating. It may connect to internal signal inside.
3. The NC pin is not connected to any internal signal. It is OK to connect it to the system ground (GND) or leave it floating.
4. The RESET# pin is a dedicated hardware reset pin regardless of device settings or operation states. If the hardware reset function is not used, it is recommended to connect it to VCC in the system but leaving it floating is OK.



BLOCK DIAGRAM



3 MEMORY ORGANIZATION

GD25LF256F

Each device has	Each block has	Each sector has	Each page has	
32M	64/32K	4K	256	Bytes
128K	256/128	16	-	pages
8K	16/8	-	-	sectors
512/1K	-	-	-	blocks

UNIFORM BLOCK SECTOR ARCHITECTURE

GD25LF256F 64K Bytes Block Sector Architecture

Block	Sector	Address range	
511	8191	1FFF000H	1FFFFFFFH
			
	8176	1FF0000H	1FF0FFFFH
510	8175	1FEF000H	1FEFFFFFH
			
	8160	1FE0000H	1FE0FFFFH
.....			
			
			
.....			
			
			
2	47	02F000H	02FFFFFFH
			
	32	020000H	020FFFFH
1	31	01F000H	01FFFFFFH
			
	16	010000H	010FFFFH
0	15	00F000H	00FFFFFFH
			
	0	000000H	000FFFFH

4 DEVICE OPERATIONS

4.1 SPI Mode

Standard SPI

The GD25LF256F features a serial peripheral interface on 4 signals bus: Serial Clock (SCLK), Chip Select (CS#), Serial Data Input (SI) and Serial Data Output (SO). Both SPI bus mode 0 and 3 are supported. Input data is latched on the rising edge of SCLK and data shifts out on the falling edge of SCLK.

Dual SPI

The GD25LF256F supports Dual SPI operation when using the “Dual Output Fast Read”, “Dual Output Fast Read with 4-Byte address”, “Dual I/O Fast Read” and “Dual I/O Fast Read with 4-Byte address” commands (3Bh, 3Ch, BBh and BCh). These commands allow data to be transferred to or from the device at twice the rate of the standard SPI. When using the Dual SPI command, the SI and SO pins become bidirectional I/O pins: IO0 and IO1.

DTR Dual SPI

The GD25LF256F supports DTR Dual SPI operation when using the “DTR Dual I/O Fast Read” (BDh/BEh) command. These commands allow data to be transferred to or from the device at four times the rate of the standard SPI, and data output will be latched on both rising and falling edges of the serial clock. When using the DTR Dual SPI command the SI and SO pins become bidirectional I/O pins: IO0 and IO1.

Quad SPI

The GD25LF256F supports Quad SPI operation when using the “Quad Output Fast Read”, “Quad I/O Fast Read”, “Quad Page Program” (6Bh/6Ch, EBh/ECh, 32h/34h) commands. These commands allow data to be transferred to or from the device at four times the rate of the standard SPI. When using the Quad SPI command the SI and SO pins become bidirectional I/O pins: IO0 and IO1, and the WP# pins become bidirectional I/O pins: IO2.

DTR Quad SPI

The GD25LF256F supports DTR Quad SPI operation when using the “DTR Quad I/O Fast Read” (EDh/EEh) command. These commands allow data to be transferred to or from the device at eight times the rate of the standard SPI, and data output will be latched on both rising and falling edges of the serial clock. When using the DTR Quad SPI command the SI and SO pins become bidirectional I/O pins: IO0 and IO1, and the WP# pins become bidirectional I/O pins: IO2.

4.2 QPI Mode

The GD25LF256F supports Quad Peripheral Interface (QPI) operations only when the device is switched from Standard/Dual/Quad SPI mode to QPI mode using the “Enable the QPI (38h)” command. The QPI mode utilizes all four IO pins to input the command code. Standard/Dual/Quad SPI mode and QPI mode are exclusive. Only one mode can be active at any given times. “Enable the QPI (38h)” and “Disable the QPI (FFh)” commands are used to switch between these two modes. Upon power-up and Hardware Reset or after Software Reset using “Enable Reset (66h) and Reset (99h)” command, the default state of the device is Standard/Dual/Quad SPI mode.

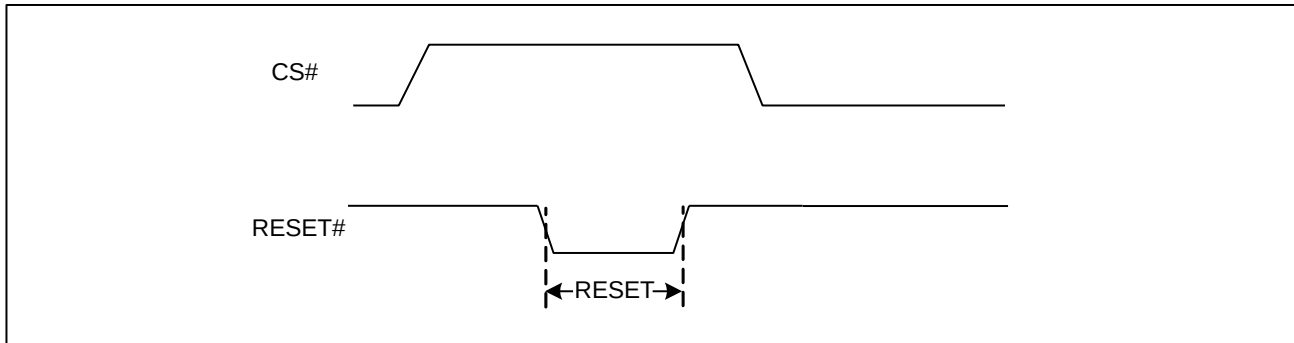
4.3 RESET Function

The RESET# pin allows the device to be reset by the control.

The RESET# pin goes low for a minimum period of tRLRH will reset the flash. After reset cycle, the flash is at the following states:

- Standby mode
- All the volatile bits will return to the default status as power on.

Figure 4 RESET Condition



4.4 Data Strobe Function

The Data Strobe function is enabled as default for this device, and the DQS pin (ball C3 of TFBGA-24 package, pin 13 of SOP16 package) is an active output pin for the Data Strobe (DQS) signal during Read operations. The DQS signal is typically used in high speed applications to indicate when the output data is ready to be fetched by the controllers. When the data strobe function is enabled, DQS signal is driven to ground once CS# goes LOW, and will start to toggle when the output data is ready on the I/O pins. The toggling frequency is the same as the CLK frequency. For STR Read operations, the data should only be latched on the rising edge of DQS signal. For DTR Read operations, the data should be latched on both rising edge and falling edge of the DQS signal.

For DTR Dual I/O Fast Read commands, DTR Quad I/O Fast Read commands and DTR Burst Read with Wrap, If DLP enable, the "Data Learning Pattern" will output with DQS signal.

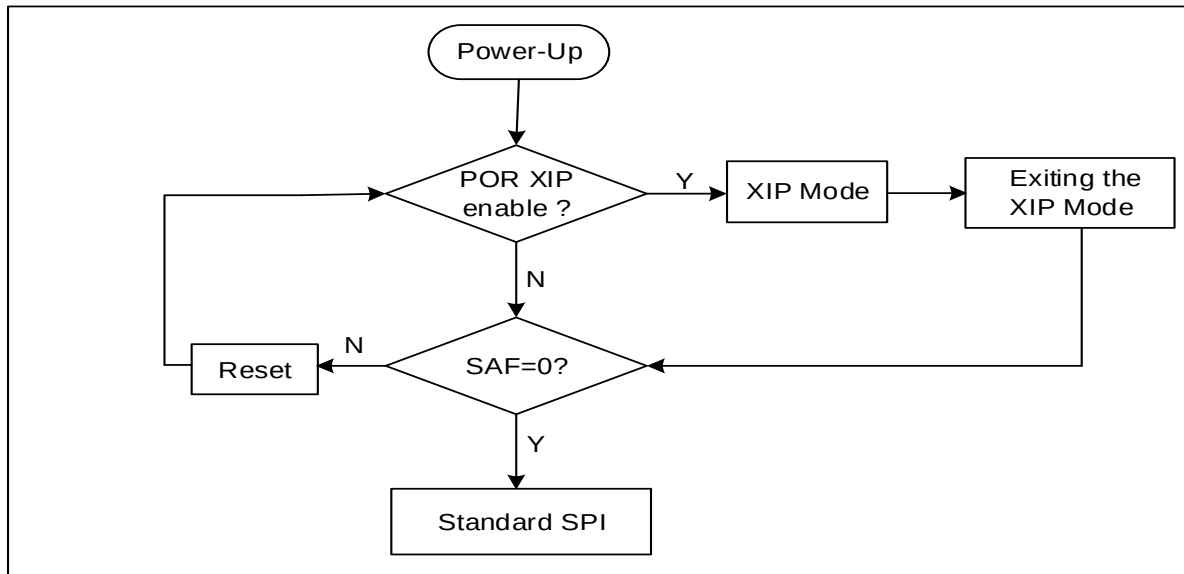
4.5 Enter XIP Mode Directly After Power-On (POR-XIP)

The device can be operated in the selected XIP mode immediately after power-on reset. The XIP mode after Power-on is set by Nonvolatile Configuration Register bits in Byte<0>. Because the device boots directly in XIP, after the power cycle, no command code is necessary. XIP is terminated by setting the "Continuous Read Mode" bits (M5-4) do not equal (1, 0).

4.6 Safe boot

The Safe boot feature allows to detect power up failure by reading SAF bit in Extended Status Register. If SAF is 1, power up is failure and Software/Hardware Reset or a power-down, power-up cycle must be executed to power up again. When the POR-XIP function is enabled, SAF can be read only after exiting the XIP mode. When the POR-XIP function is disable, SAF can be read directly. When power up is failure, Software/Hardware Reset need tRST_S busy time.

Figure 5 Safe boot flow



5 DATA PROTECTION

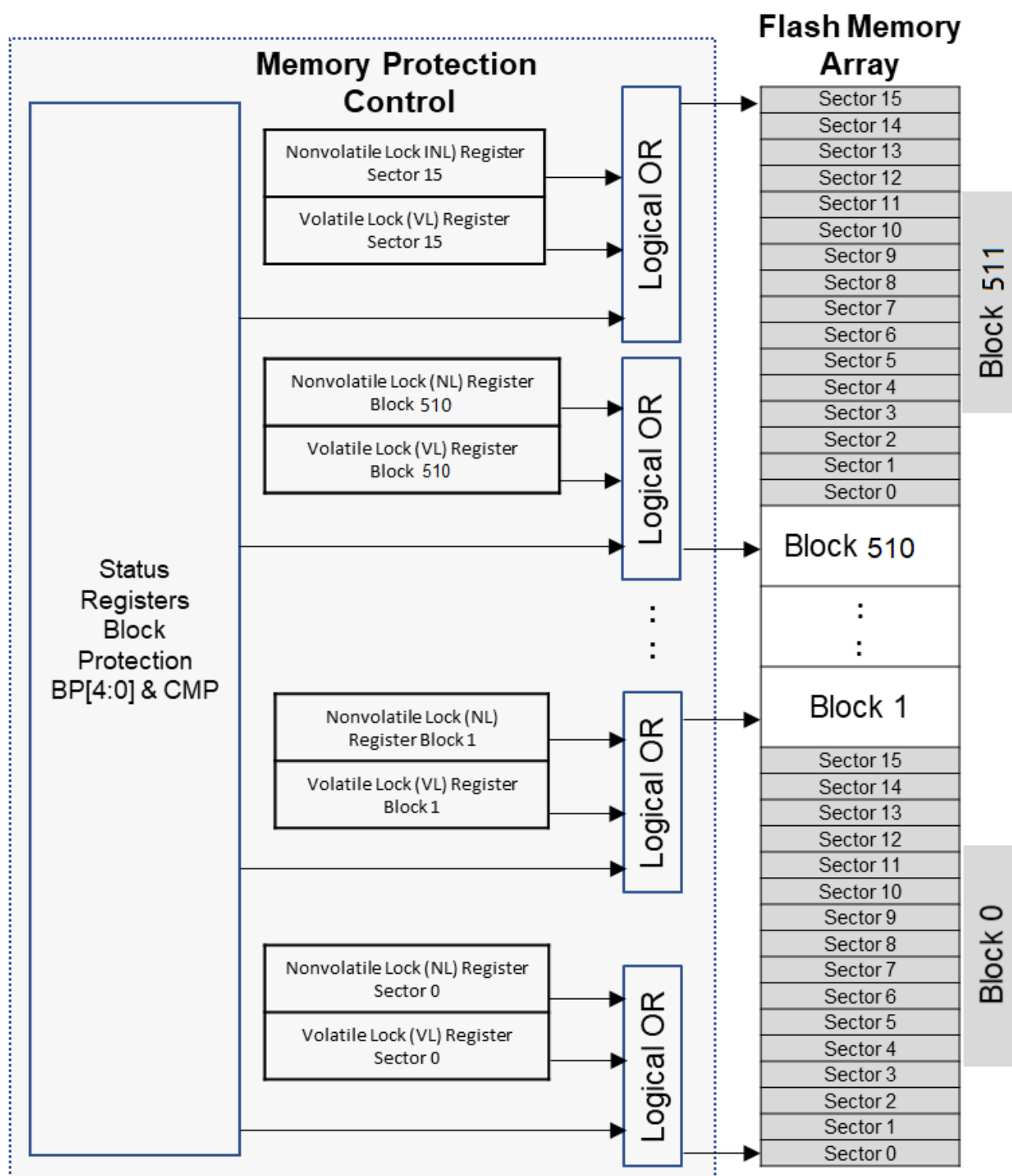
The GD25LF256F provides the following data protection methods:

- ◆ Write Enable (WREN) command: The WREN command sets the Write Enable Latch (WEL) bit to '1'. The WEL bit will reset to '0' after the following sequence or instructions:
 - Power-Up/ Software Reset (66h+99h)/Hardware Reset
 - Write Disable (WRDI)
 - Write Status Register (WRSR 1, 2 & 3)
 - Write Extended Address Register (WEAR)
 - Write Nonvolatile Configuration Register (WNVCR)
 - Write Volatile Configuration Register (WVCR)
 - Write VL Register (WVLR)
 - Page Program (PP)
 - Sector Erase (SE) / Block Erase (BE) / Chip Erase (CE)/Erase Security Registers / Program Security Registers
 - Advanced Block Protection related instructions: Set Nonvolatile Lock (NL) Register, Clear All Nonvolatile Lock (NL) Registers, Write Volatile Lock(VL)Register,Program Password Register
- ◆ Software Protection Mode:
 - The Block Protect (BP4, BP3, BP2, BP1, and BP0 along with the CMP) bits define the section of the memory array that can be read but cannot be changed.
 - Nonvolatile Lock (NL) Registers are additional memory non-volatile lock registers that can protect or unprotect a memory with greater granularity at the individual sectors/block level.
 - Volatile Lock (VL) Registers like NL Registers serve similar lock bit functionality at individual sector/block level except they are volatile bits.
- ◆ Advanced Block Protection with Password: Password Protection Mode adds another layer of write protection using a 64-bit unlock/lock Password to enable/disable writes to the legacy Status Registers Block Protection (BP[4:0] and CMP), and Nonvolatile Lock (NL) Registers. The Block Protect bits and NL Registers protect/unprotect the memory against program or erase. Unlock Password instruction must be executed successfully to change the block protection and erase/program the memory.
- ◆ Hardware Protection Mode when SRP0 is '1': WP# goes low to protect the Status Register bits (i.e., BP0~BP4, CMP bits) and Nonvolatile Lock (NL) Registers against writes.
- ◆ Deep Power-Down Mode: In Deep Power-Down Mode, all commands are ignored except the Release from Deep Power-Down Mode command and reset command (66h+99h).

The GD25LF256F device supports several memory protection methods: Block Protection via Block Protect/CMP bits, Individual Volatile Block Lock Protection via VL Register bits, Individual Nonvolatile Block Lock Protection via NL Register Bits. All these protection schemes function in a logical OR method to protect or unprotect targeted sector/block as shown on Figure 6.

An Advanced Memory Protection with Password adds an additional security layer on updating the memory block protection configuration and associated code and data memory. Once Password Protection mode is enabled, a password verification is required to unprotect and enable modification of the supported block protection setup.

Figure 6 Memory Protection Control Overview



Notes:

1. The first and last blocks will have NL/VL Registers protections at the 4KB sector level. Each 4KB sector in these blocks can be individually locked by NL/VL Registers setting.
2. Each of the middle 64KB blocks has NL/VL Registers protections at the 64KB block level. Each 64KB block can be individually locked by NL/VL Registers setting.



Status Register Block Protection (BP4, BP3, BP2, BP1, BP0, CMP)

Table 4. GD25LF256F Protected area size (CMP=0)

Status Register Content					Memory Content			
BP4	BP3	BP2	BP1	BP0	Blocks	Addresses	Density	Portion
X	0	0	0	0	NONE	NONE	NONE	NONE
0	0	0	0	1	511	01FF0000h-01FFFFFFh	64KB	Upper 1/512
0	0	0	1	0	510 to 511	01FE0000h-01FFFFFFh	128KB	Upper 1/256
0	0	0	1	1	508 to 511	01FC0000h-01FFFFFFh	256KB	Upper 1/128
0	0	1	0	0	504 to 511	01F80000h-01FFFFFFh	512KB	Upper 1/64
0	0	1	0	1	496 to 511	01F00000h-01FFFFFFh	1MB	Upper 1/32
0	0	1	1	0	480 to 511	01E00000h-01FFFFFFh	2MB	Upper 1/16
0	0	1	1	1	448 to 511	01C00000h-01FFFFFFh	4MB	Upper 1/8
0	1	0	0	0	384 to 511	01800000h-01FFFFFFh	8MB	Upper 1/4
0	1	0	0	1	256 to 511	01000000h-01FFFFFFh	16MB	Upper 1/2
1	0	0	0	1	0	00000000h-0000FFFFh	64KB	Lower 1/512
1	0	0	1	0	0 to 1	00000000h-0001FFFFh	128KB	Lower 1/256
1	0	0	1	1	0 to 3	00000000h-0003FFFFh	256KB	Lower 1/128
1	0	1	0	0	0 to 7	00000000h-0007FFFFh	512KB	Lower 1/64
1	0	1	0	1	0 to 15	00000000h-000FFFFFh	1MB	Lower 1/32
1	0	1	1	0	0 to 31	00000000h-001FFFFFFh	2MB	Lower 1/16
1	0	1	1	1	0 to 63	00000000h-003FFFFFFh	4MB	Lower 1/8
1	1	0	0	0	0 to 127	00000000h-007FFFFFFh	8MB	Lower 1/4
1	1	0	0	1	0 to 255	00000000h-00FFFFFFh	16MB	Lower 1/2
X	1	1	0	X	ALL	00000000h-01FFFFFFh	32MB	ALL
X	1	X	1	X	ALL	00000000h-01FFFFFFh	32MB	ALL

Table 5. GD25LF256F Protected area size (CMP=1)

Status Register Content					Memory Content			
BP4	BP3	BP2	BP1	BP0	Blocks	Addresses	Density	Portion
X	0	0	0	0	ALL	00000000h-01FFFFFFh	ALL	ALL
0	0	0	0	1	0 to 510	00000000h – 01FEFFFFh	32,704KB	Lower 511/512
0	0	0	1	0	0 to 509	00000000h – 01FDFFFFh	32,640KB	Lower 255/256
0	0	0	1	1	0 to 507	00000000h – 01FBFFFFh	32,512KB	Lower 127/128
0	0	1	0	0	0 to 503	00000000h – 01F7FFFFh	32,256KB	Lower 63/64
0	0	1	0	1	0 to 495	00000000h – 01EFFFFFh	31MB	Lower 31/32
0	0	1	1	0	0 to 479	00000000h – 01DFFFFFh	30MB	Lower 15/16
0	0	1	1	1	0 to 447	00000000h – 01BFFFFFh	28MB	Lower 7/8
0	1	0	0	0	0 to 383	00000000h – 017FFFFFh	24MB	Lower 3/4
0	1	0	0	1	0 to 255	00000000h – 00FFFFFFh	16MB	Lower 1/2
1	0	0	0	1	1 to 511	00010000h – 01FFFFFFh	32,704KB	Upper 511/512



1	0	0	1	0	2 to 511	00020000h – 01FFFFFFh	32,640KB	Upper 255/256
1	0	0	1	1	4 to 511	00040000h – 01FFFFFFh	32,512KB	Upper 127/128
1	0	1	0	0	8 to 511	00080000h – 01FFFFFFh	32,256KB	Upper 63/64
1	0	1	0	1	16 to 511	00100000h – 01FFFFFFh	31MB	Upper 31/32
1	0	1	1	0	32 to 511	00200000h – 01FFFFFFh	30MB	Upper 15/16
1	0	1	1	1	64 to 511	00400000h – 01FFFFFFh	28MB	Upper 7/8
1	1	0	0	0	128 to 511	00800000h – 01FFFFFFh	24MB	Upper 3/4
1	1	0	0	1	256 to 511	01000000h – 01FFFFFFh	16MB	Upper 1/2
X	1	1	0	X	NONE	NONE	NONE	NONE
X	1	X	1	X	NONE	NONE	NONE	NONE

Volatile Lock (VL) Registers

The GD25LF256F has a total of 542 Volatile Lock Registers. Each VL Register has an 8-bits. The VL Register is read independently by Read VL Register (E0h) using either a sector or block address. Each VL Register is written independently by Write VL Register (E1h) instruction using either a sector or block address. All VL Registers are cleared collectively by Global VL Registers Unlock (98h) instruction and set collectively by Global VL Registers Lock (7Eh) instruction.

The bottom and top 64KB Blocks have individual volatile protection lock registers on each 4KB sectors (total 32). The middle 64KB Blocks have their own individual volatile block lock registers (total 510). The VL register mapping assignments to the memory sectors/blocks are illustrated on Figure 6. The default value of the VL Registers from power up or reset is based on the VL Default value (NVCR Address <02> Bit 2). VL Register value of '00h' indicates the corresponding sectors or blocks are unprotected. A VL Register value of 'FFh' indicates the corresponding sectors or blocks are protected.

No.	Name	Description	Note
542 VL Registers[Sector/Block Address] = 00h or FFh ⁽¹⁾	VL Registers	Each VL Register protects or unprotect a corresponding sector or block: 00h = Unprotected Sector/Block FFh = Protected Sector/Block Others = Not Supported ⁽²⁾	Volatile Writable

Notes:

1. VL Registers mapping across the full memory is illustrated on Figure 6. NL and VL Registers Assignment on the Memory. Default value of the VL Registers from power up or reset is based on VL Default value set on NVCR Address <02> Bit 2.
2. Only valid data (00h or FFh) are valid and accepted during Write VL Register. Other data combinations are not valid data input; the Write VL Register command will be ignored; and the WEL bit will not be cleared.

Nonvolatile Lock (NL) Registers

The GD25LF256F has a total of 542 Nonvolatile Lock Registers that are non-volatile. Each NL Register has an 8-bits. The NL Register is read independently using either a sector or block address using Read NL Register (E2h) instruction. Each NL Register is also set independently by Set NL Register (E3h) instruction using either a sector or block address. All NL Registers are cleared collectively by Clear All NL Registers (E4h) instruction.

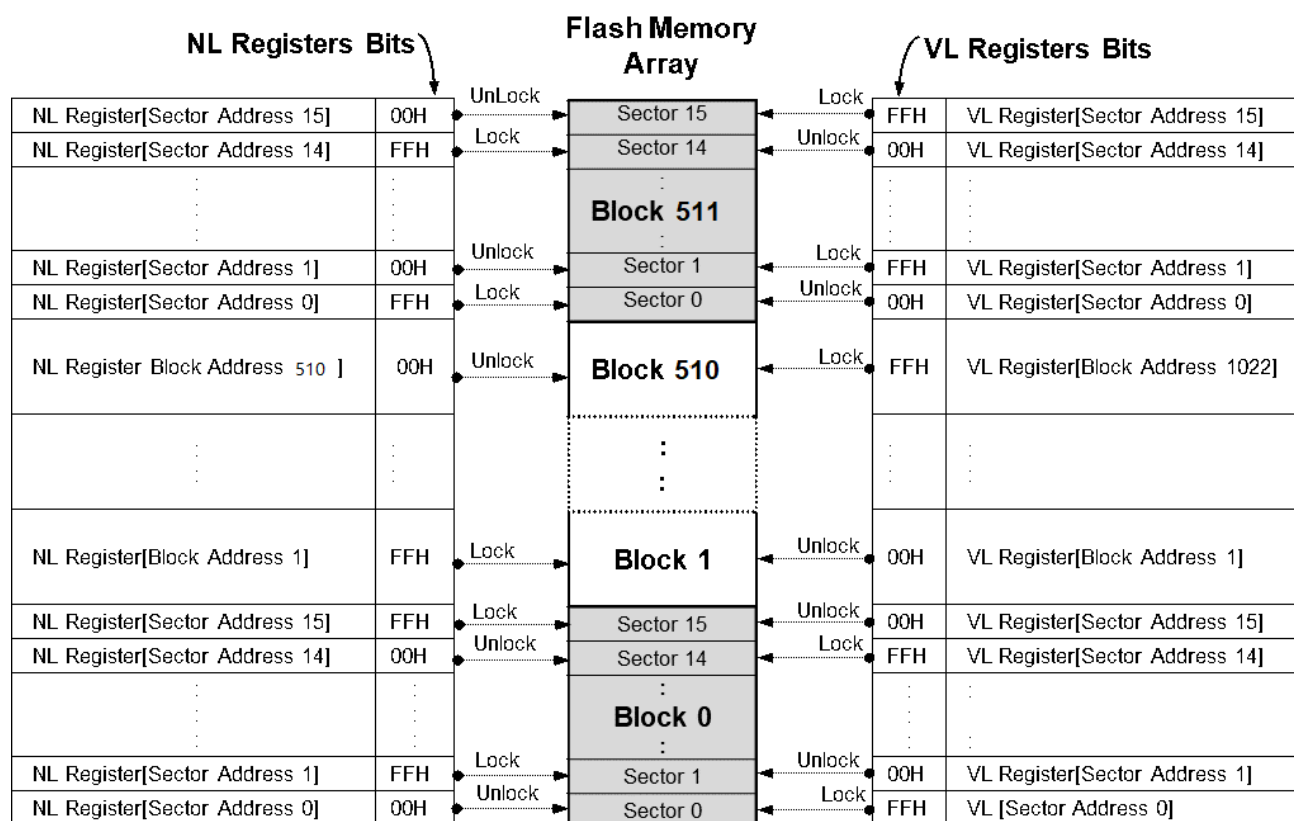
The bottom and top 64KB Blocks have individual non-volatile protection lock registers on each 4KB sectors (total 32). The middle 64KB Blocks have their own individual non-volatile block lock registers (total 510). The NL Register mapping assignments to the memory sectors/blocks are illustrated on Figure 6. The factory default value of the NL Registers is '00h' indicating the corresponding sectors or blocks are unprotected. When NL Register is set to 'FFh', the corresponding sector or block is protected against program or erase. The NL Registers are programmable and erasable as default and will require successful Unlock Password initiation if password protection mode is enabled.

No.	Name	Description	Note
542 NL Registers[Sector/Block Address] = 00h or FFh ⁽¹⁾	NL Registers	Each NL Register protects or unprotect a corresponding sector or block: 00h = Unprotected Sector/Block FFh = Protected Sector/Block Others = Not Supported	Non-Volatile Writable

Note:

1. NL Registers mapping across the full memory is illustrated on Figure 6 (NL and VL Registers Assignment on the Memory).

Figure 7. NL Registers and VL Registers Assignment on the Memory



Notes:

1. The first and last blocks will have NL/VL Registers protections at the 4KB sector level. Each 4KB sector in these blocks can be individually locked by NL/VL Registers setting.
2. Each of the middle 64KB blocks has NL/VL Registers protections at the 64KB block level. Each 64KB block can be individually locked by NL/VL Registers setting.

Password Register

Password Register is a 64-bit register that can be read or programmed when PWD bit is '0' and PWDVfy bit is '0'. When the device is in Password Protected Mode (PWD bit=1), the 64-bit Password is hidden and locked. It can't be read or programmed as it is in hidden/locked state. The instructions associated with Password Register are Read Password Register (27h), Program Password Register (28h), WRSR-3(11h) – set PWD bit to '1' and hide/lock the 64-bit Password, and Password Unlock/Lock (29h) instructions.

No.	Name	Description	Note
Password Bit[63:0]	Password Register	64-bit Password Register that is programmable and can be read while PWD bit (Status Register-3 Bit 2) is 0. It will be OTP locked and hidden once PWD bit is set to 1. Default value is 'FFFF FFFF FFFF FFFFh'.	Non-volatile Writable (OTP)

Advanced Memory Protection with Password

The GD25LF256F supports an advanced memory protection and boot-memory configuration that uses a 64-bit Password to unlock and enables write to the device Block Protect (BPx, CMP, SRP1, SRP0) bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit. The advanced memory protection adds another level of memory write protection and boot-memory configuration schemes: a) write protection using registers; b) read boot-memory option access (bottom 16Mbit boot-memory). The lists of registers associated with the advanced memory protection are as follows: Password Protection Mode (PWD) bit, Password Read Boot-Memory (PWDRD) bit, Password Verify (PWDVFY) bit, Password Register, Status Registers Block Protect related bits (BPx, CMP, SRP1, SRP0), and NL Registers.

Password Protection mode (PWD) bit must be set to enable the Advanced Memory Protection with Password. The OTP PWD bit (Status Register-3 Bit 7 – S22) must be set to '1' to enable password protection mode. Before enabling this bit, the unique 64-bit Password is needed to be program and securely stored. Successful password verification is required to disable the protection placed on Block Protect (BPx, CMP, SRP1, SRP0) bits, NL Registers, and Password Read Boot-Memory bit. When the device is placed in password protection mode (PWD =1). Once the protection is disabled or unlock (PWD, PWDVFY = 1, 1), the Block Protect (BPx, CMP, SRP1, SRP0) bits, NL Registers, and Password Read Boot-Memory bit are writable.

A bottom boot-memory block configuration is supported within Password Protection Mode. The special boot-memory configuration mode is entered by enabling Password Protection mode (PWD=1), followed by a successful Password Unlock (29h) + Valid 64-bit Password instruction sequence to set PWDVFY bit to 1. Once (PWD, PWDVFY = 1, 1), the Password Read Boot-Memory (PWDRD) bit can be set to '1', enabling access to the bottom 16Mb memory address once PWDVFY bit is cleared to '0' by "Password Lock (29h) + Invalid 64-bit Password" instruction sequence or reset. The Password Protection Mode (PWD) bit, Password Verify (PWDVFY) bit, and Password Read Boot-Memory (PWDRD) bit table below shows the Advanced Memory Protection configuration that includes the normal operation mode, the Password Protection Mode, and the accessible memory address range in Password Read Boot-Memory mode:

PWD (OTP)	PWDRD (Nonvolatile)	PWDVFY (Volatile)	Advanced Memory Protection Configuration ^(1,2)	Description
0	0	0	Normal Protection Mode; Full Memory Access	The Advanced Memory Protection with Password is disabled. Device normal protection is supported. (Default)
1	0	0	Password Protected Mode Lock; Full Memory Access	Password Protected Mode Enabled: Block Protect (BPx and CMP) bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are write protected; Full Memory is accessible.
1	0	1	Password Protected Mode Unlock; Full Memory Access	Password Protected Mode Unlock: Block Protect (BPx and CMP) bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are writable. ⁽⁴⁾ Full Memory is accessible.
1	1	0	Password Protected Mode Lock; Password Read Boot- Memory mode: Bottom 16Mbit	Password Protected Mode Enabled: Block Protect (BPx and CMP) bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are write



			Boot-Memory Access only	protected; Bottom 16Mbit Boot-Memory is only read/program/erase accessible. ⁽³⁾
1	1	1	Password-Protected Mode Unlock; Password Read Boot-Memory mode disabled: Full Memory Access	Password Protected Mode Enabled: Block Protect (BPx and CMP) bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are writable. ⁽⁴⁾ Full Memory is accessible.

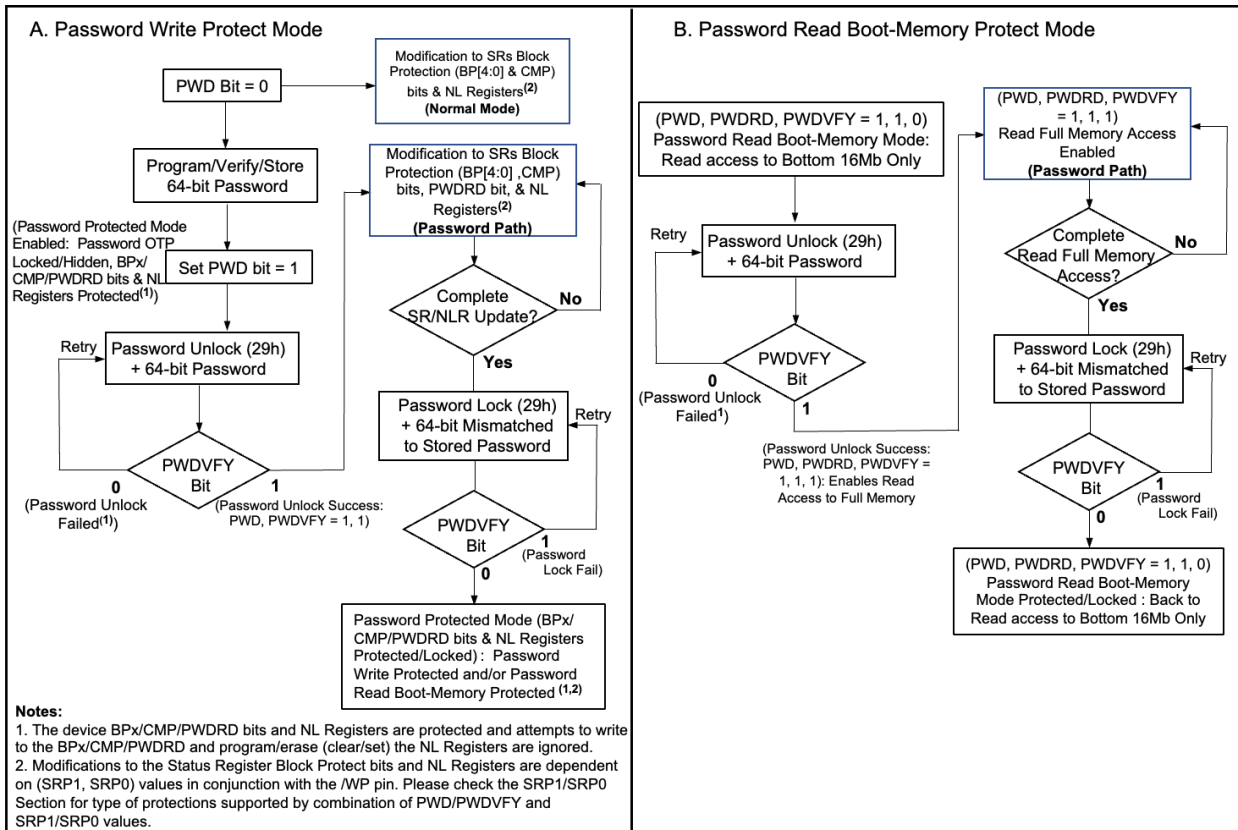
Notes:

1. When PWD bit is set to '1' the Advanced Memory Protection with Password is enabled. PWDVIFY bit must be set to '1' by a successful "Password Unlock (29h) + Valid 64-bit Password" instruction sequence to unlock and enable write to the Block Protect (BPx and CMP) bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit.
2. When (PWD, PWDVIFY = 1, 1) and if PWDRD bit is set to '1', the device full memory is still accessible. Once PWDVIFY is cleared to '0' by a "Password Lock (29h) + Invalid 64-bit Password" instruction sequence or reset, the device accessible (readable/programmable/erasable) memory will only be the bottom 16Mbit Boot-Memory range. In this mode, the >16Mb address range is programmable and erasable, however it is read inaccessible.
3. Password Read Boot-Memory Mode (PWD, PWDRD, PWDVIFY = 1, 1, 0) enables bottom 16Mb boot-memory only access. The full chip, including the read inaccessible >16Mb memory address range, is programmable or erasable, however the read inaccessible >16Mb address range is not read verifiable in this mode. PWDVIFY bit must be set to '1' by a successful "Password Unlock (29h) + Valid 64-bit Password" instruction sequence to read verify the read inaccessible memory that was programmed or erased.
4. Please check the SRP1/SRP0 Section for additional information on Write Protection including hardware write protection option.

Please check the relevant register sections (PWD/PWDVY/PWDRD bits, Password Register, BPx/CMP Bits, NL Registers, and SRP1/SRP0) for detailed operation about Advanced Memory Protection. The Advanced Memory Protection Flow (Figure 8) shows the advanced protection method with password verification.

Note: PWD bit is an OTP bit, once set PWD=1, it enters password protect mode and cannot be changed (Please remember the password before setting PWD=1). If password protect mode is not used, please do not set PWD=1. Once set, only password protect mode can be used;

Figure 8 Advanced Memory Protection Flow



6 DATA INTEGRITY CHECK

The data storage and transmission errors will cause unexpected Flash device variation that makes a harmful impact on overall system functions. To prevent these errors, GD25LF256F product provides advanced Data Integrity Check function. For the data storage and data transmission in the flash device, Data Integrity Check can check errors and correct them, allowing self-checking and preventing errors in advance.

The Data Integrity Check function includes two methods:

- **ECC (Error Checking and Correcting): to prevent the data storage errors**
- **CRC (Cyclic Redundancy Check): to prevent the data transmission errors**

The register data and software signals can also be used to associate the Data Integrity Check function to fully record the results of checking, and can also immediately feedback.

6.1 ECC (Error Checking and Correcting)

Error Correction Codes (ECC) is a commonly used technique in non-volatile memory to reduce the device Bit Error Rate (BER) during the device operation life and improve device reliability. To achieve error detection and correction, redundancy data must be added to store the ECC calculation results for a given length of data. In GD25LF256F, every aligned 8-Byte data (A[2:0] = 0, 0, 0) stored in the memory array will be checked by the internal ECC engine using SEC-DED (Single Error Correction - Double Error Detection) Hsiao Codes algorithm. With 8-Byte ECC data granularity, ECC calculation latency time can be minimized and highest level of data integrity can be preserved.

The default value of all memory data is FFh (Erased) when the device is shipped from the factory. A “Page Program (02h/12h)”, “Quad Page Program (32h/34h)” command can be used to program the user data into the memory array. When ECC is enabled, ECC calculation will be performed during the internal programming operation and the results are stored in the redundancy or spare area of the memory array. It is necessary to program every page in aligned 8-Byte granularity, so that ECC engine can store and preserve the correct ECC information. Multiple programming attempts to previously programmed 8-byte aligned memory will disable the internal ECC engine and trigger a double programmed status. Once the 8-byte aligned memory chunk internal ECC is disabled, read memory access will only be directed to the memory array without error correction. An Erase is required to recover back ECC enable of these affected 8-Byte aligned memories.

A technique previously known as “Incremental Byte/Bit Programming to the same Byte location” can be used for GD25LF256F when ECC is enabled; however, the 8-byte aligned memory locations that were attempted with multiple programming will have ECC disabled and read operation will only access the memory array without ECC. After the read operation, double programmed status can also be reflected on Volatile Configuration Register Address <03h> Bit [6:4] ECCFS = ‘1xx’ value.

A Read operation can start from any Byte address and continue through the entire memory array, so it is not necessary to align the 8-Byte granularity boundary address to start a Read command. During data read operations, the internal ECC engine will check the ECC results stored in the spare area and apply necessary error correction or error detection to the main array data being read out. It is necessary to check the ECC Status Bits (SEC and DED) in the ECCFS after every Read operation to see if the data read out contains error(s) or not. As previously stated, double programmed status is also detected by the ECCFS bits.



Additional hardware monitoring of the ECC status can also be used to observe the ECC status in real time during any data output. When configured, the ECS# (ECC Correction Signal) pin will be pulled low during any aligned 8-Byte data output if it contains SEC, DED, or Double Programmed event.

The Sec, DED, ECCFS, and ECCCNT, and ECCFA can be reset through any of the following situations:

- Sending a new Read Command
- Issuing Software Reset Command
- Hardware Reset
- Power-up cycle

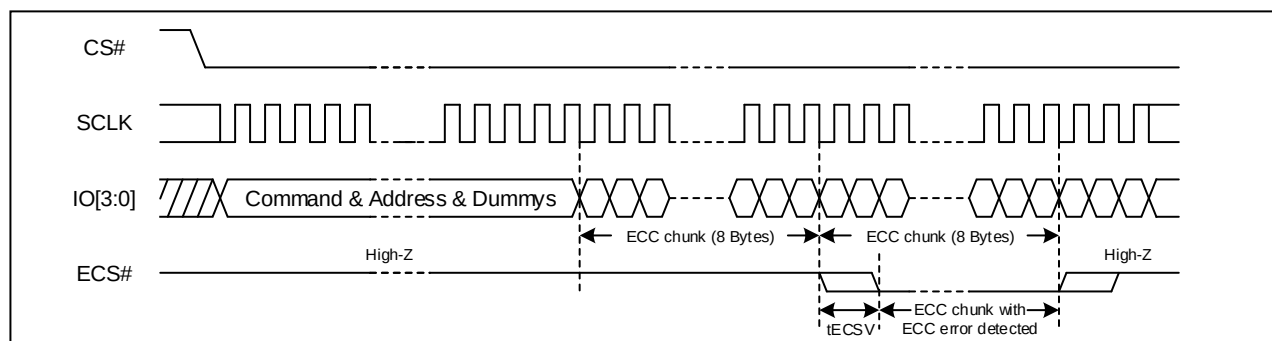
When an address range is erased, the target memory and the stored ECC data is reverted to its erased FFh state as well as clear the double programmed states.

6.2 ECS# (Error corrected Signal) Pin

The ECS# pin is a real time hardware signal to feedback the ECC correction status. It is enabled through 'ECC & ECS# Configuration (ECCECS#)' on either Nonvolatile/Volatile Configuration Register to detect ECC events during the memory read. The ECS# pin is designed as an open drain structure and a pull-up resistor (R_p) is required. In normal situation, the ECS# is kept on High-Z state. Once error correction begins, the ECS# pin will pull low during the whole ECC chunk unit after a duration of t_{ECSV} delay timing.

Depending on the ECCECS# configuration, the ECS# (ECC Correction Signal) pin will be pulled low during any aligned 8-Byte data output if it contains SEC (Single Error Correction), DED (Double Error Detection event), or Double Programmed state.

Figure 9. ECS# Timing



6.3 Interface CRC

Interface CRC performs a hardware accelerated CRC calculation on the communication between a host and the flash, ensuring the integrity of information transferred. A CRC is an error-detecting code commonly used in flash to detect accidental changes to raw data. Interface CRC protection is enable/disable by setting the non-volatile or volatile Configuration Register Byte<2>.

The Interface CRC method relies entirely on the host to verify the CRC check-value and take appropriate actions. The device calculates the CRC check-value which the host reads using the Read Interface CRC Register command (64h). The check-value calculated includes all command contents while CS# is LOW, command, address and data. This CRC checksum can be generated across either a single command or a set of commands.

The host must also calculate the CRC check-value over the same command sequence. When ready, the host can read the flash's calculated CRC check-value and compare it with its own. If there is a mismatch, the host can choose to repeat the complete command sequence.

CRC32 Polynomial(Same with AUTOSAR): $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X^1 + 1$

Notes:

1. At the end of the CRC read command, the device resets the CRC check-value and reinitializes the CRC polynomial.
2. The CRC polynomial between the host and the device must be identical.
3. Interface CRC is supported with STR/DDR protocol.
4. The Interface CRC check-value will reset to 0xFFFFFFFFh under the following conditions:
 - POR
 - Hardware Reset
 - Software Reset
 - A read of the Interface CRC check-value

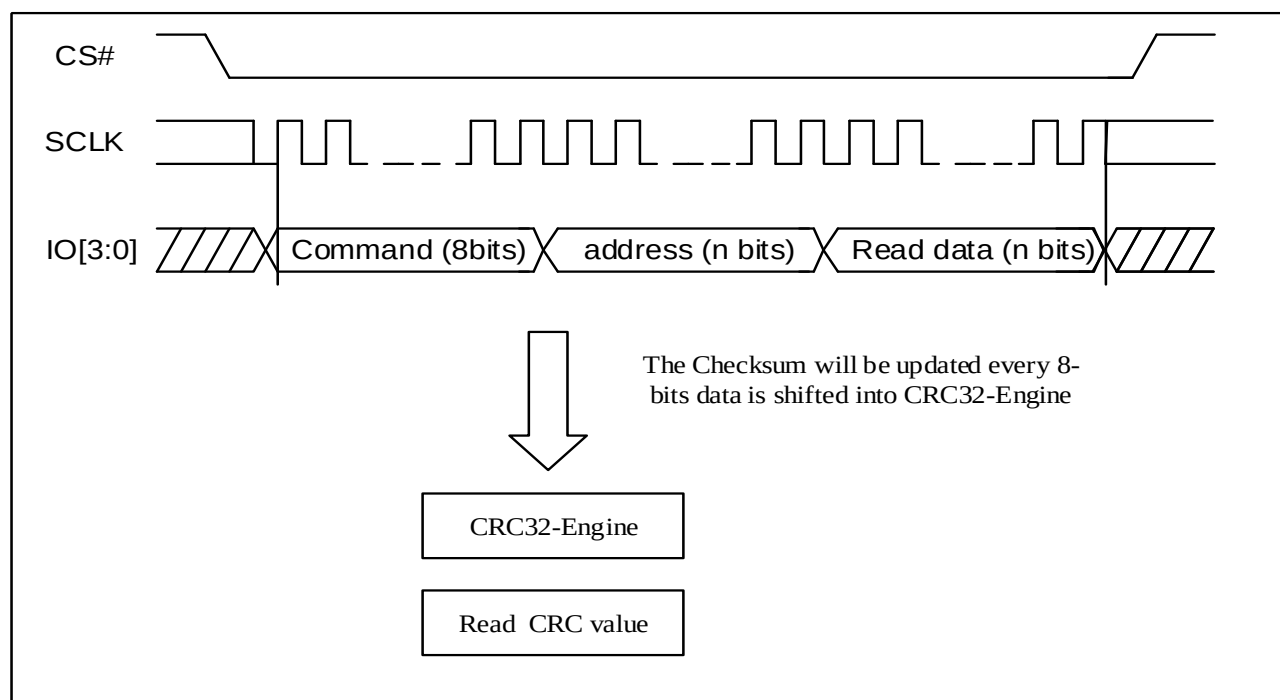
5. If a command is aborted before the command is legally received, i.e. the transfer length is cut short by CS# de-asserting early - the transferred data will still be clocked into the CRC check-value, but it is no longer guaranteed. When using Interface CRC, only valid, non-aborted commands must be used.
6. Interface CRC only support read (03h/13h, 0Bh/0Ch, 3Bh/3Ch, 6Bh/6Ch, BBh/BCh, EBh/ECh, BDh/BEh, EDh/EEh, 0Ch, 0Eh), program (02h/12h, 32h/34h), erase (20h/21h, 52h/5Ch, D8h/DCh, 60h/C7h), suspend (75h), resume (7Ah), write enable (06h), write disable (04h), enable QPI (38h), disable QPI (FFh); If other commands are used, the CRC check-value is no longer guaranteed;
7. When Interface CRC is disabled, the interface CRC register value becomes indeterminate. It is recommended to read the interface CRC register before disabling the interface CRC feature, and again after enabling the interface CRC feature to re-initialize the CRC calculation.
8. Dummy and DLP are not included in CRC calculation.

Read

The Read command is entered by driving CS# Low, then the memory contents are shifted out. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out.

The flash will calculate the CRC of all instruction sequence (dummy and DLP are not included in CRC calculation) using the proposed CRC32 polynomial. As soon as CS# is driven high, the CRC calculation will be stopped and the checksum latched into the CRC Register. If multiple Read instructions are executed, flash will continuously update the CRC checksum;

Figure 10. CRC32 Protection of Read



Note:

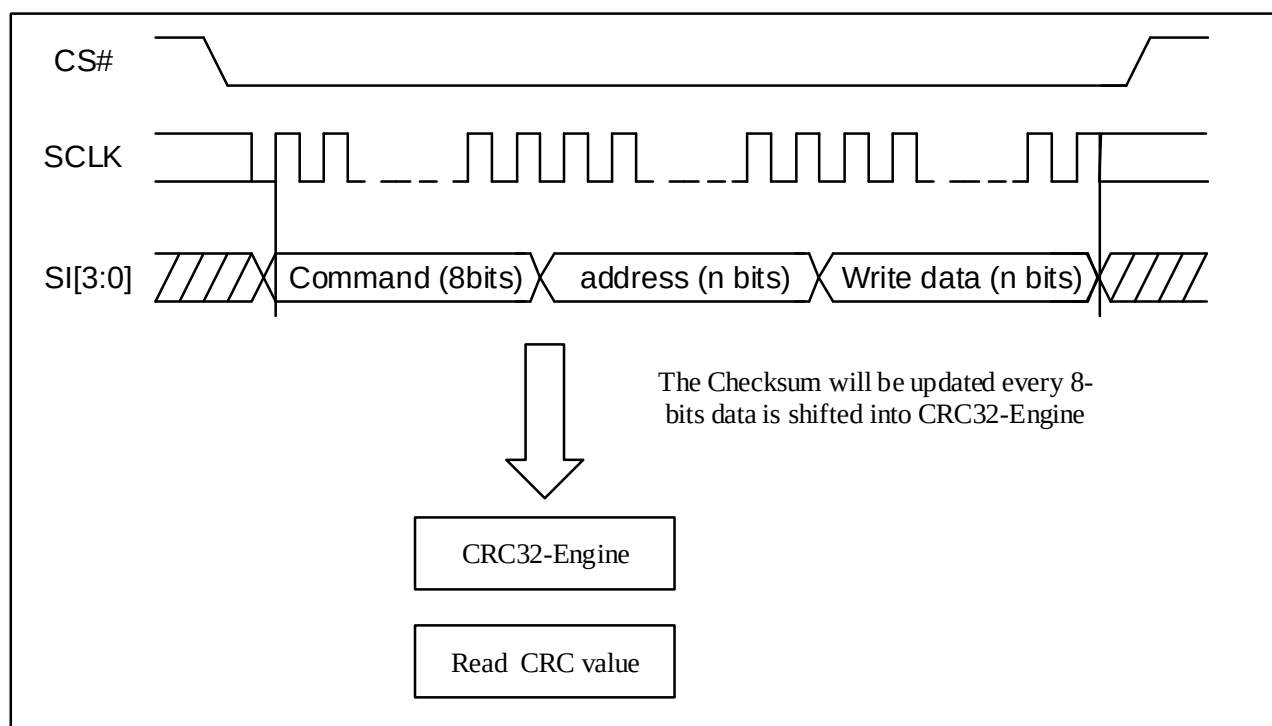
1. Back to back Interface CRC read command will not show the CRC checksum value being reset. At the end of each read interface CRC register command, the interface CRC register will get reset and updates itself with new CRC checksum value after getting a command with valid input data.

Program / Erase

The Page Program command is entered by driving CS# Low, followed by the command code, three or four address Bytes and at least one data Byte. The Sector Erase or Block Erase command is entered by driving CS# low, followed by the command code, and 3- Byte address or 4-Byte address. The Chip Erase command is entered by driving CS# Low, followed by the command code.

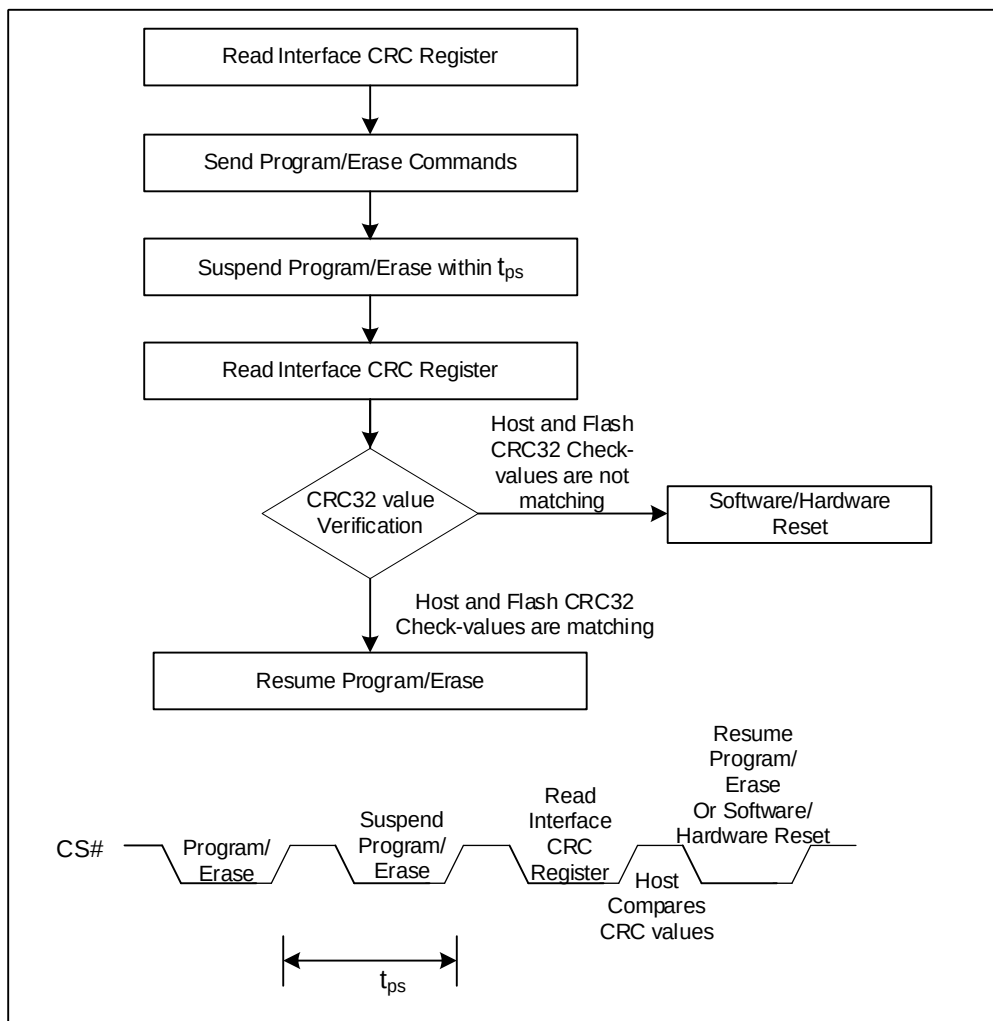
The flash will calculate the CRC of all instruction sequence using the proposed CRC32 polynomial. As soon as CS# is driven high, the CRC calculation will be stopped and the checksum latched into the CRC Register. If multiple Program/ Erase instructions are executed, flash will continuously update the CRC checksum;

Figure 11. CRC32 Protection of Program



The host will read the CRC checksum from flash using the Read Interface CRC Register command. If the CRC checksum of Flash does not match the CRC checksum calculated by the host, Program/ Erase instructions need to be resent, flash will recalculate the CRC checksum. Multiple Program / Erase to the same location due to CRC checksum errors will affect the Flash endurance. The Interface CRC32 Protection flow of Program and Erase shows the solution to this issue.

Figure 12. Interface CRC32 Protection flow of Program and Erase



Interface CRC Check-value Register

The GD25LF256F has one Interface CRC Check-value Register that are volatile. The Register has 32-bit Interface CRC Check-value Register bits. The Interface CRC Check-value Registers are read by read Interface CRC Register command (64h).

Table 6. Interface CRC Check-value Register

No.	Name	Description	Note
ITCR[31:0]	Interface CRC check-value Register	The ITCR[31:0] bits store the check-value of the Interface CRC process on the I/O. (Default, FF..FFH)	Volatile, read only

6.4 Data Integrity Check

The flash has a group of commands to perform a hardware accelerated CRC calculation over a user defined address range in the memory array. The calculation is another type of embedded operation similar to programming or erase in which the flash is busy while the calculation is in progress. The CRC operation uses the same CRC32 polynomial as Interface CRC to determine the CRC check-value.

CRC32 Polynomial(Same with AUTOSAR): $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X^1 + 1$

The check-value generation sequence is started by entering the 5Bh command. The command includes loading the beginning address into the CRC Start Address Register and identifying the beginning of the address range that will be covered by the CRC calculation. The command also includes loading the ending address into the CRC End Address Register. Bringing CS# HIGH starts the CRC calculation. The CRC process calculates the check-value on the data contained at the starting address through the ending address.

During the calculation period the device goes into the Busy state. Once the check-value calculation is completed, the device returns to the Ready state and the calculated check-value is available to be read. The check-value is stored in the Data Integrity CRC Register (DCRV[31:0]) and can be read using Read Volatile Configuration Register.

The check-value calculation can only be initiated when the device is in Standby State; and once started it can be suspended with the Suspend command to read data from the memory array. During the Suspended state the CRC Suspend Status Bit in the Flag Status Register will be set (SUS3=1). Once suspended, the host can read the Status Register, read data from the array and can resume the CRC calculation by using the Resume command.

The Ending Address (ENDADD) must be not less than the Starting Address (STRADD). If $ENDADD < STRADD$, the check-value calculation will abort and the device will return to the Ready state. Data Integrity CRC abort status bit will be set (CRCA=1) to indicate the aborted condition. The CRCA bit can be cleared, once set, by Software/Hardware Reset or Clear Flag Status Register(30h) or a valid subsequent CRC command execution. If $ENDADD < STRADD$, the check-value will hold indeterminate data.

Note:

Any invalid command during CRC check-value calculation can corrupt the check-value data.

7 REGISTERS

7.1 Status Register

Table 7. Status Register-SR No.1

No.	Bit Name	Description	Note
S7	SRP0	Status Register Protection Bit	Non-volatile writable
S6	BP4	Block Protect Bit	Non-volatile writable
S5	BP3	Block Protect Bit	Non-volatile writable
S4	BP2	Block Protect Bit	Non-volatile writable
S3	BP1	Block Protect Bit	Non-volatile writable
S2	BP0	Block Protect Bit	Non-volatile writable
S1	WEL	Write Enable Latch	Volatile, read only
S0	WIP	Erase/Write/Data Integrity Check In Progress	Volatile, read only

Table 8. Status Register-SR No.2

No.	Bit Name	Description	Note
S15	SUS1	Erase Suspend Bit	Volatile, read only
S14	CMP	Complement Protect	Non-volatile writable
S13	LB3	Security Register Lock Bit	Non-volatile writable (OTP)
S12	LB2	Security Register Lock Bit	Non-volatile writable (OTP)
S11	LB1	Security Register Lock Bit	Non-volatile writable (OTP)
S10	SUS2	Program Suspend Bit	Volatile, read only
S9	QE	Quad Enable Bit	QE = 1 permanently
S8	SRP1	Status Register Protection Bit	Non-volatile writable

Table 9. Status Register-SR No.3

No.	Bit Name	Description	Note
S23	Reserved	Reserved	Reserved
S22	Reserved	Reserved	Reserved
S21	PWDRD	Password Read Mode Bit	Non-volatile writable
S20	ADP	Power Up Address Mode Bit	Non-volatile writable
S19	ADS	Current Address Mode Bit	Volatile, read only
S18	PWD	Password Protection Mode Bit	Non-volatile writable (OTP)
S17	DC1	Dummy Configuration Bit	Non-volatile writable
S16	DC0	Dummy Configuration Bit	Non-volatile writable

The status and control bits of the Status Register are as follows:

WIP bit

The Write in Progress (WIP) bit indicates whether the memory is busy in program/erase/write status register or configuration register progress. When WIP bit sets to 1, means the device is busy in program/erase/write status register or configuration register progress, when WIP bit sets 0, means the device is not in program/erase/write status register or configuration register progress.

WEL bit

The Write Enable Latch (WEL) bit indicates the status of the internal Write Enable Latch. When set to 1 the internal Write Enable Latch is set, when set to 0 the internal Write Enable Latch is reset and no Write, Program or Erase command is accepted.

BP4, BP3, BP2, BP1, BP0 bits

The Block Protect (BP4, BP3, BP2, BP1, and BP0) bits are non-volatile. They define the size of the area to be software protected against Program and Erase commands. These bits are written with the Write Status Register (WRSR) command. When the Block Protect (BP4, BP3, BP2, BP1, and BP0) bits are set to 1, the relevant memory area becomes protected against Page Program (PP), Sector Erase (SE) and Block Erase (BE) commands. The Block Protect (BP4, BP3, BP2, BP1, and BP0) bits can be written provided that the Hardware Protected mode has not been set. The Chip Erase (CE) command is executed only if none sector or block is protected.

In advanced memory protection with password, successful password verification is required to disable the protection placed on Block Protect (BPx, CMP, SRP1, SRP0) bits, then the Block Protect (BPx, CMP, SRP1, SRP0) bits are writable.

SRP0, SRP1 bits

The Status Register Protect (SRP1 and SRP0) bits are non-volatile Read/Write bits in the status register. The SRP bits control the method of write protection: software protection, hardware protection, power supply lock-down or one-time programmable protection.

In advanced memory protection with password, successful password verification is required to disable the protection placed on Status Register Protect (SRP1 and SRP0) bits, then the Status Register Protect (SRP1 and SRP0) bits are writable.

SRP1	SRP0	#WP	Status Register/NL Register	Description
0	0	X	Software Protected	The Status Register and NL Registers can be written to after a Write Enable command, WEL=1. (Default)
0	1	0	Hardware Protected	WP#=0, the Status Register and NL Registers are locked and cannot be written to.
0	1	1	Hardware Unprotected ⁽¹⁾	WP#=1, the Status Register and NL Registers are unlocked and can be written to after a Write Enable command, WEL=1.
1	X	X	Power Supply Lock-Down ⁽²⁾	Status Register and NL Registers are protected and cannot be written to again until the next Power-Down, Power-Up cycle, Software Reset(66h+99h), Hardware Reset.
1	X	X	One Time Program ⁽³⁾	Status Register and NL Registers are permanently protected and cannot be written to. (Enabled by adding prefix command AAh, 55h)

Notes:

1. In advanced memory protection with password, successful password verification is required to disable the protection placed on Status Registers (BP[4:0] and CMP) and NL Registers.
2. When SRP1 =1, a power-down, power-up cycle, Software reset (66H+99H), Hardware Reset will change SRP1 =0 state.
3. Please contact GigaDevice for details regarding the special instruction sequence.

QE bit

The Quad Enable (QE) bit is a non-volatile bit in the Status Register that allows Quad operation. The default value of QE bit is 1 and it cannot be changed, so that the IO2 and IO3 pins are enabled all the time.

SUS1, SUS2, SUS3 bits

The SUS1 and SUS2 and SUS3 bits are read only bits in the status register (S15 and S10) and flag status register (FS5) that are set to 1 after executing a Program/Erase/Data Integrity Check Suspend (75h) command (The Erase Suspend will set the SUS1 bit to 1, and the Program Suspend will set the SUS2 bit to 1, and the Data Integrity Check Suspend will set the SUS3 bit to 1). The SUS1 and SUS2 and SUS3 bits are cleared to 0 by Program/Erase/Data Integrity Check Resume (7AH) command, Software Reset (66h+99h) command, Hardware Reset as well as a power-down, power-up cycle.

LB3, LB2, LB1 bits

The LB3, LB2 and LB1 bits are non-volatile One Time Program (OTP) bits in Status Register (S13, S12 and S11) that provide the write protect control and status to the Security Registers. The default state of LB3, LB2 and LB1 bits are 0, the security registers are unlocked. The LB3, LB2 and LB1 bits can be set to 1 individually using the Write Register instruction. The LB3, LB2 and LB1 bits are One Time Programmable, once they are set to 1, the Security Registers will become read-only permanently.

CMP bit

The CMP bit is a non-volatile Read/Write bit in the Status Register (S14). It is used in conjunction with the BP4-BP0 bits to provide more flexibility for the array protection. Please see the Status registers Memory Protection table for details. The default setting is CMP=0.

In advanced memory protection with password, successful password verification is required to disable the protection placed on Block Protect (BPx, CMP, SRP1, SRP0) bits, then the Block Protect (BPx, CMP, SRP1, SRP0) bits are writable.

DC1, DC0 bits

The Dummy Configuration (DC) bits are non-volatile, which select the number of dummy cycles between the end of address and the start of read data output. Dummy cycles provide additional latency that is needed to complete the initial read access of the flash array before data can be returned to the host system. Some read commands require additional dummy cycles as the SCLK frequency increases.

The following dummy cycle tables provide different dummy cycle settings that are configured.

Command	DC1, DC0	Dummy Cycles	Freq.(MHz)
0Bh, 0Ch 3Bh, 3Ch 6Bh, 6Ch	00 (default)	8	166
	01	8	166
	10	8	166
	11	8	166



BBh, BCh	00 (default)	4	104
	01	8	166
	10	4	104
	11	8	166
EBh, ECh	00 (default)	6	120
	01	6	120
	10	8	133
	11	10	166
BDh, BEh EDh, EEh	00 (default)	10	104
	01	8	80
	10	10	104
	11	10	104

Password Protection Mode (PWD) bit

The PWD bit is an OTP bit that controls whether the device is either in normal protection (default – using BPx/ CMP bits, VL Registers, or NL Registers) or Advanced Password Protection Mode. When PWD bit is '0' (default), password protection is disabled (normal protection mode), and the 64-bit Password is readable and programmable. In parallel, the Block Protect/CMP bits and NL Registers are writable.

When PWD bit is set to 1 (and PWDVFY = 0), the Password Register is locked/hidden, programmed permanently (OTP), and can no longer be read; password protection is enabled permanently; and writes to the BPx/CMP bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are ignored.

To modify/write to the Block Protect (BPx and CMP), NL Registers, and PWDRD bit in Password Protection Mode, the Unlock Password (29h) instruction using the valid 64-bit Password data must be executed successfully to set the PWDVFY Bit to '1'. Once PWDVFY bit is set to '1', then the Block Protect bits, NL Registers, and PWDRD bit are enabled (unprotected) for modification or writes.

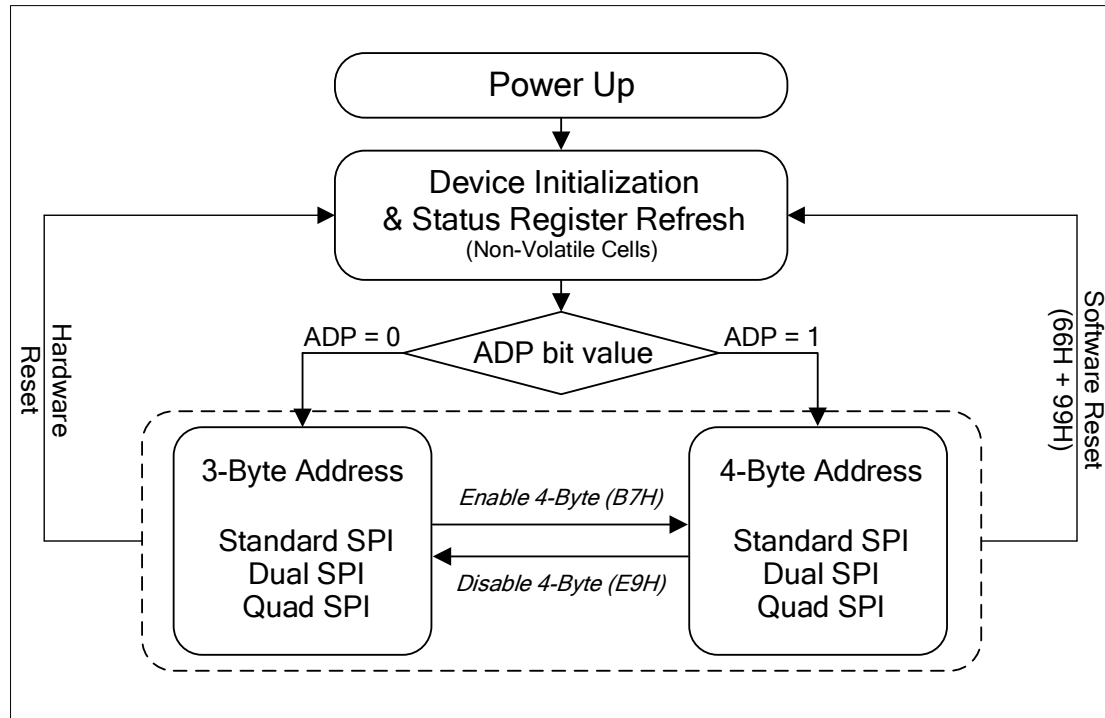
In Password Protection Mode, the device supports a boot-memory feature called Password Read Boot-Memory mode. In this mode, the device memory is configured to bottom 16Mbit address range accessible only for boot-up. The boot-memory feature is entered when PWDRD bit is set to '1' and (PWD, PWDVFY = 1, 0), by clearing the PWDVFY bit to '0' or reset after setting the PWDRD bit. The Password Read Boot-Memory, Password Verify Bit, and Advanced Memory Protection with Password sections provide more detail information on Password Read Boot-Memory mode.

ADS bit

The Address Status (ADS) bit is a read only bit that indicates the current address mode the device is operating in. The device is in 3-Byte address mode when ADS=0 (default), and in 4-Byte address mode when ADS=1.

ADP bit

The Address Power-up (ADP) bit is a non-volatile writable bit that determines the initial address mode when the device is powered on or reset. This bit is only used during the power on or device reset initialization period. When ADP=0 (factory default), the device will power up into 3-Byte address mode, the Extended Address Register must be used to access memory regions beyond 128Mb. When ADP=1, the device will power up into 4-Byte address mode directly.



Password Read Boot-Memory (PWDRD) bit

In Password Protection Mode, the device supports a boot-memory feature called Password Read Boot-Memory mode. In this mode, the device memory is configured to bottom 16Mbit address range accessibility only for boot-up. The accessible bottom 16Mbit address range in Password Read Boot-Memory mode is from 00000000h to 001FFFFFFh. This memory address range is readable, programmable, and erasable as default. The >16Mb address range from 00200000h to 03FFFFFFh is read inaccessible. In this mode, all read instructions address input are redirected to the bottom 16Mb Boot-Memory until the password is unlocked (PWDVfy=1) and the full memory becomes accessible. Additionally, the full memory including the >16Mb address range is programmable or erasable as default, however read verification of the >16Mb read inaccessible memory address range will require the setting of the PWDVfy bit to '1' (via Password Unlock instruction sequence). If protection of the memory is needed in Password Read Boot-Memory mode, please use the associated Block Protect (BPx & CMP) bits and Nonvolatile/Volatile Lock Registers.

The PWDRD bit is a nonvolatile bit (S21) in Status Register-3. It can only be written to '1' or '0' in Password Protection mode when Password Protection Mode (PWD) bit is '1' and Password Verify (PWDVfy) bit is 1 (PWD, PWDRD, PWDVfy = 1, x, 1) after a successful Password Unlock instruction sequence. The Password Read Boot-Memory Mode is entered once the PWDVfy bit is cleared from '1' to '0' by a "Password Lock (29h) + Invalid 64-bit Password" instruction sequence or reset (PWD, PWDRD, PWDVfy = 1, 1, 0).

To exit Password Read Boot-Memory mode, the PWDVfy bit must be set to '1' by "Password Unlock(29h) + Valid 64-bit Password" instruction sequence (PWD, PWDRD, PWDVfy = 1, 1, 1). This will enable access to the previously read inaccessible >16Mb address range (00200000h to 03FFFFFFh). A successful "Password Lock (29h) + Invalid 64-bit Password" instruction sequence or reset will return the device to Password Read Boot-Memory mode (PWD, PWDRD, PWDVfy = 1, 1, 0) and the device returns to bottom 16Mb address range (00000000H to 001FFFFFFh) accessible only.

Reserved bit

It is recommended to set the value of the reserved bit as "0".

7.2 Extended Address Register

Table 10 Extended Address Register

No.	Name	Description	Note
EA7	SAF	Safe Boot Bit	Volatile, read only
EA6	Reserved	Reserved	Reserved
EA5	Reserved	Reserved	Reserved
EA4	Reserved	Reserved	Reserved
EA3	Reserved	Reserved	Reserved
EA2	Reserved	Reserved	Reserved
EA1	Reserved	Reserved	Reserved
EA0	A24	Address bit	Volatile writable

The extended address register is only used when the address mode is 3-Byte mode, as to set the higher address. The default value of the address bit is "0".

For the read operation, the whole array can be continually read out with one command. Data output starts from the selected 128Mb, and it can cross the boundary. When the last Byte of the segment is reached, the next Byte (in a continuous reading) is the first Byte of the next segment. However, the EAR (Extended Address Register) value does not change. The random access reading can only be operated in the selected segment.

The Chip erase command will erase the whole chip and is not limited by EAR selected segment. However, the sector erase, block erase, program operation are limited in selected segment and will not cross the boundary.

A24 bits

The Extended Address Bits are used only when the device is operating in the 3-Byte Address Mode (ADS=0), which is volatile writable by C5h command.

If the device powers up with ADP bit set to 1, or an "Enter 4-Byte Address Mode (B7h)" instruction is issued, the device will require 4-Byte address input for all address related instructions, and the Extended Address Bit setting will be ignored.

A24	Address Range
0	0000 0000h-00FF FFFFh
1	0100 0000h-01FF FFFFh

SAF bits

The SAF bit is read only bit in the extended address register (EA7) that is set to 1 after POR failure. If SAF = 1, Software/Hardware Reset or a power-down, power-up cycle must be executed to power up again. Only if SAF = 0, the flash can be used.

Reserved bit

It is recommended to set the value of the reserved bit as "0".

7.3 Flag Status Register

Table 11 Flag Status Register

No.	Name	Description	Note
FS7	RY/BY#	Ready/Busy# Bit	Volatile, read only
FS6	CRCA	Data Integrity Check Abort Bit	Volatile, read only
FS5	SUS3	Data Integrity Check Suspend Bit	Volatile, read only
FS4	Reserved	Reserved	Reserved
FS3	Reserved	Reserved	Reserved
FS2	PWDVFY	Password Verify Bit	Volatile, read only
FS1	PE	Program Error Bit	Volatile, read only
FS0	EE	Erase Error bit	Volatile, read only

The status and control bits of the Flag Status Register are as follows:

RY/BY# bit

The RY/BY# bit is a read only bit that indicates Program or Erase Status bit. Indicates whether one of the following command cycles is in progress: WRITE STATUS REGISTER, WRITE NONVOLATILE CONFIGURATION REGISTER, PROGRAM, or ERASE.

CRCA bit

The Data Integrity Check Abort Bit (CRCA) bit is a read only bit that indicates a Data Integrity Check calculation operation was aborted. If $ENDADD < STRADD$, then CRCA will be set and the flash will return to the Standby state. CRCA flag is cleared to '0' at the next Data Integrity Check calculation operation when $ENDADD \geq STRADD$.

The CRCA bit also can be cleared by Software/Hardware Reset or Clear Flag Status Register(30h).

SUS3 bit

The SUS3 bit is read only bits in the flag status register (FS5) that is set to 1 after executing an Program/Erase/Data Integrity Check Suspend (75h) command (The Erase Suspend will set the SUS1 bit to 1, and the Program Suspend will set the SUS2 bit to 1, and the CRC Suspend will set the SUS3 bit to 1). The SUS1 and SUS2 and SUS3 bits are cleared to 0 by Erase/Program/CRC Resume (7Ah) command, software reset (66h+99h) command as well as a power-down, power-up cycle.

Password Verify (PWDVFY) Bit

PWDVFY is a status volatile bit identification when the device is in Password Protection mode (PWD=1). When PWD bit is '1', the PWDVFY bit identifies if the device's BPx/CMP bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are write protected or not. When PWDVFY bit is '0', the BPx/CMP bits, SRP1/SRP0 bits, NL Registers, and PWDRD bit are password protected and locked (default and after successful Password Lock). When PWDVFY bit is '1', the BPx/CMP bits, NL Registers, and PWDRD bit are unprotected and writable.

To set the PWDVFY bit to '1' in Password Protection mode, a successful Password Unlock (29h) instruction must be executed using the valid 64-bit password for verification. Using the invalid password as a data input will keep the device in password protected mode. After completing the BPx/CMP bits, NL Registers, and PWDRD bit changes and associated tasks while in (PWD, PWDVFY = 1, 1), initiating Password Lock (29h) with invalid 64-bit Password data is required to return



the device in Password Protected Mode (PWD, PWDVFY = 1, 0).

The device supports a Password Read Boot-Memory mode configuration (bottom 16Mbit Boot-Memory access only) in Password Protection mode (PWD=1) when PWDRD bit is set to '1' (PWD, PWDRD = 1, 1). After a successful 'Password Unlock (29h) + Valid 64-bit Password' instruction sequence, the PWDVY bit is set to 1. While PWDVFY is '1', PWDRD bit can be set to '1'. The device's BPx/CMP bits, NL Registers, and Password Read Boot-Memory (PWDRD) bit are unprotected and writable when (PWD, PWDRD, PWDVFY = 1, 1, 1). To complete the Password Read Boot-Memory mode setting, the 'Password Lock (29h) + Invalid 64-bit Password' instruction sequence or reset must be executed to clear the PWDVFY bit to '0' (PWD, PWDRD, PWDVFY = 1, 1, 0). When PWDVFY bit is '0', besides the protection set on the relevant registers in the Password Protection Mode (PWD=1), the access to the device memory address is limited to the bottom 16Mb address range (PWD, PWDRD, PWDVFY = 1, 1, 0). The >16Mb memory address range is read inaccessible.

PE bit

The Program Error (PE) bit is a read only bit that indicates a program failure. It will also be set when the user attempts to program a protected array sector or access the locked OTP space. PE is cleared to "0" after program operation resumes or by Clear Flag Status Register command (30h).

EE bit

The Erase Error (EE) bit is a read only bit that indicates an erase failure. It will also be set when the user attempts to erase a protected array sector or access the locked OTP space. EE is cleared to "0" after erase operation resumes or by Clear Flag Status Register command (30h).

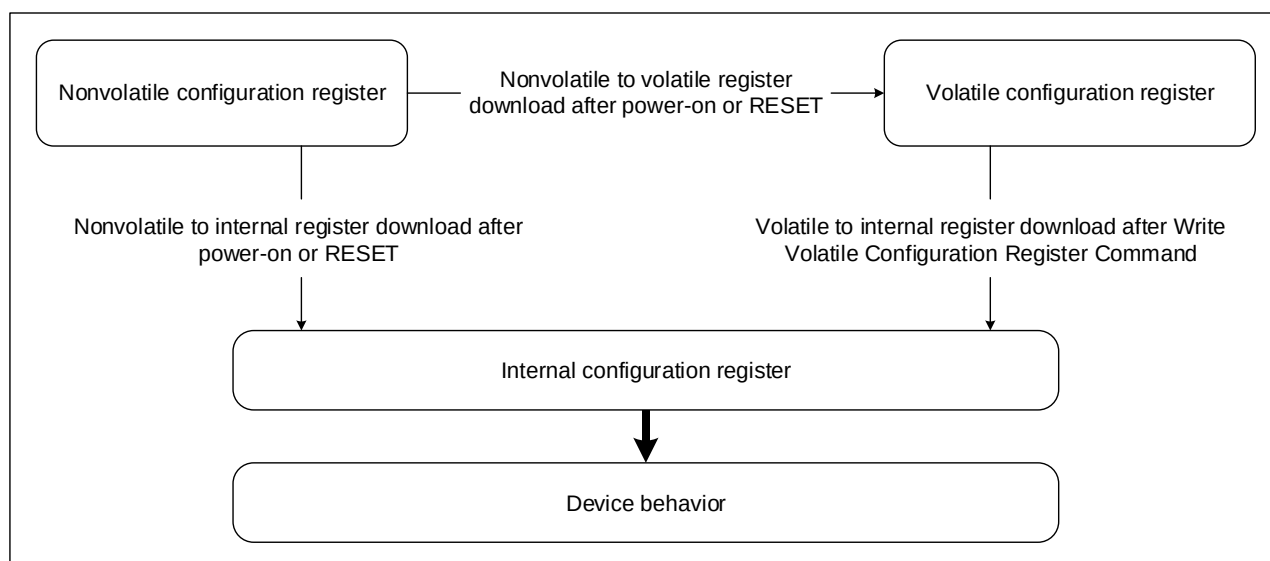
Reserved bit

It is recommended to set the value of the reserved bit as "0".

8 INTERNAL CONFIGURATION REGISTER

The memory configuration is set by an internal configuration register that is not directly accessible to users. The user can change the default configuration at power up by using the WRITE NONVOLATILE CONFIGURATION REGISTER. Information from the nonvolatile configuration register overwrites the internal configuration register during power on or after a reset.

The user can change the configuration during device operation using the WRITE VOLATILE CONFIGURATION REGISTER command. Information from the volatile configuration registers overwrite the internal configuration register immediately after the WRITE command completes.



8.1 Nonvolatile Configuration Register

Nonvolatile Configuration Register bits set the device configuration after power-up or reset. All bits are erased (FFh) unless stated otherwise. This register is read from and written to using the READ NONVOLATILE CONFIGURATION REGISTER and the WRITE NONVOLATILE CONFIGURATION REGISTER commands, respectively. The commands use the main array address scheme, but only the LSB is used to access different register settings, thereby providing up to 256 Bytes of registers (See the table below for the details). A WRITE command to a reserved address will set the device to the default status of the corresponding Byte.

Table 12 Nonvolatile Configuration Register

Addr	Settings	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	Description
<0>	XIP mode after Power on reset	1	1	1	1	1	1	0	0	XIP: Dual I/O fast read
		1	1	1	1	1	1	0	1	XIP: DTR Dual I/O Fast
		1	1	1	1	1	1	1	0	XIP: Quad I/O fast read
		1	1	1	1	1	0	1	1	XIP: DTR Quad I/O Fast Read
		1	1	1	1	1	1	1	1	Disabled (Default)
		Others								Reserved
<1>	Driver Strength configuration	x	x	x	x	x	x	0	0	100%
		x	x	x	x	x	x	0	1	75% (Default)
		x	x	x	x	x	x	1	0	50%
		x	x	x	x	x	x	1	1	25%
		Others								Reserved
<2>	Interface CRC enable	Others								Reserved
		x	x	x	1	x	x	x	x	CRC Enabled (Default)
		x	x	x	0	x	x	x	x	CRC Disabled
	DLP enable	x	x	x	x	1	x	x	x	DLP Disabled (Default)
		x	x	x	x	0	x	x	x	DLP Enabled
	VL default value	x	x	x	x	x	1	x	x	all VL=1 after POR or Reset
		x	x	x	x	x	0	x	x	all VL=0 after POR or Reset(Default)
	ECC & ECS# configuration	x	x	x	x	x	x	1	1	ECC Enable ECS#: 2-bit Error or Double Programmed (Default)
		x	x	x	x	x	x	1	0	ECC Enable ECS#: 2 bit Error only
		x	x	x	x	x	x	0	1	ECC Enable ECS#: 1- or 2-bit Error
		x	x	x	x	x	x	0	0	ECC Disabled
<3>	Reserved									
<4>	Reserved									
<5>	Reserved									
<6>	Reserved									
<7>	Reserved									
<8>	Reserved									
<9>	Reserved									
<A>	Reserved									
	Reserved									

8.2 Volatile Configuration Register

Volatile Configuration Register bits temporarily set the device configuration after power-up or reset. All bits are erased (FFh) unless stated otherwise. This register is read from and written to using the READ VOLATILE CONFIGURATION REGISTER and the WRITE VOLATILE CONFIGURATION REGISTER commands, respectively. The commands use the main array address scheme; however, only the LSB is used to access different register settings to provide up to 256 Bytes of registers (See the table below for the details). A WRITE command to a reserved address will set the device to the default status of the corresponding Byte.

Table 13 Volatile Configuration Register

Addr	Settings	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	Description
<0>	Reserved	x	x	x	x	x	x	x	x	Reserved
<1>	Driver Strength configuration	x	x	x	x	x	x	0	0	100%
		x	x	x	x	x	x	0	1	75% (Default)
		x	x	x	x	x	x	1	0	50%
		x	x	x	x	x	x	1	1	25%
		Others								Reserved
<2>	Interface CRC Enable	Others								Reserved
		x	x	x	1	x	x	x	x	CRC Enabled (Default)
		x	x	x	0	x	x	x	x	CRC Disabled
	DLP enable	x	x	x	x	1	x	x	x	DLP Disabled (Default)
		x	x	x	x	0	x	x	x	DLP Enabled
	VL default value ⁽¹⁾	x	x	x	x	x	1	x	x	all VL=1 after POR or Reset
		x	x	x	x	x	0	x	x	all VL=0 after POR or Reset(Default)
	ECC & ECS# configuration	x	x	x	x	x	x	1	1	ECC Enable ECS#: 2-bit Error or Double Programmed (Default)
		x	x	x	x	x	x	1	0	ECC Enable ECS#: 2 bit Error only
		x	x	x	x	x	x	0	1	ECC Enable ECS#: 1- or 2-bit Error
		x	x	x	x	x	x	0	0	ECC Disabled
<3>	ECCFAVLD (ECC fail address valid indicator)	0	x	x	x	x	x	x	x	ECC failure address invalid (no fail address recorded)
		1	x	x	x	x	x	x	x	ECC failure address valid (there's fail address recorded)
	ECCFS (ECC fail status)	x	0	0	0	x	x	x	x	None
		x	x	x	1	x	x	x	x	1 bit corrected
		x	x	1	x	x	x	x	x	2 bits detected
		x	1	x	x	x	x	x	x	Double programmed page detected



	ECCCNT (ECC failure chunk counter)	x	x	x	x	0	0	0	0	No chunk ECC failure
		x	x	x	x	...	...	...	...	1~15 chunk ECC failure
<4>	ECCFA (ECC failure chunk address)	A7	A6	A5	A4	A3	x	x	x	ECC 1st failure chunk address (A7:A3)
	Reserved	Others								Reserved
<5>	ECCFA (ECC failure chunk address)	A15	A14	A13	A12	A11	A10	A9	A8	ECC 1st failure chunk address (A15:A8)
<6>	ECCFA (ECC failure chunk address)	A23	A22	A21	A20	A19	A18	A17	A16	ECC 1st failure chunk address (A23:A16)
<7>	Reserved	Others								Reserved
	ECCFA (ECC failure chunk address)	x	x	x	x	x	x	x	A24	ECC 1st failure chunk address (A24)
<8>	DCRV[7:0]									Data Integrity Check Value DCRV[7:0]
<9>	DCRV[15:8]									Data Integrity Check Value DCRV[15:8]
<A>	DCRV[23:16]									Data Integrity Check Value DCRV[23:16]
	DCRV[31:24]									Data Integrity Check Value DCRV[31:24]

Note:

1. Volatile Configuration Register's VL default value download from Nonvolatile Configuration register after power-on or RESET; Write Volatile Configuration Register (81h) can change Volatile Configuration Register VL default value, but it is invalid.

DLP bit

The DLP bit is Data Learning Pattern Enable bit, which is writable by B1h/81h command. For Dual I/O DTR Fast Read commands, and Quad I/O DTR Fast Read commands, a pre-defined “Data Learning Pattern” can be used by the flash memory controller to determine the flash data output timing on 2 I/O pins and 4 I/O pins. When DLP=0, before the data output, the flash will output “00110100” Data Learning Pattern sequence on each of the 2 I/O or 4 I/O pins. If the dummy clock is not enough for the output of the whole Data Learning Pattern, the last several bit of the Data Learning Pattern would be cut-off. During this period, controller can fine tune the data latching timing for each I/O pins to achieve optimum system performance. DLP=1 will disable the Data Learning Pattern output.

Figure 13. Data Learning Pattern Sequence Diagram (BDh/BEh, Dummy Clock ≥ 8)

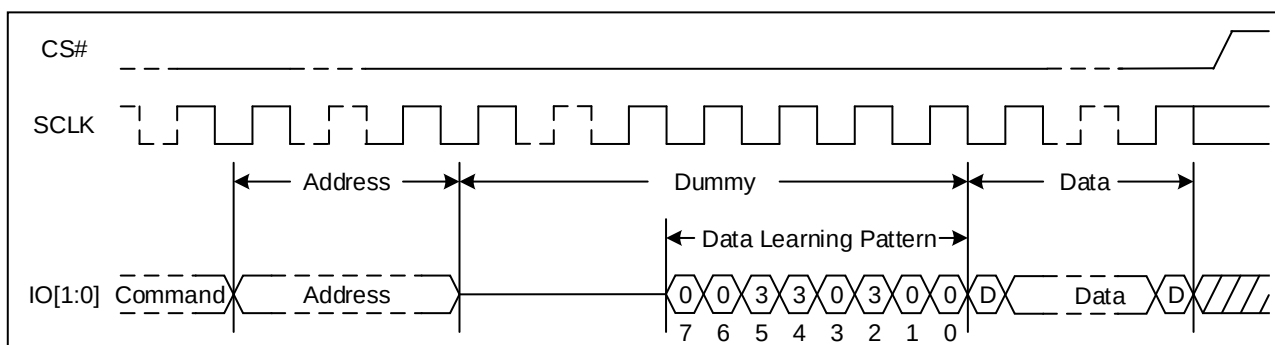
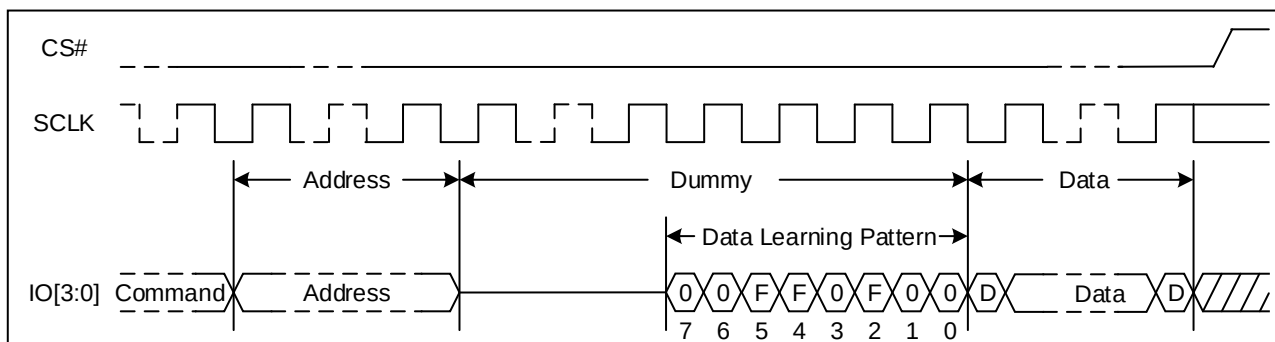


Figure 14. Data Learning Pattern Sequence Diagram (EDh/EEh, 0Eh, Dummy Clock ≥ 8)



ECC & ECS# Configuration (ECCECS#)

The two bit ECC & ECS# Configuration bits are writable and located on Nonvolatile/Volatile Configuration Registers Address <02> Bit[1:0]. The Volatile Configuration Register's ECCECS# value will be based on the Nonvolatile Configuration Register's ECCECS# value from power up or reset. The ECCECS# bits disable (ECCECS# = 00b) or enable (ECCECS# = '01b, 10b, or 11b') the on chip ECC engine along with setting up the ECC status functionality of the ECS# pin when ECC is enabled. Based on the associated value setting of the ECCECS# bits and if that specified ECC event occurs during the read memory access, the ECS# pin will be driven low for the duration of the 8-byte aligned ECC affected memory chunk. Additionally, the ECC Fail Status (ECCFS) bits on Volatile Configuration Register can also provide the ECC status as Single Error Correction (SEC), Double Error Detected (DED), or Double Program state after the previous read memory.

The ECCECS# value determines the ECS# pin functionality. When ECCECS# = '01b', the ECS# pin is setup to detect 1- or 2-bit Error. When ECCECS# = '10b', the ECS# pin is setup to detect 2-bit Error only. When ECCECS# = '11b', the ECS# pin is setup to detect 2-bit Error or Double Programmed (default). A Double Programmed state indicates the affected 8-byte aligned memory chunk was programmed more than 1 time without an erase. The multiple programming without an

erase disables the error correction engine of that particular 8-byte aligned memory chunk and read is directly to the memory array only without the ECC. An erase operation will revert the memory along with ECC stored data and double programmed states to its default erase FFh state.

ECC Fail Address Valid Indicator (ECCFAVLD)

The ECC Fail Address Valid Indicator (ECCFAVLD) flag is a bit indicator that identifies either there is a valid captured address on ECCFA[24:3] with an ECC failure status (ECCFAVLD=1) or not (ECCFAVLD=0). The combination of the captured ECCFAVLD=1 and the first ECC failure address ECCFA[24:3] value confirms that there was an ECC event from the previous read. The ECCFAVLD is located on Volatile Configuration Register Address <00H> Bit 7. ECCFAVLD is cleared to '0' from power up, reset, or at the start of every read memory operation.

ECC Fail Status Bits (ECCFS)

The ECC Fail Status bits can provide the ECC status captured from the previous memory read. The 3-bit ECCFS status bits are located on Volatile Configuration Register Address <03h> Bits[6:4]. The ECC Fail Status type can either be No Error (ECCFS=000b), Single (1-Bit) Error Corrected/SEC (ECCFS=xx1b), Double (2-Bit) Error Detected/DED (ECCFS=x1xb), or Double Programmed (ECCFS=1xxb). There can be multiple types of ECC errors captured from the last read. The ECCFS bits are cleared to 000b from power-up cycle, reset, or on every start of a new read memory instruction.

ECC Failure Chunk Counter (ECCCNT)

The 4-bit ECC Failure Chunk Counter (ECCCNT) tracks the count of the 1-bit and 2-bit error events from a previous read memory operation. The ECCCNT default value from power up reset, or at the start of every read operation is 0000b and can count up to 15 ECC events. The ECCCNT is located on Volatile Configuration Register Address <04h> Bit[3:0]. ECCCNT is reset by a power-up cycle, reset, or on every start of a new read memory instruction.

ECC Failure Chunk Address (ECCFA[24:3])

The ECC Failure Chunk address (ECCFA[24:3]) is the captured first address with 1-bit error correction or 2-bit error detection from a previous read memory operation. The ECCFA[24:3] are spread on Volatile Configuration Register Addresses 07H to 04H: A[24] on Bit[0] of Address 07H, A[23:16] and A[15:8] on Address 06H and 05H, and A[7:3] on Address 04H Bit[7:3]. The captured ECC Failure Chunk Address is valid when ECCFAVLD bit is '1'. ECCFA[24:3] is reset by a power-up cycle, reset, or on every start of a new read memory instruction.

9 COMMAND DESCRIPTIONS

All commands, addresses and data are shifted in and out of the device, beginning with the most significant bit on the first rising edge of SCLK after CS# is driven low. Then, the one-Byte command code must be shifted in to the device, with most significant bit first on SI, and each bit is latched on the rising edges of SCLK.

Every command sequence starts with a one-Byte command code. Depending on the command, this might be followed by address Bytes, or by data Bytes, or by both or none. CS# must be driven high after the last bit of the command sequence has been completed. For the command of Read, Fast Read, Read Status Register or Release from Deep Power-Down, and Read Device ID, the shifted-in command sequence is followed by a data-out sequence. All read instruction can be completed after any bit of the data-out sequence is being shifted out, and then CS# must be driven high to return to deselected status.

For the command of Page Program, Sector Erase, Block Erase, Chip Erase, Write Status Register, Write Enable, Write Disable or Deep Power-Down command, CS# must be driven high exactly at a Byte boundary, otherwise the command is rejected, and is not executed. That is CS# must be driven high when the number of clock pulses after CS# being driven low is an exact multiple of eight. For Page Program, if at any time the input Byte is not a full Byte, nothing will happen and WEL will not be reset.

Table 14. Commands (SPI, 3- or 4-Byte Addr. Mode)

Command Name	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9
Write Enable	06h								
Write Disable	04h								
Read Status Register-1	05h	(S7-S0)	(cont.)						
Read Status Register-2	35h	(S15-S8)	(cont.)						
Read Status Register-3	15h	(S23-S16)	(cont.)						
Read Flag Status Register	70h	FS7~FS0	(cont.)						
Write Status Register-1&2	01h	S7-S0	S15-S8						
Write Status Register-3	11h	S23-S16							
Read Extended Addr. Register	C8h	(EA7-EA0)							
Write Extended Addr. Register	C5h	EA7-EA0							
Volatile SR write Enable	50h								
Clear SR Flags	30h								
Set Burst with Wrap	77h	dummy ⁽¹⁾	dummy ⁽¹⁾	dummy ⁽¹⁾	W7-W0 ⁽¹⁾				
Chip Erase	60h/C7h								



Enter 4-Byte Address Mode	B7h								
Exit 4-Byte Address Mode	E9h								
Read Manufacturer/Device ID	90h	00H	00H	00H	(MID7-MID0)	(DID7-DID0)	(cont.)		
Read Identification	9Fh	(M7-M0)	(JDID15-JDID8)	(JDID7-JDID0)	(cont.)				
Enable Reset	66h								
Reset	99h								
Program/Erase/Data Integrity Check Suspend	75h								
Program/Erase/Data Integrity Check Resume	7Ah								
Deep Power-Down	B9h								
Release From Deep Power-Down	ABh								
Release From Deep Power-Down and Read Device ID	ABh	dummy	dummy	dummy	(DID7-DID0)	(cont.)			
Enable QPI	38h								
Read Serial Flash Discoverable Parameter	5Ah	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)		
Read Data with 4-Byte Address	13h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)	(cont.)		
Fast Read with 4-Byte Address	0Ch	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)	
Fast Read Dual Output with 4-Byte Address	3Ch	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0) ⁽²⁾	(cont.)	
DTR Dual I/O Fast Read with 4-Byte Address	BEh	A31-A24 ⁽⁴⁾	A23-A16 ⁽⁴⁾	A15-A8 ⁽⁴⁾	A7-A0 ⁽⁴⁾	M7-M0 ⁽⁵⁾	dummy	(D7-D0) ⁽²⁾	(cont.)
Fast Read Quad Output with 4-Byte Address	6Ch	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0) ⁽³⁾	(cont.)	
Fast Read Dual I/O with 4-Byte Address	BCh	A31-A24 ⁽⁴⁾	A23-A16 ⁽⁴⁾	A15-A8 ⁽⁴⁾	A7-A0 ⁽⁴⁾	M7-M0 ⁽⁵⁾	(D7-D0) ⁽²⁾	(cont.)	



Fast Read Quad I/O with 4-Byte Address	ECh	A31-A24 ⁽⁶⁾	A23-A16 ⁽⁶⁾	A15-A8 ⁽⁶⁾	A7-A0 ⁽⁶⁾	M7-M0 ⁽⁷⁾	dummy	dummy	(D7-D0) ⁽³⁾
DTR Quad I/O Fast Read with 4-Byte Address	EEh	A31-A24 ⁽⁶⁾	A23-A16 ⁽⁶⁾	A15-A8 ⁽⁶⁾	A7-A0 ⁽⁶⁾	M7-M0 ⁽⁷⁾	n-CLK dummy	(D7-D0) ⁽³⁾	(cont.)
Page Program with 4-Byte Address	12h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte		
Quad Page Program with 4-Byte Address	34h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte		
Sector Erase with 4-Byte Address	21h	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (32K) with 4-Byte Address	5Ch	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (64K) with 4-Byte Address	DCh	A31-A24	A23-A16	A15-A8	A7-A0				
Read VL Register	E0h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)	(cont.)		
Write VL Register	E1h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0			
Read NL Register	E2h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)	(cont.)		
NL bit Program	E3h	A31-A24	A23-A16	A15-A8	A7-A0				
All NL bit Erase	E4h								
Read Password Register	27h	P7~0	P15~8	P23~16	P31~24	P39~32	P47~40	P55~48	P63~56
Program Password Register	28h	P7~0	P15~8	P23~16	P31~24	P39~32	P47~40	P55~48	P63~56
Password Unlock	29h	P7~0	P15~8	P23~16	P31~24	P39~32	P47~40	P55~48	P63~56
Global Block Lock	7Eh								
Global Block Unlock	98h								
Read Interface CRC Register	64h	A31-A24	A23-A16	A15-A8	A7-A0	V7-0	V15-8	V23-16	V31-24
Data Integrity Check	5Bh	Start ADDR [31:24]	Start ADDR [23:16]	Start ADDR [15:8]	Start ADDR [7:0]	End ADDR [31:24]	End ADDR [23:16]	End ADDR [15:8]	End ADDR [7:0]



Table 15. Commands (SPI, 3-Byte Addr. Mode)

Command Name	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9
Read Data	03h	A23-A16	A15-A8	A7-A0	(D7-D0)	(cont.)			
Fast Read	0Bh	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)		
Dual Output Fast Read	3Bh	A23-A16	A15-A8	A7-A0	dummy	(D7-D0) ⁽²⁾	(cont.)		
Quad Output Fast Read	6Bh	A23-A16	A15-A8	A7-A0	dummy	(D7-D0) ⁽³⁾	(cont.)		
Dual I/O Fast Read	BBh	A23-A16 ⁽⁹⁾	A15-A8 ⁽⁹⁾	A7-A0 ⁽⁹⁾	M7-M0 ⁽⁵⁾	(D7-D0) ⁽²⁾	(cont.)		
DTR Dual I/O Fast Read	BDh	A23-A16 ⁽⁹⁾	A15-A8 ⁽⁹⁾	A7-A0 ⁽⁹⁾	M7-M0 ⁽⁵⁾	dummy	(D7-D0) ⁽²⁾	(cont.)	
Quad I/O Fast Read	EBh	A23-A16 ⁽¹⁰⁾	A15-A8 ⁽¹⁰⁾	A7-A0 ⁽¹⁰⁾	M7-M0 ⁽⁷⁾	dummy	dummy	(D7-D0) ⁽³⁾	(cont.)
DTR Quad I/O Fast Read	EDh	A23-A16 ⁽¹⁰⁾	A15-A8 ⁽¹⁰⁾	A7-A0 ⁽¹⁰⁾	M7-M0 ⁽⁷⁾	n-CLK dummy	(D7-D0) ⁽³⁾	(cont.)	
Page Program	02h	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte			
Quad Page Program	32h	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte			
Sector Erase	20h	A23-A16	A15-A8	A7-A0					
Block Erase (32K)	52h	A23-A16	A15-A8	A7-A0					
Block Erase (64K)	D8h	A23-A16	A15-A8	A7-A0					
Read Unique ID	4Bh	00H	00H	00H	dummy	(UID7-UID0)	(cont.)		
Erase Security Registers ⁽¹¹⁾	44h	A23-A16	A15-A8	A7-A0					
Program Security Registers ⁽¹¹⁾	42h	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte			
Read Security Registers ⁽¹¹⁾	48h	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)		
Write Nonvolatile Configuration Register	B1h	A23-A16	A15-A8	A7-A0	(D7-D0)				
Write Volatile Configuration Register	81h	A23-A16	A15-A8	A7-A0	(D7-D0)				
Read Nonvolatile Configuration Register	B5h	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)			
Read Volatile Configuration Register	85h	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)			



Table 16. Commands (SPI, 4-Byte Addr. Mode)

Command Name	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9
Read Data	03h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)	(cont.)		
Fast Read	0Bh	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)	
Dual Output Fast Read	3Bh	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0) ⁽²⁾	(cont.)	
Quad Output Fast Read	6Bh	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0) ⁽³⁾	(cont.)	
Dual I/O Fast Read	BBh	A31-A24 ⁽⁴⁾	A23-A16 ⁽⁴⁾	A15-A8 ⁽⁴⁾	A7-A0 ⁽⁴⁾	M7-M0 ⁽⁵⁾	(D7-D0) ⁽²⁾	(cont.)	
DTR Dual I/O Fast Read	BDh	A31-A24 ⁽⁴⁾	A23-A16 ⁽⁴⁾	A15-A8 ⁽⁴⁾	A7-A0 ⁽⁴⁾	M7-M0 ⁽⁵⁾	dummy	(D7-D0) ⁽²⁾	(cont.)
Quad I/O Fast Read	EBh	A31-A24 ⁽⁶⁾	A23-A16 ⁽⁶⁾	A15-A8 ⁽⁶⁾	A7-A0 ⁽⁶⁾	M7-M0 ⁽⁷⁾	dummy	dummy	(D7-D0) ⁽³⁾
DTR Quad I/O Fast Read	EDh	A31-A24 ⁽⁶⁾	A23-A16 ⁽⁶⁾	A15-A8 ⁽⁶⁾	A7-A0 ⁽⁶⁾	n-CLK dummy	(D7-D0) ⁽³⁾	(cont.)	
Page Program	02h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte		
Quad Page Program	32h	A31-A24 ⁽⁶⁾	A23-A16 ⁽⁶⁾	A15-A8 ⁽⁶⁾	A7-A0 ⁽⁶⁾	D7-D0	Next Byte		
Sector Erase	20h	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (32K)	52h	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (64K)	D8h	A31-A24	A23-A16	A15-A8	A7-A0				
Read Unique ID	4Bh	00H	00H	00H	00H	dummy	(UID7-UID0)	(cont.)	
Erase Security Registers ⁽¹¹⁾	44h	A31-A24	A23-A16	A15-A8	A7-A0				
Program Security Registers ⁽¹¹⁾	42h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte		
Read Security Registers ⁽¹¹⁾	48h	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)	
Write Nonvolatile Configuration Register	B1h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)			
Write Volatile Configuration Register	81h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)			
Read Nonvolatile Configuration Register	B5h	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)		
Read Volatile Configuration Register	85h	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)		



Table 17. Commands (QPI, 3- or 4-Byte Addr. Mode)

Command Name	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9
Clock Number	(0,1)	(2,3)	(4,5)	(6,7)	(8,9)	(10,11)	(12,13)	(14,15)	(16,17)
Write Enable	06h								
Write Disable	04h								
Read Status Register-1	05h	(S7~S0)	(cont.)						
Read Status Register-2	35h	(S15-S8)	(cont.)						
Read Status Register-3	15h	(S23~S16)	(cont.)						
Read Flag Status Register	70h	FS7~FS0	(cont.)						
Write Status Register-1&2	01h	S7-S0	S15-S8						
Write Status Register-3	11h	S23-S16							
Read Extended Addr. Register	C8h	(EA7-EA0)							
Write Extended Addr. Register	C5h	EA7-EA0							
Volatile SR Write Enable	50h								
Clear SR Flags	30h								
Chip Erase	60h/C7h								
Enter 4-Byte Address Mode	B7h								
Exit 4-Byte Address Mode	E9h								
Manufacturer/Device ID	90h	00H	00H	00H	(M7-M0)	(ID7-ID0)	(cont.)		
Read Identification	9Fh	(M7-M0)	(JDID15-JDID8)	(JDID7-JDID0)	(cont.)				
Enable Reset	66h								
Reset	99h								
Program/Erase/Data Integrity Check Suspend	75h								
Program/Erase/Data Integrity Check Resume	7Ah								
Deep Power-Down	B9h								
Release From Deep Power-Down	ABh								
Release From Deep Power-Down, And Read Device ID	ABh	dummy	dummy	dummy	(ID7-ID0)	(cont.)			
Read Serial Flash	5Ah	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)		



Discoverable Parameter									
Fast Read Quad Output with 4-Byte Address	ECh	A31-A24	A23-A16	A15-A8	A7-A0	M7-M0	dummy	(D7-D0)	
DTR Fast Read Quad Output with 4-Byte Address	EEh	A31-A24	A23-A16	A15-A8	A7-A0	M7-M0	n-CLK dummy	(D7-D0)	
Page Program with 4-Byte Address	12h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte		
Sector Erase with 4-Byte Address	21h	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (32K) with 4-Byte Address	5Ch	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (64K) with 4-Byte Address	DCh	A31-A24	A23-A16	A15-A8	A7-A0				
Set Read Parameters	C0h	P7-P0							
Disable QPI	FFh								
Read VL Register	E0h	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)	
Write VL Register	E1h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0			
Read NL Register	E2h	A31-A24	A23-A16	A15-A8	A7-A0	dummy	(D7-D0)	(cont.)	
NL bit Program	E3h	A31-A24	A23-A16	A15-A8	A7-A0				
All NL bit Erase	E4h								
Read Password Register	27h ⁽¹³⁾	dummy	P7~0	P15~8	P23~16	P31~24	P39~32	P47~40	P55~48
Program Password Register	28h	P7~0	P15~8	P23~16	P31~24	P39~32	P47~40	P55~48	P63~56
Password Unlock	29h	P7~0	P15~8	P23~16	P31~24	P39~32	P47~40	P55~48	P63~56
Global Block Lock	7Eh								
Global Block Unlock	98h								
Read Interface CRC Register	64h	A31-A24	A23-A16	A15-A8	A7-A0	V7-0	V15-8	V23-16	V31-24
Data Integrity Check	5Bh	Start ADDR [31:24]	Start ADDR [23:16]	Start ADDR [15:8]	Start ADDR [7:0]	End ADDR [31:24]	End ADDR [23:16]	End ADDR [15:8]	End ADDR [7:0]



Table 18. Commands (QPI, 3-Byte Addr. Mode)

Command Name	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9
Clock Number	(0,1)	(2,3)	(4,5)	(6,7)	(8,9)	(10,11)	(12,13)	(14,15)	(16,17)
Fast Read	0Bh	A23-A16	A15-A8	A7-A0	dummy	dummy	(D7-D0)		
Quad I/O Fast Read	EBh	A23-A16	A15-A8	A7-A0	M7-M0	dummy	(D7-D0)		
DTR Quad I/O Fast Read	EDh	A23-A16	A15-A8	A7-A0	M7-M0	n-CLK dummy	(D7-D0)		
Page Program	02h	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte			
Sector Erase	20h	A23-A16	A15-A8	A7-A0					
Block Erase (32K)	52h	A23-A16	A15-A8	A7-A0					
Block Erase (64K)	D8h	A23-A16	A15-A8	A7-A0					
Burst Read with Wrap	0Ch	A23-A16	A15-A8	A7-A0	dummy	dummy	(D7-D0)		
DTR Burst Read with Wrap	0Eh	A23-A16	A15-A8	A7-A0	dummy	dummy	(D7-D0)		
Write Nonvolatile Configuration Register	B1h	A23-A16	A15-A8	A7-A0	(D7-D0)				
Write Volatile Configuration Register	81h	A23-A16	A15-A8	A7-A0	(D7-D0)				
Read Nonvolatile Configuration Register	B5h	A23-A16	A15-A8	A7-A0	n-CLK dummy	(D7-D0)			
Read Volatile Configuration Register	85h	A23-A16	A15-A8	A7-A0	n-CLK dummy	(D7-D0)			

Table 19. Commands (QPI, 4-Byte Addr. Mode)

Command Name	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9
Clock Number	(0,1)	(2,3)	(4,5)	(6,7)	(8,9)	(10,11)	(12,13)	(14,15)	(16,17)
Fast Read	0Bh	A31-A24	A23-A16	A15-A8	A7-A0	dummy	dummy	(D7-D0)	
Quad I/O Fast Read	EBh	A31-A24	A23-A16	A15-A8	A7-A0	M7-M0	dummy	(D7-D0)	
DTR Quad I/O Fast Read	EDh	A31-A24	A23-A16	A15-A8	A7-A0	M7-M0	n-CLK dummy	(D7-D0)	
Page Program	02h	A31-A24	A23-A16	A15-A8	A7-A0	D7-D0	Next Byte		
Sector Erase	20h	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (32K)	52h	A31-A24	A23-A16	A15-A8	A7-A0				
Block Erase (64K)	D8h	A31-A24	A23-A16	A15-A8	A7-A0				
Burst Read with Wrap	0Ch	A31-A24	A23-A16	A15-A8	A7-A0	dummy	dummy	(D7-D0)	
DTR Burst Read with Wrap	0Eh	A31-A24	A23-A16	A15-A8	A7-A0	dummy	dummy	(D7-D0)	
Write Nonvolatile Configuration Register	B1h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)			
Write Volatile Configuration Register	81h	A31-A24	A23-A16	A15-A8	A7-A0	(D7-D0)			



Read Nonvolatile Configuration Register	B5h	A31-A24	A23-A16	A15-A8	A7-A0	n-CLK dummy	(D7-D0)		
Read Volatile Configuration Register	85h	A31-A24	A23-A16	A15-A8	A7-A0	n-CLK dummy	(D7-D0)		

Notes:

1. Dummy bits and Wrap Bits

IO0 = (x, x, x, x, x, x, W4, x)

IO1 = (x, x, x, x, x, x, W5, x)

IO2 = (x, x, x, x, x, x, W6, x)

IO3 = (x, x, x, x, x, x, x, x)

2. Dual Output data

IO0 = (D6, D4, D2, D0)

IO1 = (D7, D5, D3, D1)

3. Quad Output Data

IO0 = (D4, D0, ...)

IO1 = (D5, D1, ...)

IO2 = (D6, D2, ...)

IO3 = (D7, D3, ...)

4. Dual Input 4-Byte Address

IO0 = A30, A28, A26, A24, A22, A20, A18, A16, A14, A12, A10, A8, A6, A4, A2, A0

IO1 = A31, A29, A27, A25, A23, A21, A19, A17, A15, A13, A11, A9, A7, A5, A3, A1

5. Dual Input Mode bit

IO0 = M6, M4, M2, M0

IO1 = M7, M5, M3, M1

6. Quad Input 4-Byte Address

IO0 = A28, A24, A20, A16, A12, A8, A4, A0

IO1 = A29, A25, A21, A17, A13, A9, A5, A1

IO2 = A30, A26, A22, A18, A14, A10, A6, A2

IO3 = A31, A27, A23, A19, A15, A11, A7, A3

7. Quad Input Mode bit

IO0 = M4, M0

IO1 = M5, M1

IO2 = M6, M2

IO3 = M7, M3

8. Quad Output Data

IO0 = D4, D0, ...

IO1 = D5, D1, ...

IO2 = D6, D2, ...

IO3 = D7, D3, ...

9. Dual Input Address

IO0 = A22, A20, A18, A16, A14, A12, A10, A8, A6, A4, A2, A0

IO1 = A23, A21, A19, A17, A15, A13, A11, A9, A7, A5, A3, A1

10. Quad Input Address

IO0 = A20, A16, A12, A8, A4, A0

IO1 = A21, A17, A13, A9, A5, A1

IO2 = A22, A18, A14, A10, A6, A2

IO3 = A23, A19, A15, A11, A7, A3

11. Security Registers Address

Security Register1: A23-A16=00H, A15-A12=1H, A11--A0= Byte Address;

Security Register2: A23-A16=00H, A15-A12=2H, A11- A0= Byte Address;

Security Register3: A23-A16=00H, A15-A12=3H, A11- A0= Byte Address;

12. QPI Command, Address, Data input/output format:

CLK #0 1 2 3 4 5 6 7 8 9 10 11

IO0= C4, C0, A20, A16, A12, A8, A4, A0, D4, D0, D4, D0,

IO1= C5, C1, A21, A17, A13, A9, A5, A1, D5, D1, D5, D1

IO2= C6, C2, A22, A18, A14, A10, A6, A2, D6, D2, D6, D2

IO3= C7, C3, A23, A19, A15, A11, A7, A3, D7, D3, D7, D3

13. In QPI mode, Read Password Register(27H) need 8dummy cycle.

Table of ID Definitions

GD25LF256F

Operation Code	MID7-MID0	ID15-ID8	ID7-ID0
9Fh	C8	63	19
90h	C8		18
ABh			18

9.1 Write Enable (WREN) (06h)

The Write Enable (WREN) command is for setting the Write Enable Latch (WEL) bit. The Write Enable Latch (WEL) bit must be set prior to every Page Program (PP), Sector Erase (SE), Block Erase (BE), Chip Erase (CE), Write Status Register (WRSR), Write Extended Address Register (WEAR), Write Nonvolatile/Volatile configure register and Erase/Program Security Registers command. The Write Enable (WREN) command sequence: CS# goes low → sending the Write Enable command → CS# goes high.

Figure 15 Write Enable Sequence Diagram (SPI)

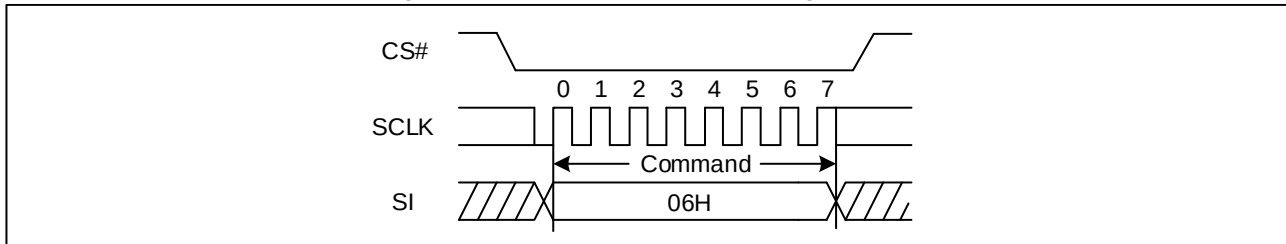
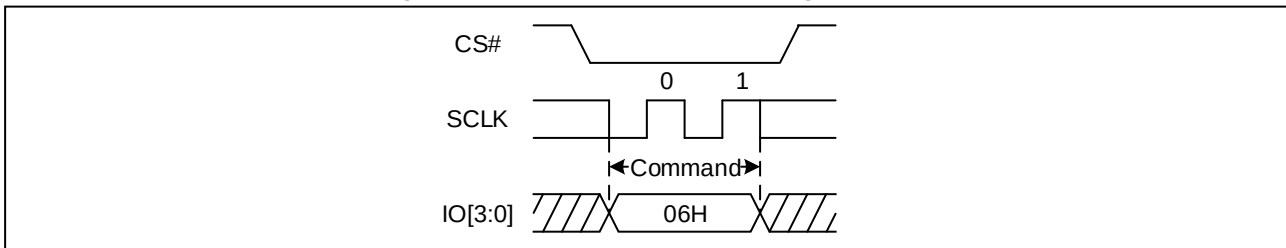


Figure 16 Write Enable Sequence Diagram (QPI)



9.2 Write Disable (WRDI) (04h)

The Write Disable command is for resetting the Write Enable Latch (WEL) bit. The Write Disable command sequence: CS# goes low → Sending the Write Disable command → CS# goes high. The WEL bit is reset by following condition: Power-up and upon completion of the Write Status Register, Write Extended Address Register (WEAR), Write Nonvolatile/Volatile configure register, Page Program, Sector Erase, Block Erase, Chip Erase, Erase/Program Security Registers and Reset commands.

Figure 17 Write Disable Sequence Diagram (SPI)

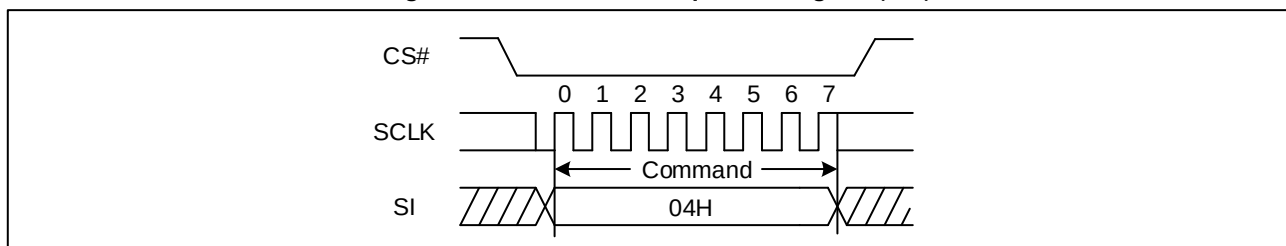
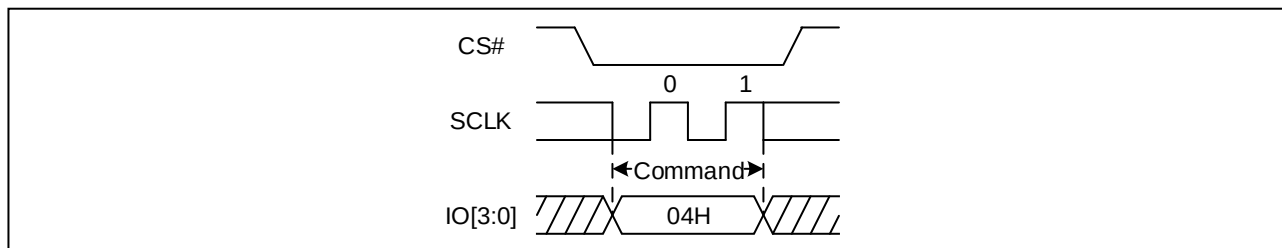


Figure 18 Write Disable Sequence Diagram (QPI)



9.3 Read Status Register (05h/35h/15h)

The Read Status Register (RDSR) command is for reading the Status Register. The Status Register may be read at any time, even while a Program, Erase or Write Status Register cycle is in progress. When one of these cycles is in progress, it is recommended to check the Write in Progress (WIP) bit before sending a new command to the device. It is also possible to read the Status Register continuously. For command code of "05h" / "35h" / "15h", the SO will output Status Register bits S7~S0 / S15-S8 / S23-S16.

Figure 19 Read Status Register Sequence Diagram (SPI)

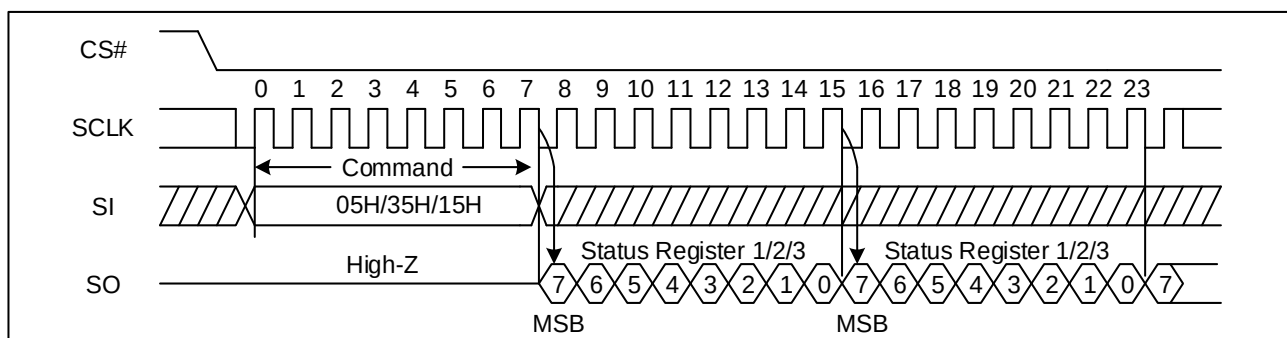
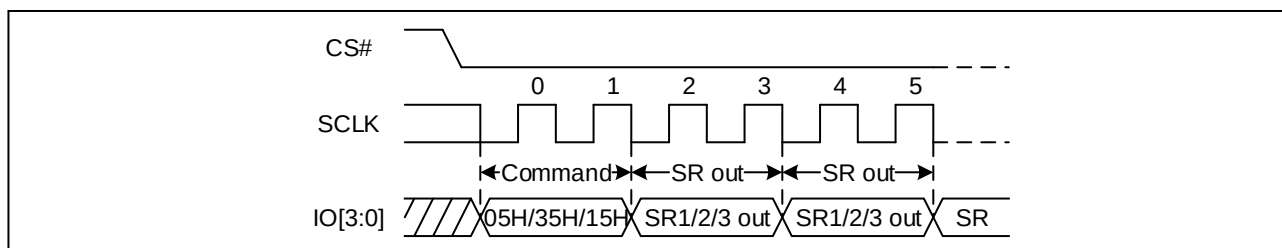


Figure 20 Read Status Register Sequence Diagram (QPI)



9.4 Read Flag Status Register (70h)

The Read Flag Status Register command is for reading the Flag Status Register. The Flag Status Register may be read at any time, even while a Program, Erase or Write Status Register cycle is in progress. When one of these cycles is in progress, it is also possible to read the Flag Status Register continuously.

Figure 21 Read Flag Status Register Sequence Diagram (SPI)

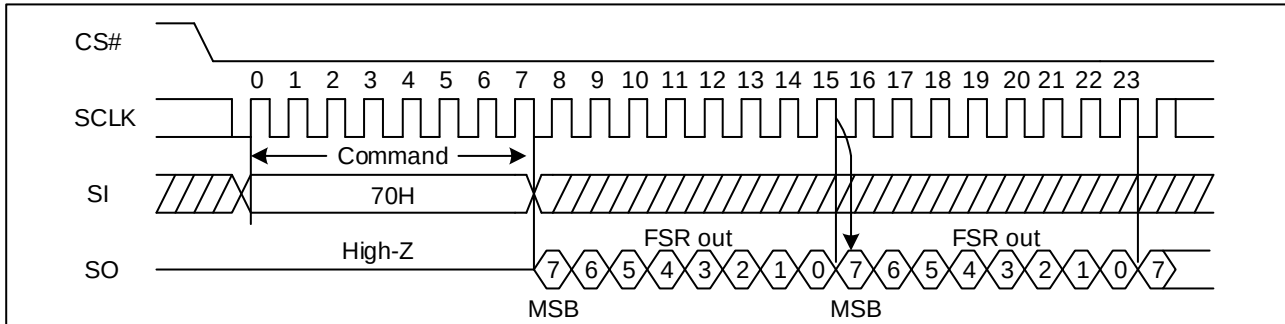
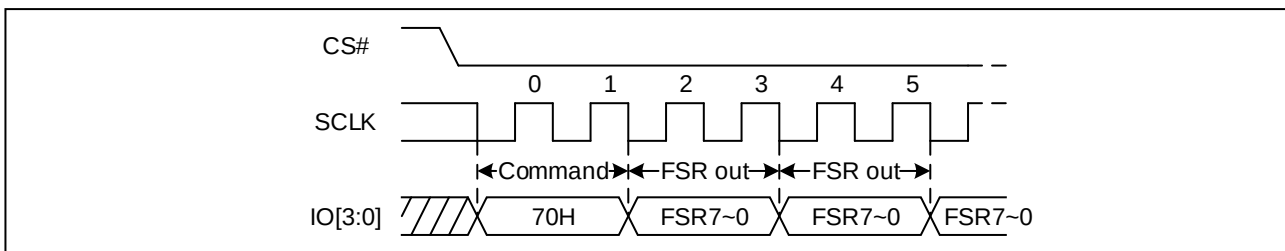


Figure 22 Read Flag Status Register Sequence Diagram (QPI)



9.5 Write Status Register (WRSR) (01h/11h)

The Write Status Register (WRSR) command allows new values to be written to the Status Register. Before it can be accepted, a Write Enable (WREN) command must previously have been executed. After the Write Enable (WREN) command has been decoded and executed, the device sets the Write Enable Latch (WEL).

The Write Status Register (WRSR) command has no effect on S19, S15, S10, S9, S1 and S0 of the Status Register. CS# must be driven high after the eighth or sixteen bit of the data byte has been latched in. Otherwise, the Write Status Register (WRSR) command is not executed. If CS# is driven high after eighth bit of the data byte, the alterable bits in Status Register-2 (S15~S8) will be cleared to 0. As soon as CS# is driven high, the self-timed Write Status Register cycle (whose duration is t_W) is initiated. While the Write Status Register cycle is in progress, the Status Register may still be read to check the value of the Write In Progress (WIP) bit. The Write In Progress (WIP) bit is 1 during the self-timed Write Status Register cycle, and is 0 when it is completed. When the cycle is completed, the Write Enable Latch (WEL) is reset.

The Write Status Register (WRSR) command allows the user to change the values of the Block Protect (BP4, BP3, BP2, BP1, and BP0) bits, to define the size of the area that is to be treated as read-only.

01h command is used to write Status Register-1&2. 11h command is used to write Status Register-3.

Figure 23. Write Status Register-1&2 Sequence Diagram (SPI)

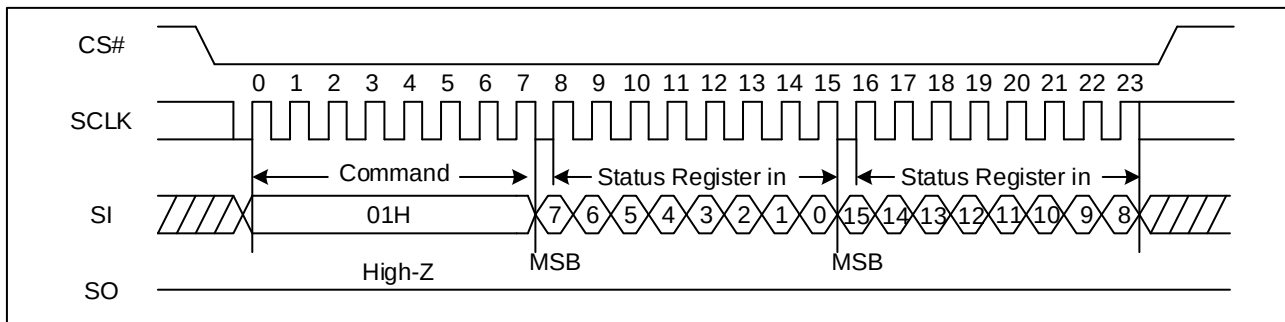


Figure 24. Write Status Register-1&2 Sequence Diagram (QPI)

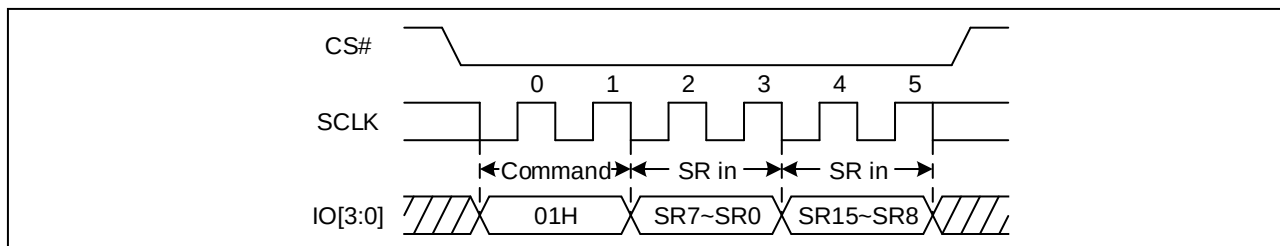


Figure 25. Write Status Register-3 Sequence Diagram (SPI)

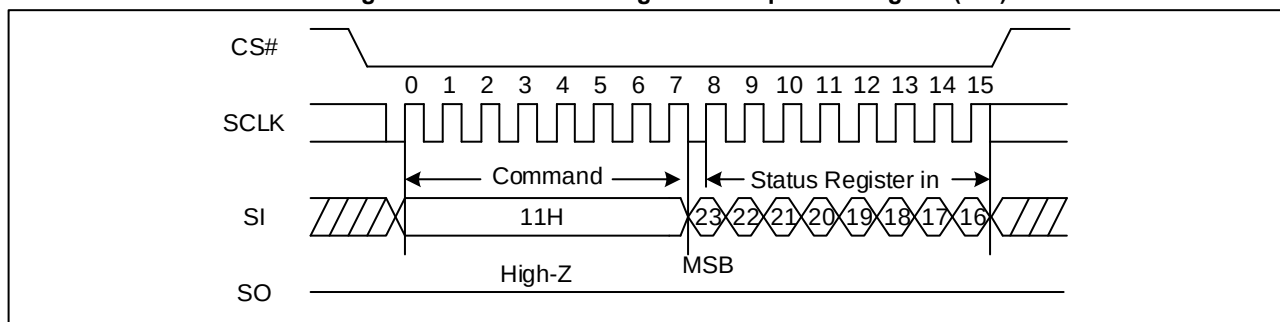
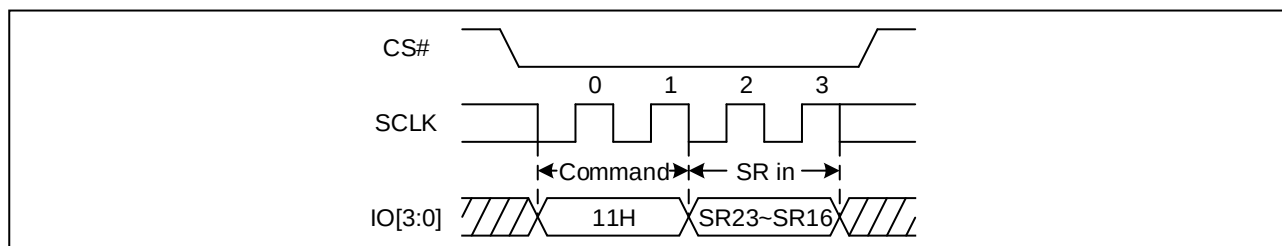


Figure 26. Write Status Register-3 Sequence Diagram





9.6 Read Extended Address Register (C8h)

The Read Extended Address Register instruction is entered by driving CS# low and shifting the instruction code “C8h” into the SI pin on the rising edge of SCLK. The Extended Address Register bits are then shifted out on the SO pin at the falling edge of SCLK with most significant bit (MSB) first.

When the device is in the 4-Byte Address Mode, the value of the address bits is ignored.

Figure 27 Read Extended Address Register Sequence Diagram (SPI)

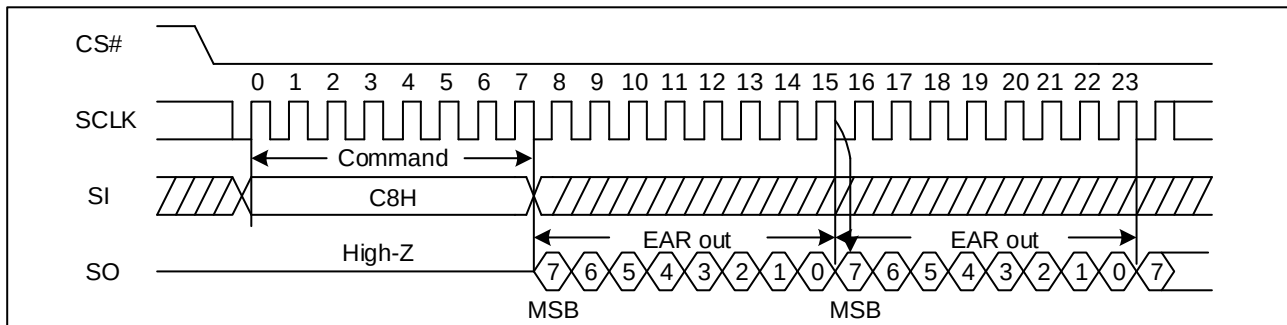
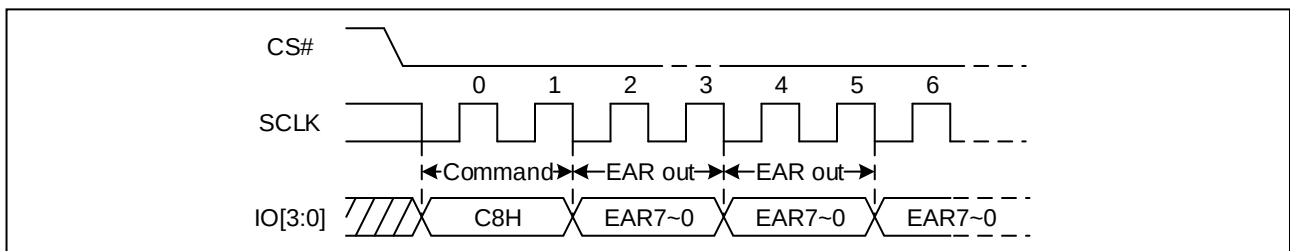


Figure 28 Read Extended Address Register Sequence Diagram (QPI)



9.7 Write Extended Address Register (C5h)

The Extended Address Register is a volatile register that stores the 4th Byte address (A31-A24) when the device is operating in the 3-Byte Address Mode (ADS=0). To write the Extended Address Register bits, a Write Enable (06h) instruction must previously have been executed for the device to accept the Write Extended Address Register instruction (Status Register bit WEL must equal 1). Once write enabled, the instruction is entered by driving CS# low, sending the instruction code “C5h”, and then writing the Extended Address Register data Byte.

Upon power up or the execution of a Software/Hardware Reset, the Extended Address Register bit values will be cleared to 0.

The Extended Address Bit is only effective when the device is in the 3-Byte Address Mode. When the device operates in the 4-Byte Address Mode (ADS=1), any command with address input of A31-A24 will replace the Extended Address Register values. It is recommended to check and update the Extended Address Register if necessary when the device is switched from 4-Byte to 3-Byte Address Mode.

Figure 29 Write Extended Address Register Sequence Diagram (SPI)

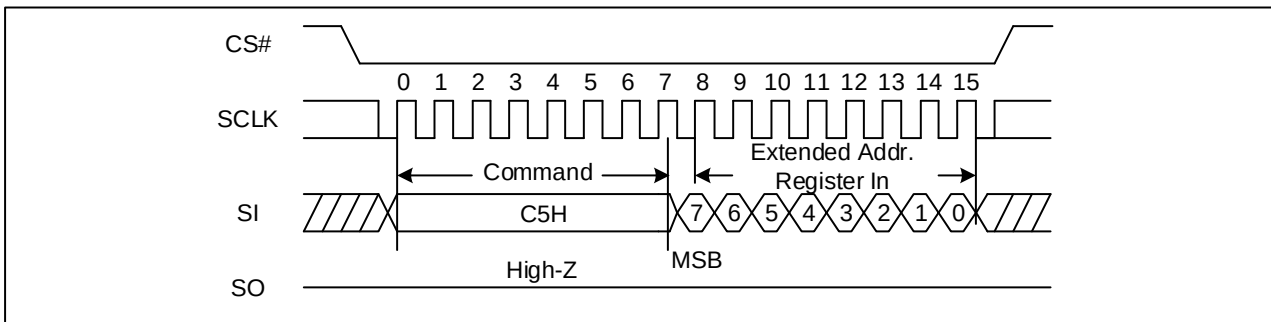
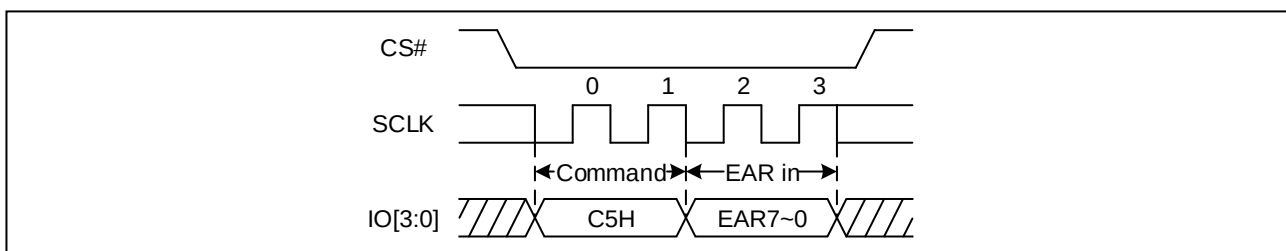


Figure 30 Write Extended Address Register Sequence Diagram (QPI)



9.8 Write Enable for Volatile Status Register (50h)

The non-volatile Status Register bits can also be written to as volatile bits. This gives more flexibility to change the system configuration and memory protection schemes quickly without waiting for the typical non-volatile bit write cycles or affecting the endurance of the Status Register non-volatile bits. The Write Enable for Volatile Status Register command must be issued prior to a Write Status Register command, and any other commands cannot be inserted between them. Otherwise, Write Enable for Volatile Status Register will be cleared. The Write Enable for Volatile Status Register command will not set the Write Enable Latch bit, it is only valid for the Write Status Register command to change the volatile Status Register bit values.

Figure 31 Write Enable for Volatile Status Register Sequence Diagram (SPI)

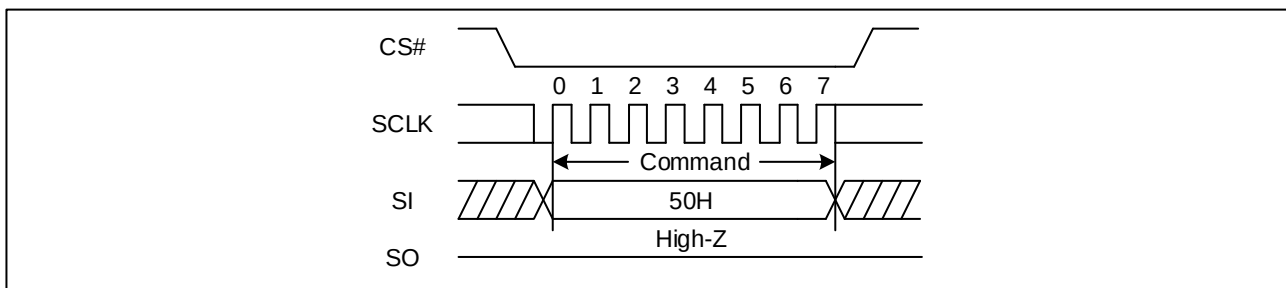
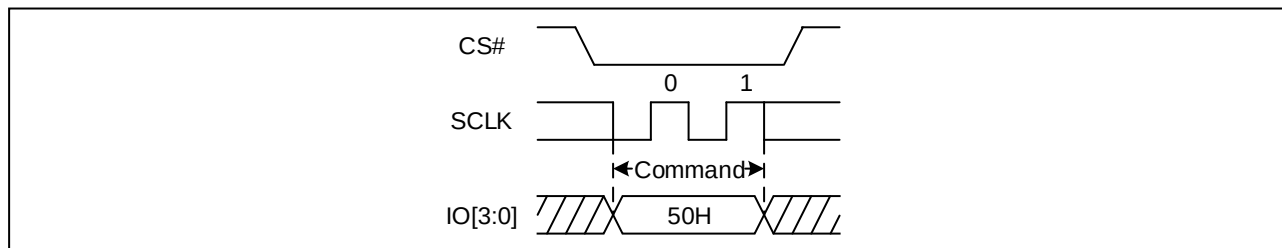


Figure 32 Write Enable for Volatile Status Register Sequence Diagram (QPI)



9.9 Clear Flag Status Register (30h)

The Clear Status Register Flags command resets bit FS0 (Erase Error bit) and FS1 (Program Error bit) and FS6 (Data Integrity Check Abort bit) in status register. It is not necessary to set the WEL bit before the Clear Status Register command is executed. The Clear Flag Status Register command will not be accepted when the device remains busy with WIP set to 1. The WEL bit will be unchanged after this command is executed.

Figure 33. Clear Status Register Flags Diagram (SPI)

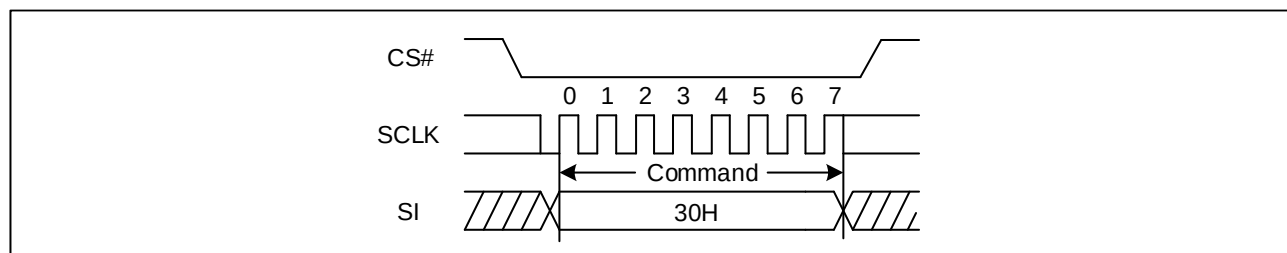
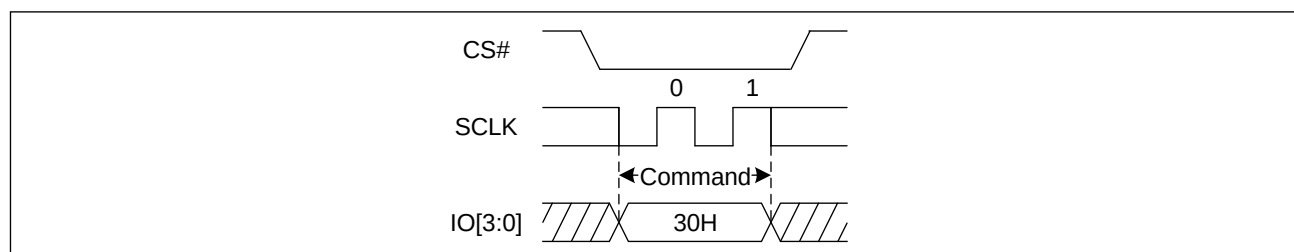


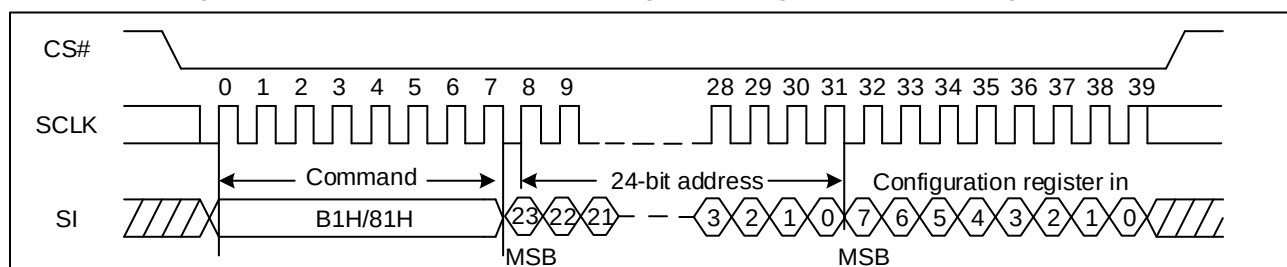
Figure 34. Clear Status Register Flags Diagram (QPI)



9.10 Write Nonvolatile/Volatile Configuration Register (B1h/81h)

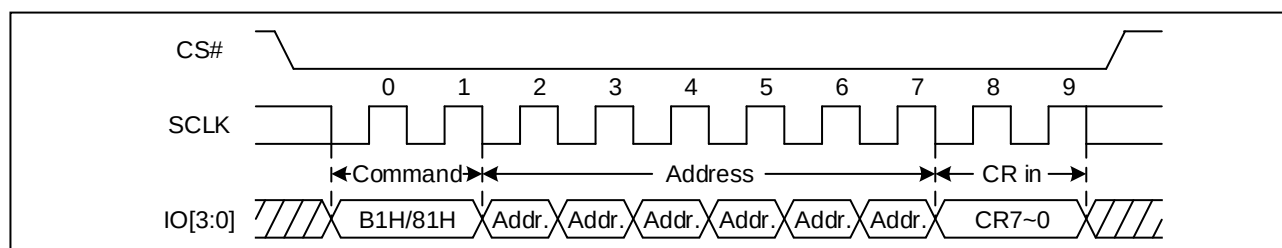
The Write Nonvolatile/Volatile Configuration Register command allows new values to be written to the Nonvolatile/Volatile Configuration Register. Before it can be accepted, a Write Enable (WREN) command must previously have been executed. After the Write Enable (WREN) command has been decoded and executed, the device sets the Write Enable Latch (WEL). CS# must be driven high after the data Byte has been latched in. If not, the Write Configuration Register command is not executed. As soon as CS# is driven high, the self-timed Write Configuration Register cycle (whose duration is t_W for B1h) is initiated. The Write In Progress (WIP) bit is 1 during the self-timed Write Configuration Register cycle, and is 0 when it is completed. When the cycle is completed, the Write Enable Latch (WEL) is reset.

Figure 35 Write Nonvolatile/Volatile Configuration Register Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 36 Write Nonvolatile/Volatile Configuration Register Sequence Diagram (QPI)

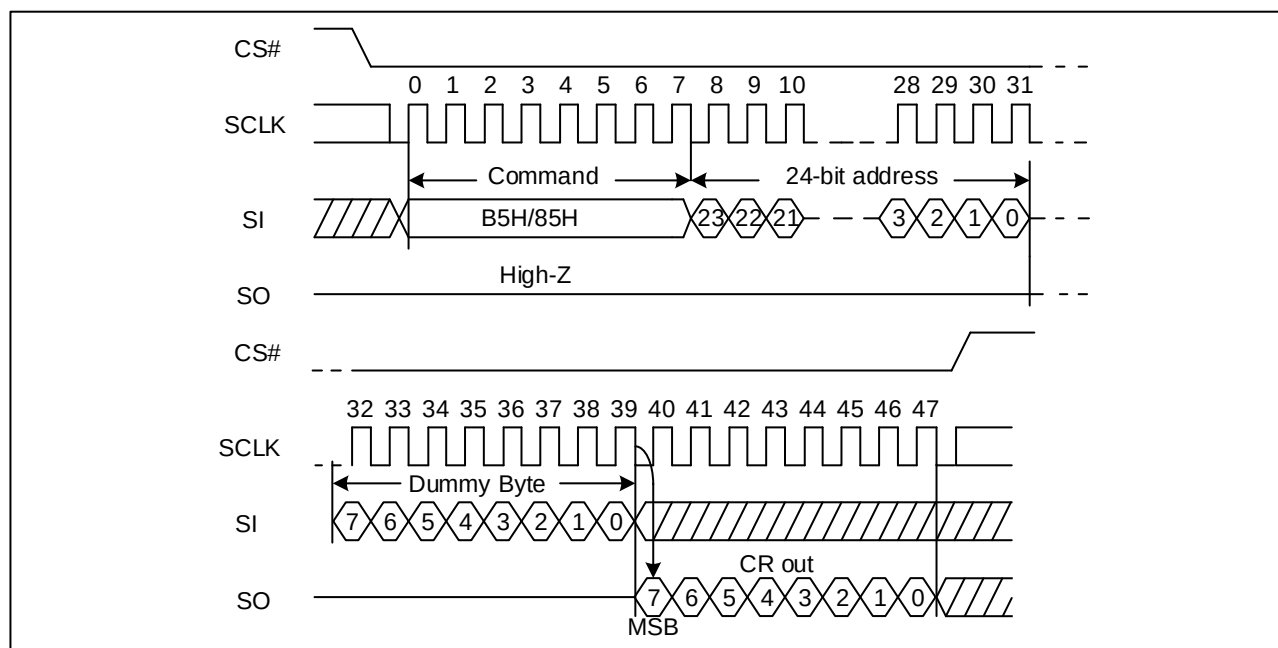


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.11 Read Nonvolatile/Volatile Configuration Register (B5h/85h)

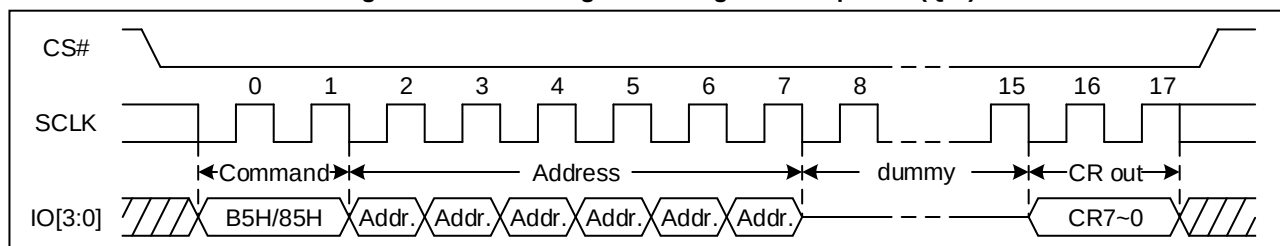
The Read Nonvolatile/Volatile Configuration Register command is for reading the Nonvolatile/Volatile Configuration Registers. It is followed by a 3-Byte address (A23-A0) or a 4-Byte address (A31-A0) and a dummy Byte, and each bit is latched-in on the rising edge of SCLK. Then the Configuration Register, at that address, is shifted out on SO, and each bit is shifted out, at a Max frequency f_c , on the falling edge of SCLK. Read Nonvolatile/Volatile Configuration Register command, while an Erase, Program or Write cycle is in progress, is rejected without having any effects on the cycle that is in progress.

Figure 37 Read Configuration Registers Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 38 Read Configuration Registers Sequence (QPI)

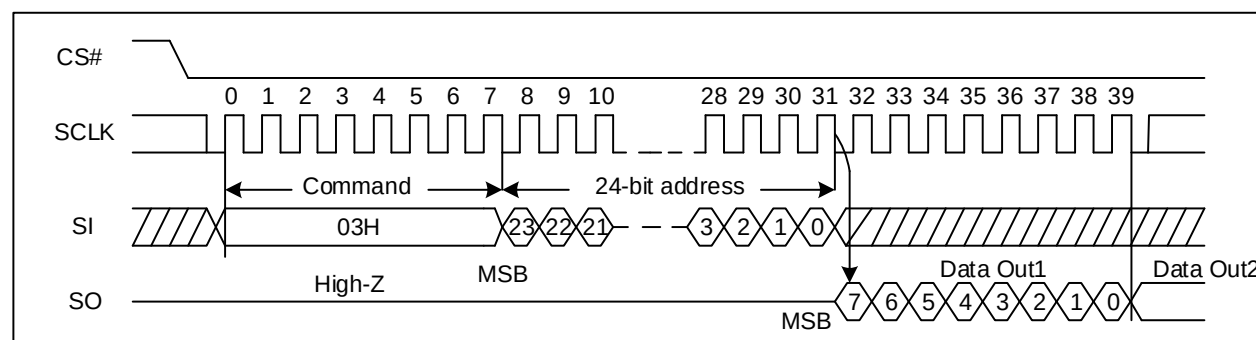


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.12 Read Data Bytes (03h/13h)

The Read Data Bytes (READ) command is followed by a 3-Byte address (A23-A0) or a 4-Byte address (A31-A0), and each bit is latched-in on the rising edge of SCLK. Then the memory content, at that address, is shifted out on SO, and each bit is shifted out, at a Max frequency f_R , on the falling edge of SCLK. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out. The whole memory can, therefore, be read with a single Read Data Bytes (READ) command. Any Read Data Bytes (READ) command, while an Erase, Program or Write cycle is in progress, is rejected without having any effects on the cycle that is in progress.

Figure 39 Read Data Bytes Sequence Diagram

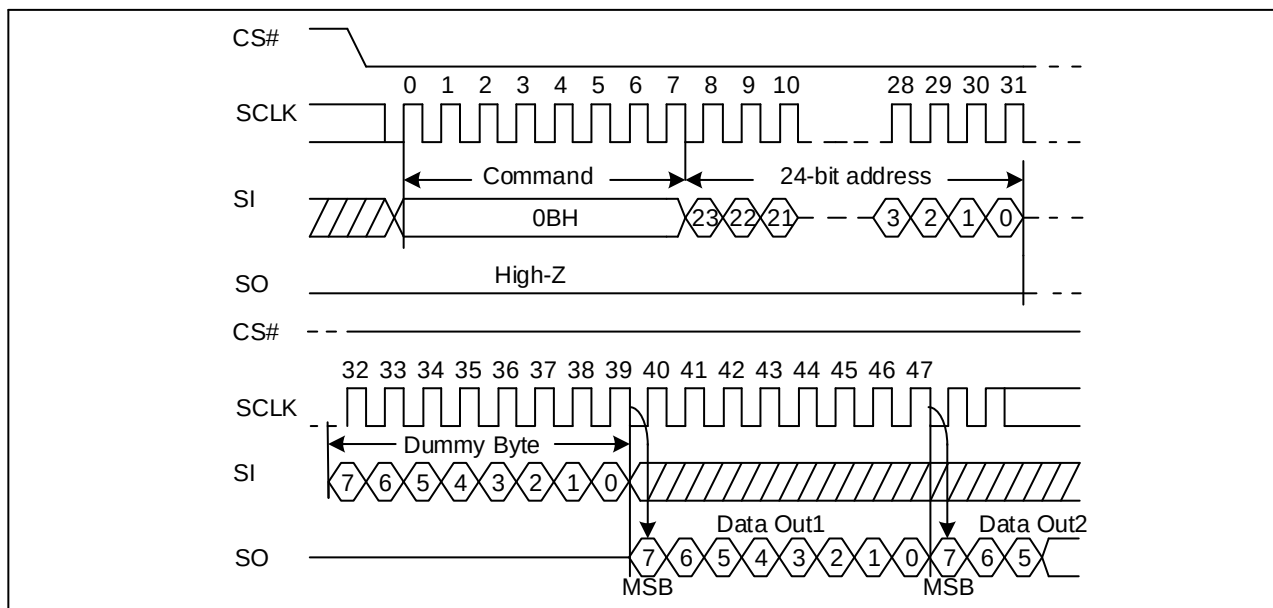


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.13 Read Data Bytes at Higher Speed (0Bh/0Ch)

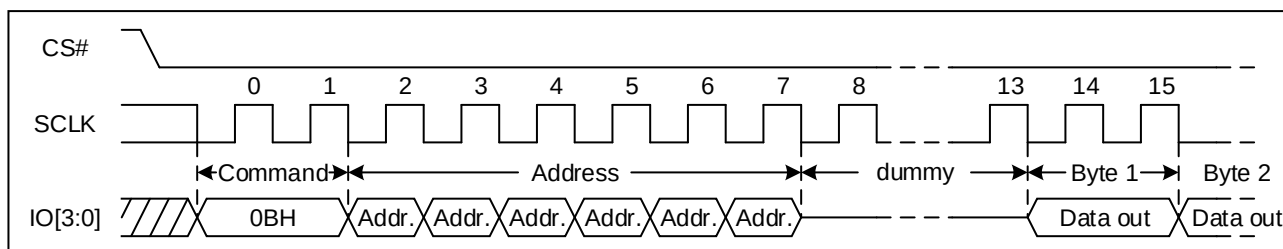
The Read Data Bytes at Higher Speed (Fast Read) command is for quickly reading data out. It is followed by a 3-Byte address (A23-A0) or a 4-Byte address (A31-A0) and dummy clocks, and each bit is latched-in on the rising edge of SCLK. Then the memory content, at that address, is shifted out on SO, and each bit is shifted out, at a Max frequency f_C , on the falling edge of SCLK. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out.

Figure 40 Read Data Bytes at Higher Speed Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 41 Read Data Bytes at Higher Speed Sequence Diagram (QPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

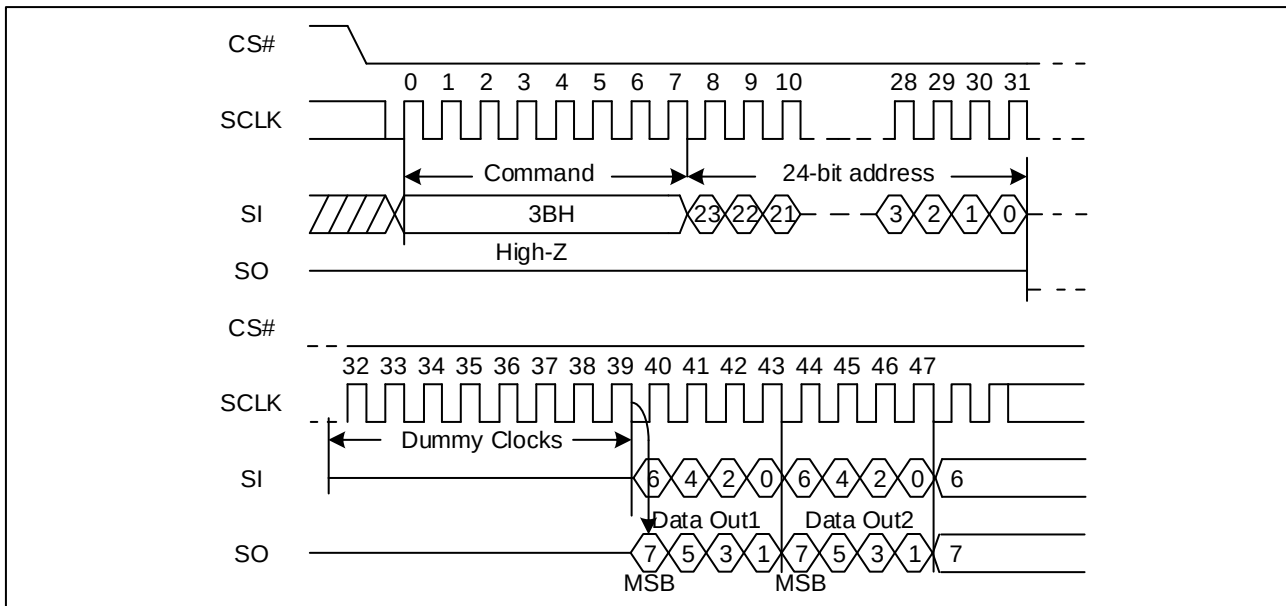


9.14 Dual Output Fast Read (3Bh/3Ch)

The Dual Output Fast Read command is followed by 3/4-Byte address and a dummy Byte, and each bit being latched in on the rising edge of SCLK, then the memory contents are shifted out 2-bit per clock cycle from SI and SO.

The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out.

Figure 42. Dual Output Fast Read Sequence Diagram

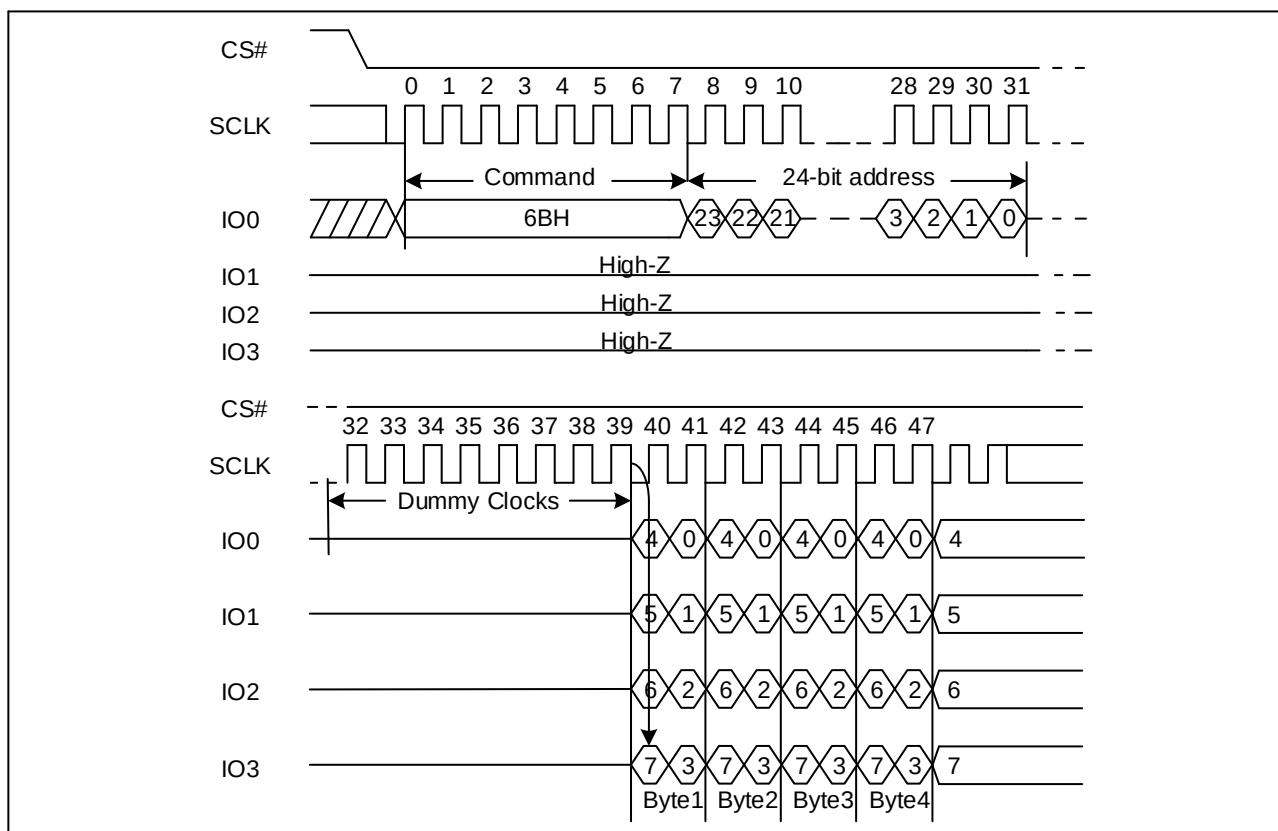


Note: The device default is in 24-bit address mode. For 4-Byte mode or 4-Byte command, the address length is 32-bit.

9.15 Quad Output Fast Read (6Bh/6Ch)

The Quad Output Fast Read command is followed by 3-Byte address (A23-A0) or a 4-Byte address (A31-A0) and dummy clocks, and each bit is latched in on the rising edge of SCLK, then the memory contents are shifted out 4-bit per clock cycle from IO3, IO2, IO1 and IO0. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out.

Figure 43 Quad Output Fast Read Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

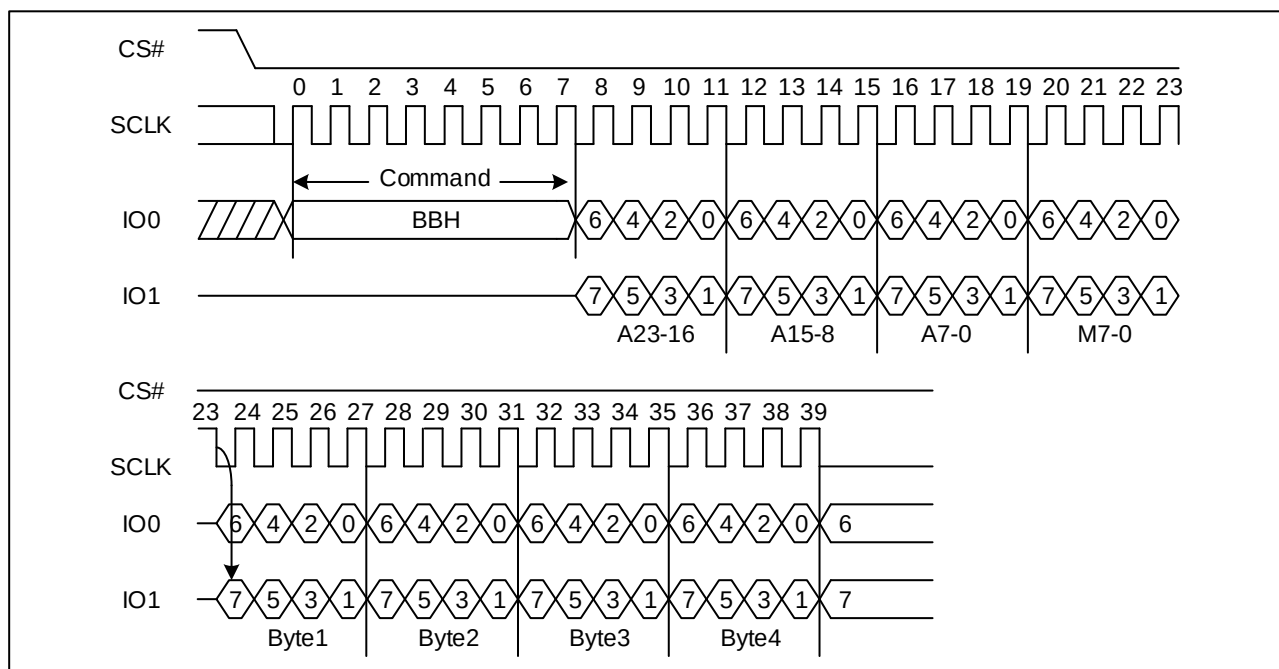
9.16 Dual I/O Fast Read (BBh/BCh)

The Dual I/O Fast Read command is similar to the Dual Output Fast Read command but with the capability to input the 3/4-Byte address and a “Continuous Read Mode” Byte 2-bit per clock by SI and SO, and each bit being latched in on the rising edge of SCLK, then the memory contents are shifted out 2-bit per clock cycle from SI and SO. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out.

Dual I/O Fast Read with “Continuous Read Mode”

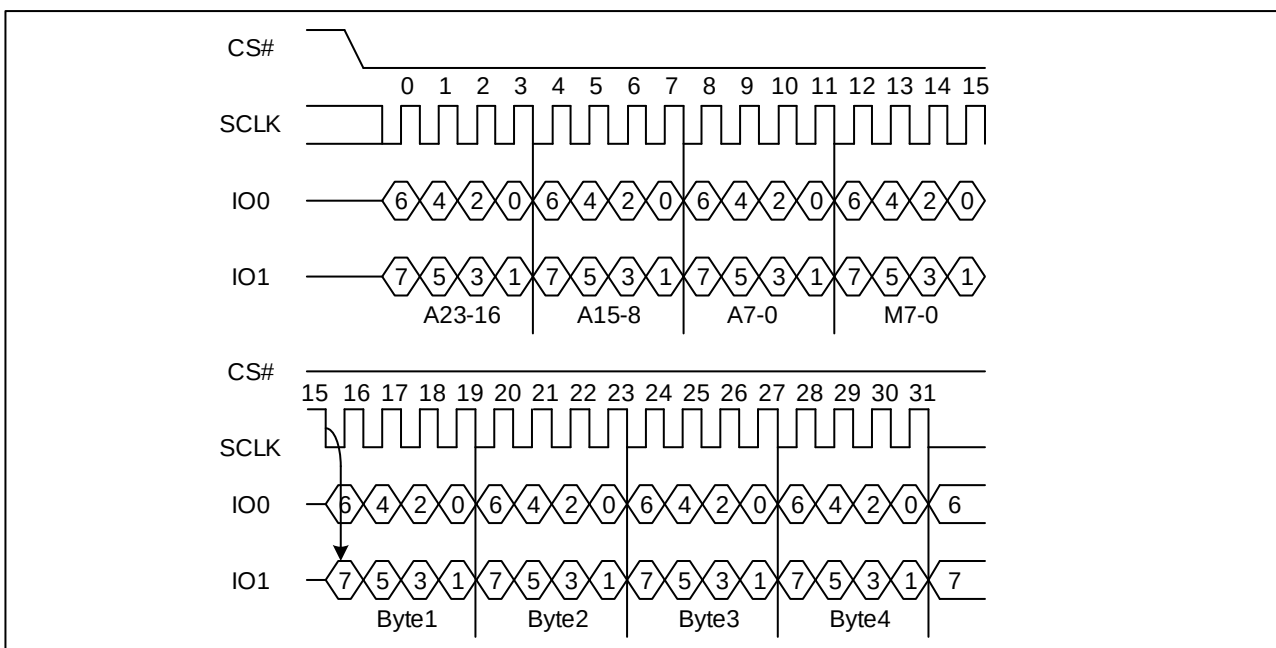
The Dual I/O Fast Read command can further reduce command overhead through setting the “Continuous Read Mode” bits (M7-0) after the input 3- or 4-Byte address (A23-A0 or A31-A0). If the “Continuous Read Mode” bits (M5-4) = (1, 0), then the next Dual I/O Fast Read command (after CS# is raised and then lowered) does not require the BBh/BCh command code. If the “Continuous Read Mode” bits (M5-4) do not equal (1, 0), the next command requires the command code, thus returning to normal operation. A Reset command can be also used to reset (M7-0) before issuing normal command.

Figure 44 Dual I/O Fast Read Sequence Diagram ((M5-4) ≠ (1, 0))



Note: The device default is in 24-bit address mode. For 4-Byte mode or 4-Byte command, the address length is 32-bit.

Figure 45 Dual I/O Fast Read Sequence Diagram ((M5-4) = (1, 0))



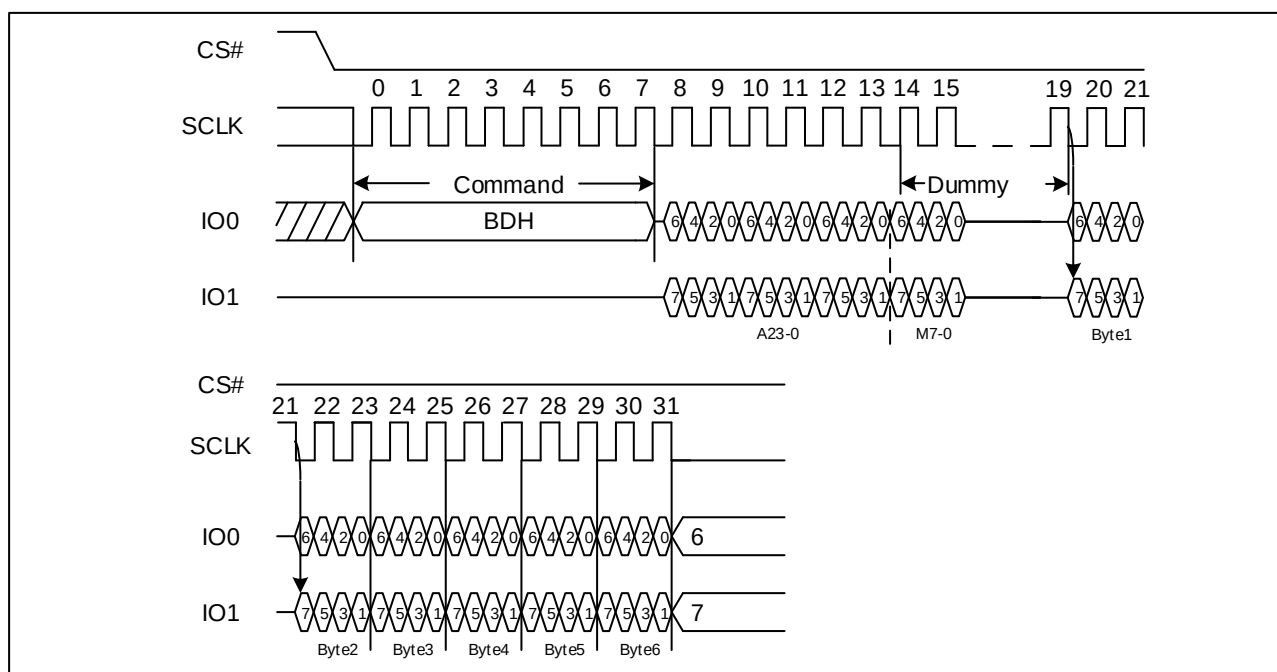
Note: The device default is in 24-bit address mode. For 4-Byte mode or 4-Byte command, the address length is 32-bit.

9.17 Dual I/O DTR Read (BDh/BEh)

The Dual I/O DTR Read instruction enables Double Transfer Rate throughput on dual I/O of Serial Flash in read mode. The address (interleave on 2 I/O pins, IO0 and IO1) is latched on both rising and falling edge of SCLK, and data (interleave on 2 I/O pins, IO0 and IO1) shift out on both rising and falling edge of SCLK. The 4-bit address can be latched-in at one clock, and 4-bit data can be read out at one clock, which means two bits at rising edge of clock, the other two bits at falling edge of clock. The first address Byte can be at any location. The address is automatically increased to the next higher address after each Byte data is shifted out, so the whole memory can be read out at a single Dual I/O DTR Read command. The address counter rolls over to 0 when the highest address has been reached.

While Program/Erase/Write Status Register cycle is in progress, Dual I/O DTR Read command is rejected without any impact on the Program/Erase/Write Status Register current cycle.

Figure 46. DTR Dual I/O Fast Read Sequence Diagram (M5-4 ≠ (1, 0))

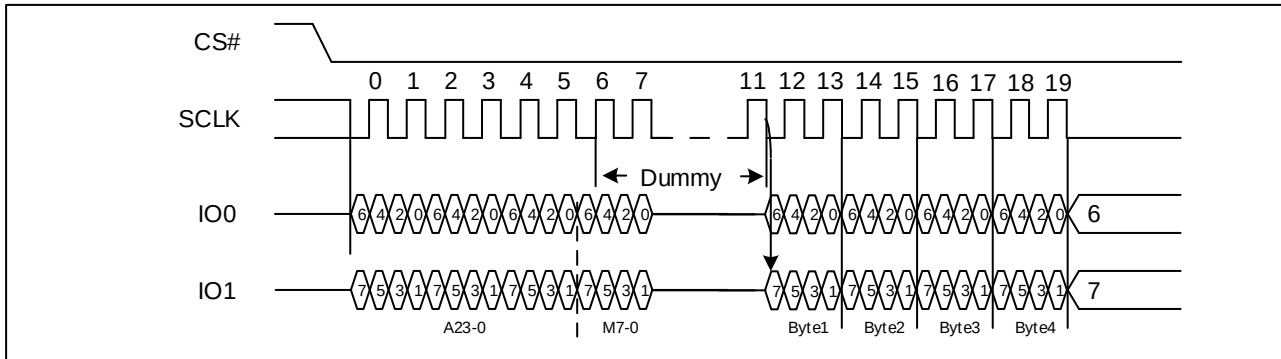


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Dual I/O DTR Read with “Continuous Read Mode”

The Dual I/O Fast Read command can further reduce command overhead through setting the “Continuous Read Mode” bits (M7-4) after the input 3- or 4-Byte address (A23-A0 or A31-A0). If the “Continuous Read Mode” bits (M5-4) = (1, 0), then the next Dual I/O Fast Read command (after CS# is raised and then lowered) does not require the BDh/BEh command code. If the “Continuous Read Mode” bits (M5-4) do not equal (1, 0), the next command requires the command code, thus returning to normal operation. A Reset command can be also used to reset (M7-0) before issuing normal command.

Figure 47. DTR Dual I/O Fast Read Sequence Diagram (M5-4 = (1, 0))



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

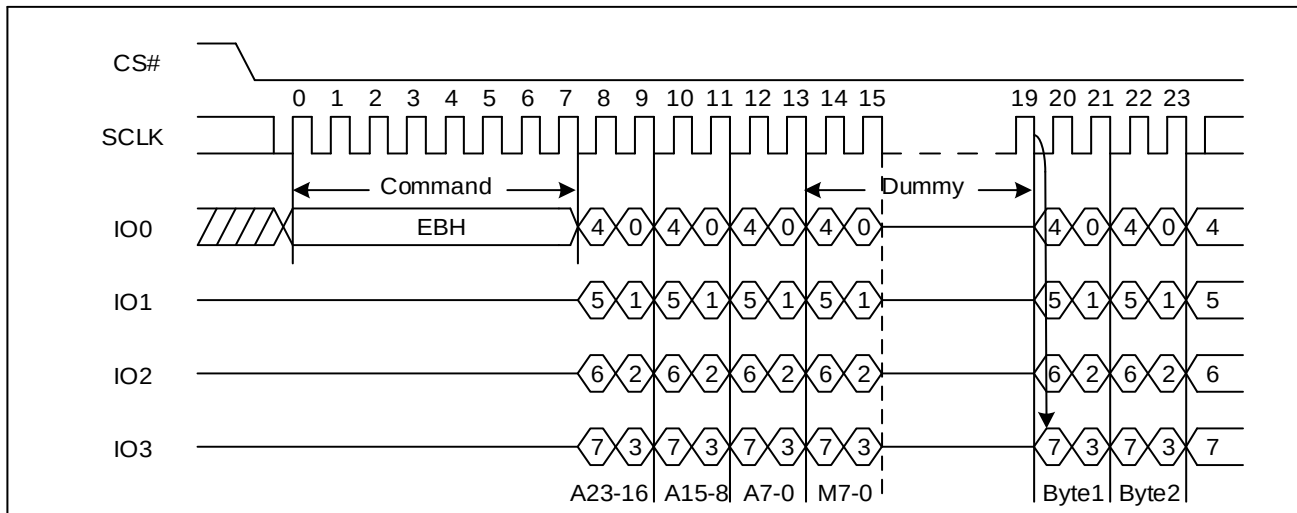
9.18 Quad I/O Fast Read (EBh/ECh)

The Quad I/O Fast Read command is similar to the Quad Output Fast Read command but with the capability to input the 3-Byte address (A23-A0) or a 4-Byte address (A31-A0) and a “Continuous Read Mode” Byte and dummy clocks. 4-bit per clock is transferred by IO0, IO1, IO2, IO3, and each bit is latched in on the rising edge of SCLK, then the memory contents are shifted out 4-bit per clock cycle from IO0, IO1, IO2, IO3. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out.

Quad I/O Fast Read with “Continuous Read Mode”

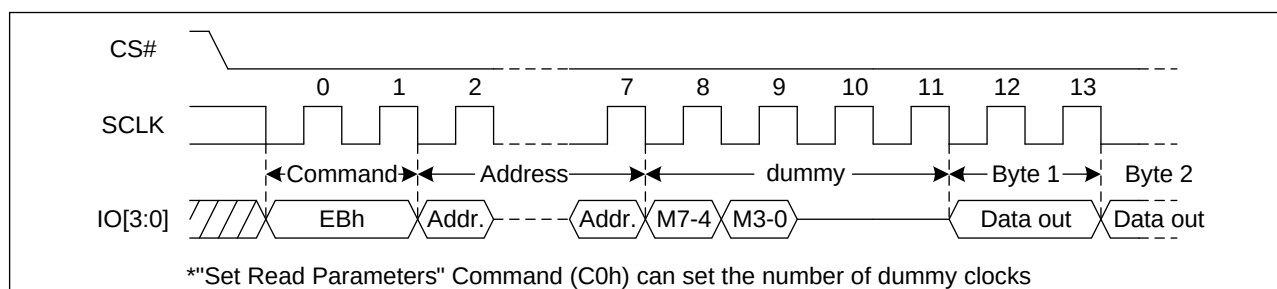
The Quad I/O Fast Read command can further reduce command overhead through setting the “Continuous Read Mode” bits (M7-0) after the input 3-Byte address (A23-A0) or 4-Byte address (A31-A0). If the “Continuous Read Mode” bits (M5-4) = (1, 0), then the next Quad I/O Fast Read command (after CS# is raised and then lowered) does not require the EBh/ECh command code. If the “Continuous Read Mode” bits (M5-4) do not equal to (1, 0), the next command requires the command code, thus returning to normal operation. A Reset command can be also used to reset (M7-0) before issuing normal command.

Figure 48 Quad I/O Fast Read Sequence Diagram (SPI, M5-4≠ (1, 0))



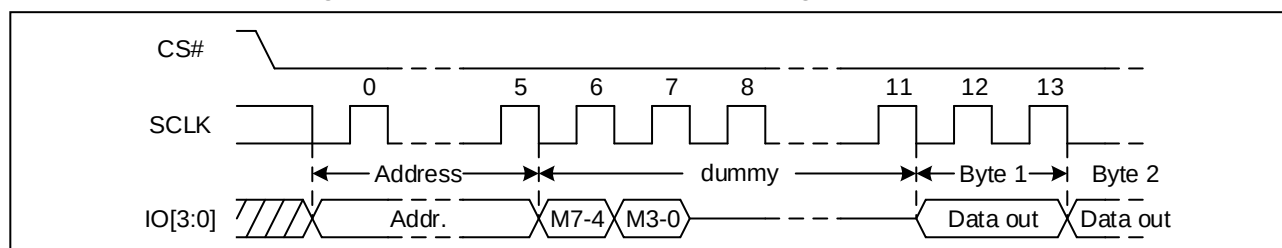
Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 49 Quad I/O Fast Read Sequence Diagram (QPI, M5-4≠ (1, 0))



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 50 Quad I/O Fast Read Sequence Diagram (M5-4= (1, 0))



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Quad I/O Fast Read with “8/16/32/64-Byte Wrap Around”

The Quad I/O Fast Read command can be used to access a specific portion within a page by issuing Wrap configuration register Byte prior to EBh/ECh. The data being accessed can be limited to either a 8/16/32/64-Byte section of a 256-Byte page. The output data starts at the initial address specified in the command, once it reaches the ending boundary of the 8/16/32/64-Byte section, the output will wrap around the beginning boundary automatically until CS# is pulled high to terminate the command.

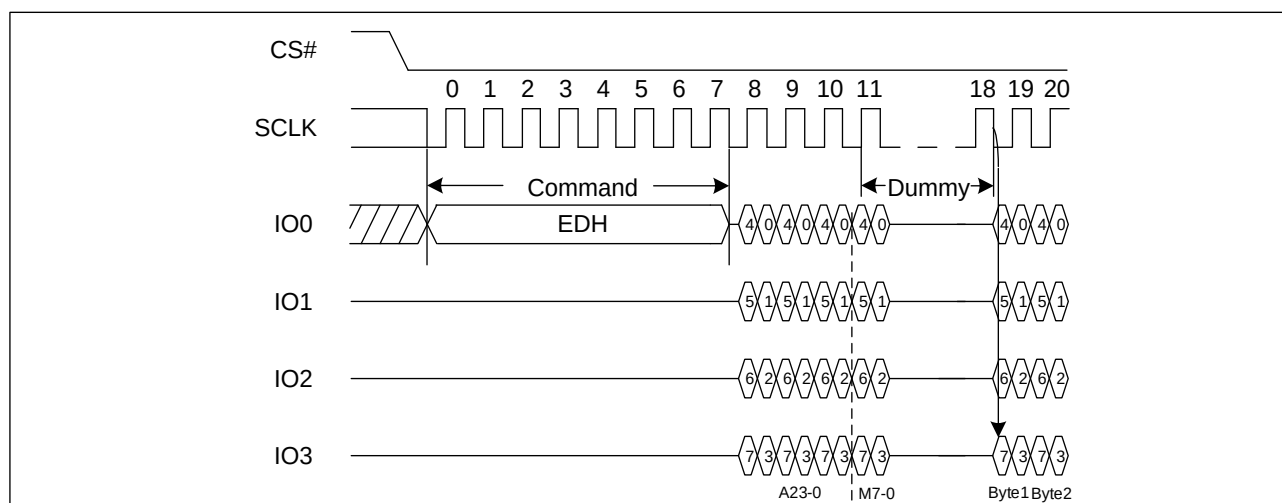
The Burst with Wrap feature allows applications that use cache to quickly fetch a critical address and then fill the cache afterwards within a fixed length (8/16/32/64-Byte) of data without issuing multiple read commands.

9.19 Quad I/O DTR Read (EDh/EEh)

The Quad I/O DTR Read instruction enables Double Transfer Rate throughput on quad I/O of Serial Flash in read mode. The address (interleave on 4 I/O pins) is latched on both rising and falling edge of SCLK, and data (interleave on 4 I/O pins) shift out on both rising and falling edge of SCLK. The 8-bit address can be latched-in at one clock, and 8-bit data can be read out at one clock, which means four bits at rising edge of clock, the other four bits at falling edge of clock. The first address Byte can be at any location. The address is automatically increased to the next higher address after each Byte data is shifted out, so the whole memory can be read out at a single Quad I/O DTR Read command. The address counter rolls over to 0 when the highest address has been reached.

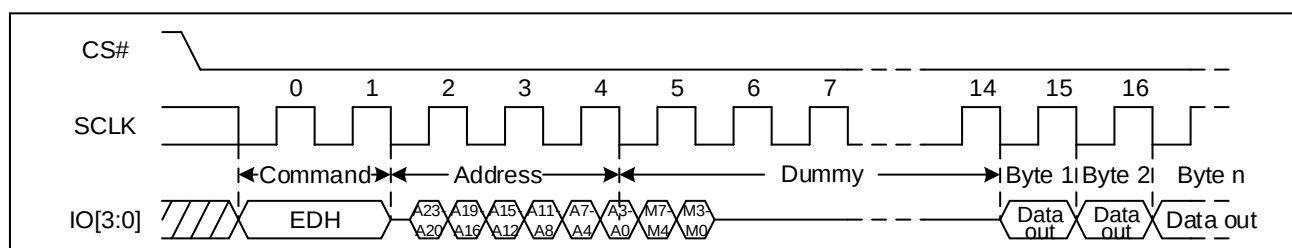
While Program/Erase/Write Status Register cycle is in progress, Quad I/O DTR Read command is rejected without any impact on the Program/Erase/Write Status Register current cycle.

Figure 51. DTR Quad I/O Fast Read Sequence Diagram (SPI, M5-4 ≠ (1, 0))



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 52. DTR Quad I/O Fast Read Sequence Diagram (QPI, M5-4 ≠ (1, 0))

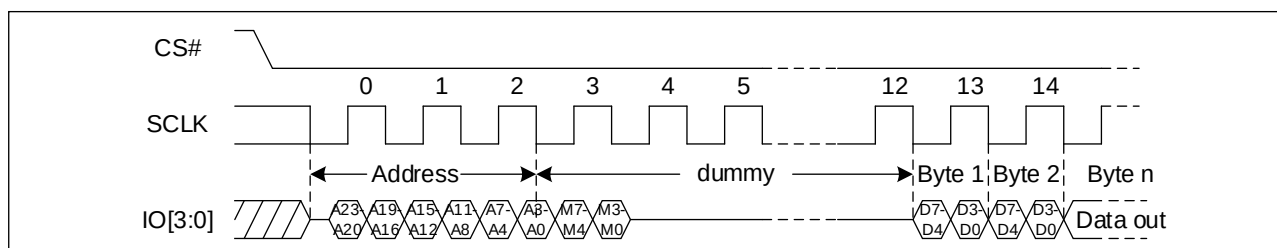


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Quad I/O DTR Read with “Continuous Read Mode”

The Quad I/O DTR Read command can further reduce command overhead through setting the “Continuous Read Mode” bits (M7-0) after the input address. If the “Continuous Read Mode” bits (M5-4) = (1, 0), then the next Quad I/O DTR Read command (after CS# is raised and then lowered) does not require the EDh/EEh command code. If the “Continuous Read Mode” bits (M5-4) do not equal to (1, 0), the next command requires the first EDh/EEh command code, thus returning to normal operation. The only way to quit the Quad I/O DTR Continuous Read Mode” is to set the “Continuous Read Mode” bits (M5-4) not equal to (1, 0).

Figure 53. DTR Quad I/O Fast Read Sequence Diagram (M5-4 = (1, 0))



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Quad I/O DTR Fast Read with “8/16/32/64-Byte Wrap Around”

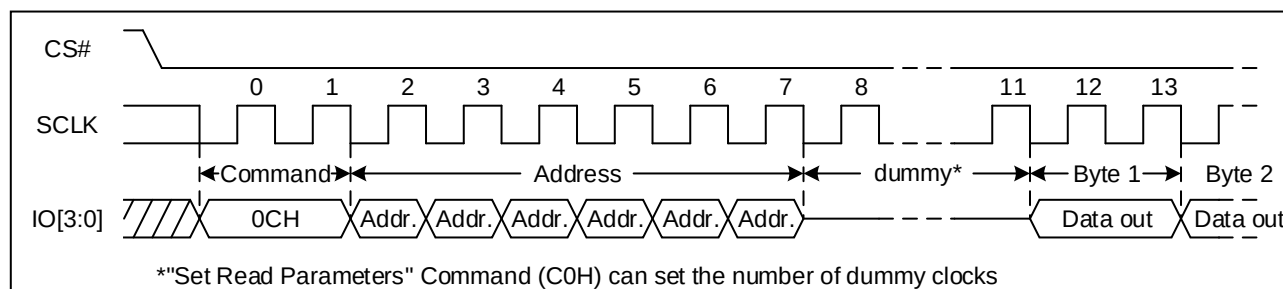
The Quad I/O DTR Fast Read command can be used to access a specific portion within a page by issuing Wrap configuration register Byte prior to EDh/EEh. The data being accessed can be limited to either a 8/16/32/64-Byte section of a 256-Byte page. The output data starts at the initial address specified in the command, once it reaches the ending boundary of the 8/16/32/64-Byte section, the output will wrap around the beginning boundary automatically until CS# is pulled high to terminate the command.

The Burst with Wrap feature allows applications that use cache to quickly fetch a critical address and then fill the cache afterwards within a fixed length (8/16/32/64-Byte) of data without issuing multiple read commands.

9.20 Burst Read with Wrap (0Ch)

The “Burst Read with Wrap (0Ch)” command provides an alternative way to perform the read operation with “Wrap Around” in QPI mode. This command is similar to the “Fast Read (0Bh)” command in QPI mode, except the addressing of the read operation will “Wrap Around” to the beginning boundary of the “Wrap Around” once the ending boundary is reached. The “Wrap Length” and the number of dummy clocks can be configured by the “Set Read Parameters (C0h)” command.

Figure 54. Burst Read with Wrap Sequence Diagram

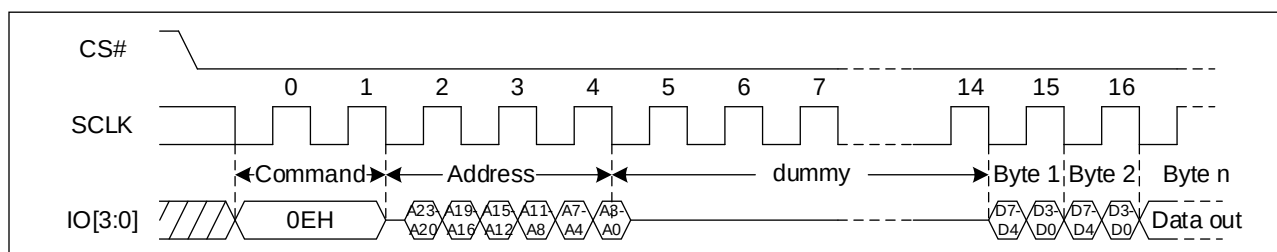


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.21 DTR Burst Read with Wrap (0Eh)

The “DTR Burst Read with Wrap (0Eh)” instruction provides an alternative way to perform the read operation with “Wrap Around” in QPI mode. The instruction is similar to the “Fast Read (0Bh)” instruction in QPI mode, except the addressing of the read operation will “Wrap Around” to the beginning boundary of the “Wrap Around” once the ending boundary is reached. The “Wrap Length” can be configured by the “Set Read Parameters (C0h)” command.

Figure 55. DTR Burst Read with Wrap Sequence Diagram



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.22 Set Burst with Wrap (77h)

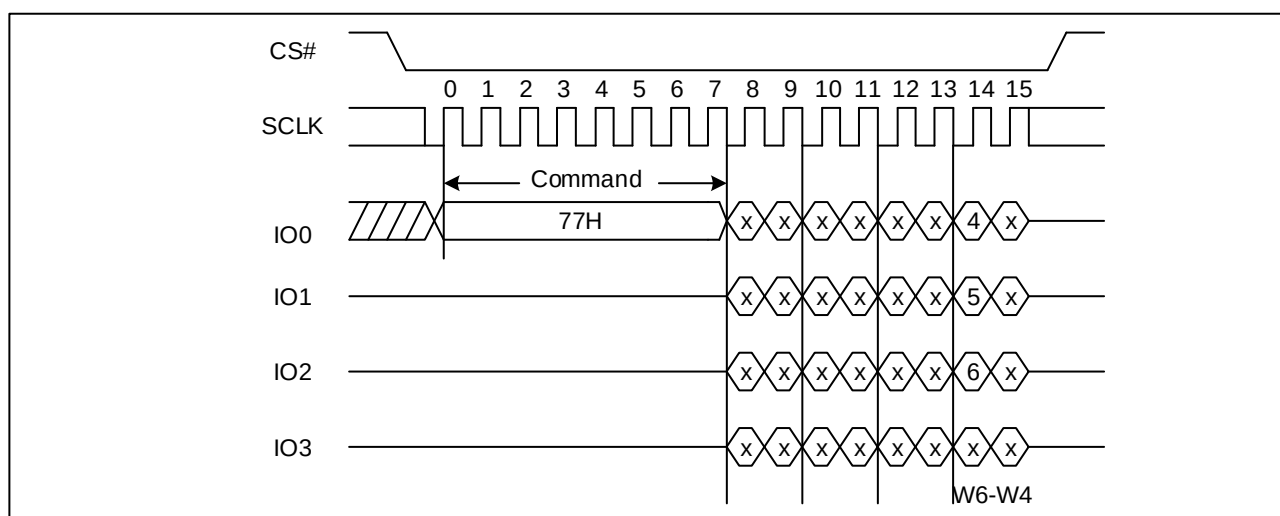
The Set Burst with Wrap command is used in conjunction with “Quad I/O Fast Read” command to access a fixed length of 8/16/32/64-byte section within a 256-byte page, in standard SPI mode.

The Set Burst with Wrap command sequence: CS# goes low → Send Set Burst with Wrap command → Send 24 dummy bits → Send 8 bits “Wrap bits” → CS# goes high.

W6,W5	W4=0		W4=1 (default)	
	Wrap Around	Wrap Length	Wrap Around	Wrap Length
0, 0	Yes	8-byte	No	N/A
0, 1	Yes	16-byte	No	N/A
1, 0	Yes	32-byte	No	N/A
1, 1	Yes	64-byte	No	N/A

If the W6-W4 bits are set by the Set Burst with Wrap command, all the following “Quad I/O Fast Read” command will use the W6-W4 setting to access the 8/16/32/64-byte section within any page. To exit the “Wrap Around” function and return to normal read operation, another Set Burst with Wrap command should be issued to set W4=1.

Figure 56. Set Burst with Wrap Sequence Diagram



9.23 Set Read Parameters (C0h)

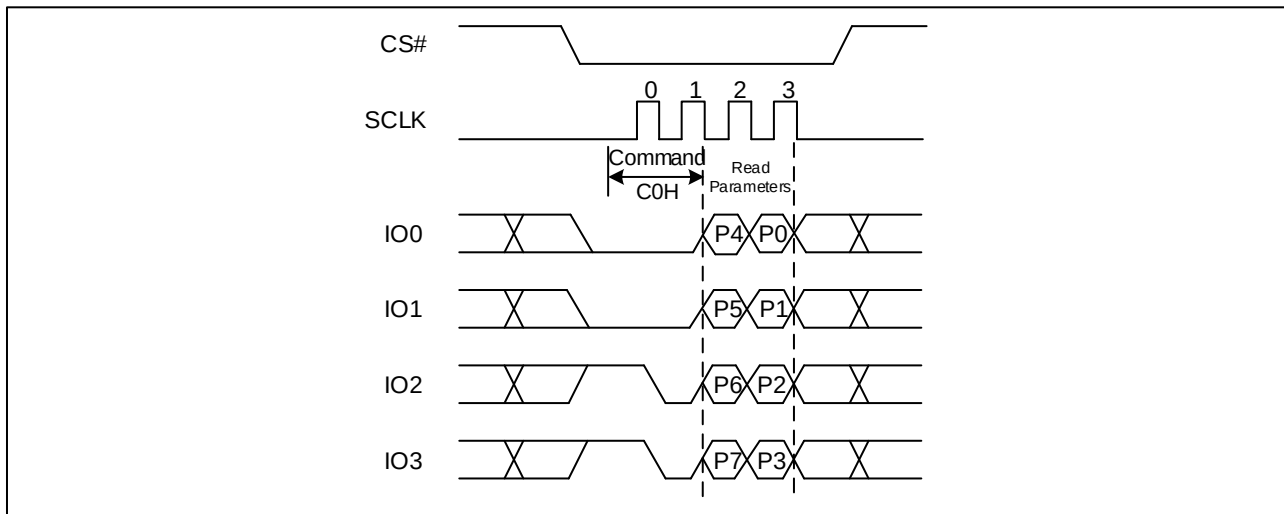
In QPI mode the “Set Read Parameters (C0h)” command can be used to configure the number of dummy clocks for “Fast Read (0Bh)”, “Quad I/O Fast Read (EBh)”, “DTR Quad I/O Fast Read(EDh)”, “Burst Read with Wrap (0Ch)” and “DTR Burst Read with Wrap (0Eh)” command, and to configure the number of bytes of “Wrap Length” for the “Burst Read with Wrap (0Ch)” and “DTR Burst Read with Wrap (0Eh)” command. The “Wrap Length” is set by W5-6 bit in the “Set Burst with Wrap (77H)” command. This wrap setting will remain unchanged when the device is switched from Standard SPI mode to QPI mode.



P5-P4	STR FAST READ		DTR FAST READ		P1-P0	Wrap Length
	Dummy Clocks	Maximum Read Freq.	Dummy Clocks	Maximum Read Freq.		
0 0(Default)	4	80MHz	10	104MHz	0 0(Default)	8-Byte
0 1	6	104MHz	8	80MHz	0 1	16-Byte
1 0	8	133MHz	10	104MHz	1 0	32-Byte
1 1	10	166MHz	10	104MHz	1 1	64-Byte

Note: Default from power up or reset.

Figure 57. Set Read Parameters command Sequence Diagram



9.24 Page Program (PP) (02h/12h)

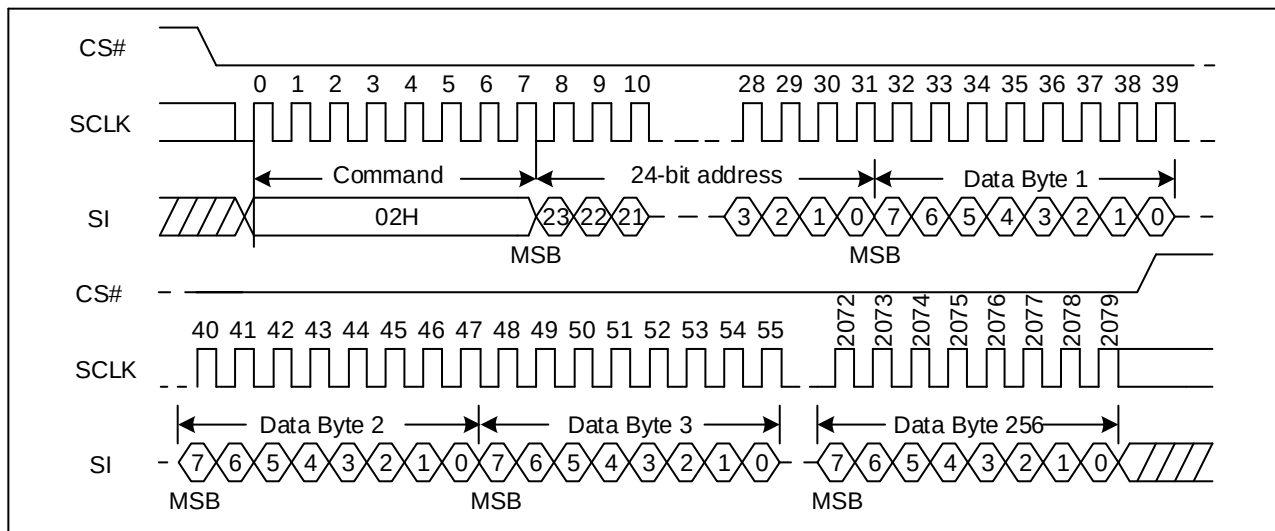
The Page Program (PP) command is for programming the memory. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit before sending the Page Program command.

The Page Program (PP) command is entered by driving CS# Low, followed by the command code, three or four address Bytes and at least one data Byte on SI. If the 8 least significant address bits (A7-A0) are not all zero, all transmitted data that goes beyond the end of the current page are programmed from the start address of the same page (from the address whose 8 least significant bits (A7-A0) are all zero). CS# must be driven low for the entire duration of the sequence. The Page Program command sequence: CS# goes low → sending Page Program command → 3-Byte address or 4-Byte address on SI → at least 1 Byte data on SI → CS# goes high. If more than 256 Bytes are sent to the device, previously latched data are discarded and the last 256 data Bytes are guaranteed to be programmed correctly within the same page. If less than 256 data Bytes are sent to device, they are correctly programmed at the requested addresses without having any effects on the other Bytes of the same page. CS# must be driven high after the eighth bit of the last data Byte has been latched in; otherwise the Page Program (PP) command is not executed.

As soon as CS# is driven high, the self-timed Page Program cycle (whose duration is t_{PP}) is initiated. While the Page Program cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Page Program cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset.

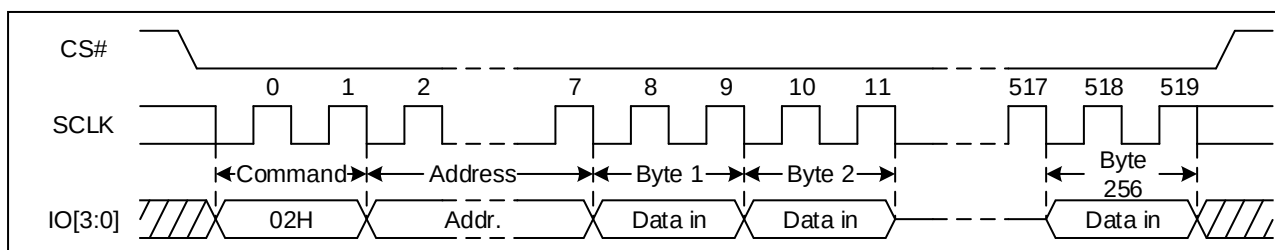
A Page Program (PP) command applied to a page which is protected by the Block Protect (BP4, BP3, BP2, BP1, and BP0) is not executed.

Figure 58 Page Program Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 59 Page Program Sequence Diagram (QPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.25 Quad Page Program (32h/34h)

The Quad Page Program command is for programming the memory using four pins: IO0, IO1, IO2, and IO3. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit before sending the Page Program command. The quad Page Program command is entered by driving CS# Low, followed by the command code (32h/34h), three or four address Bytes and at least one data Byte on IO pins.

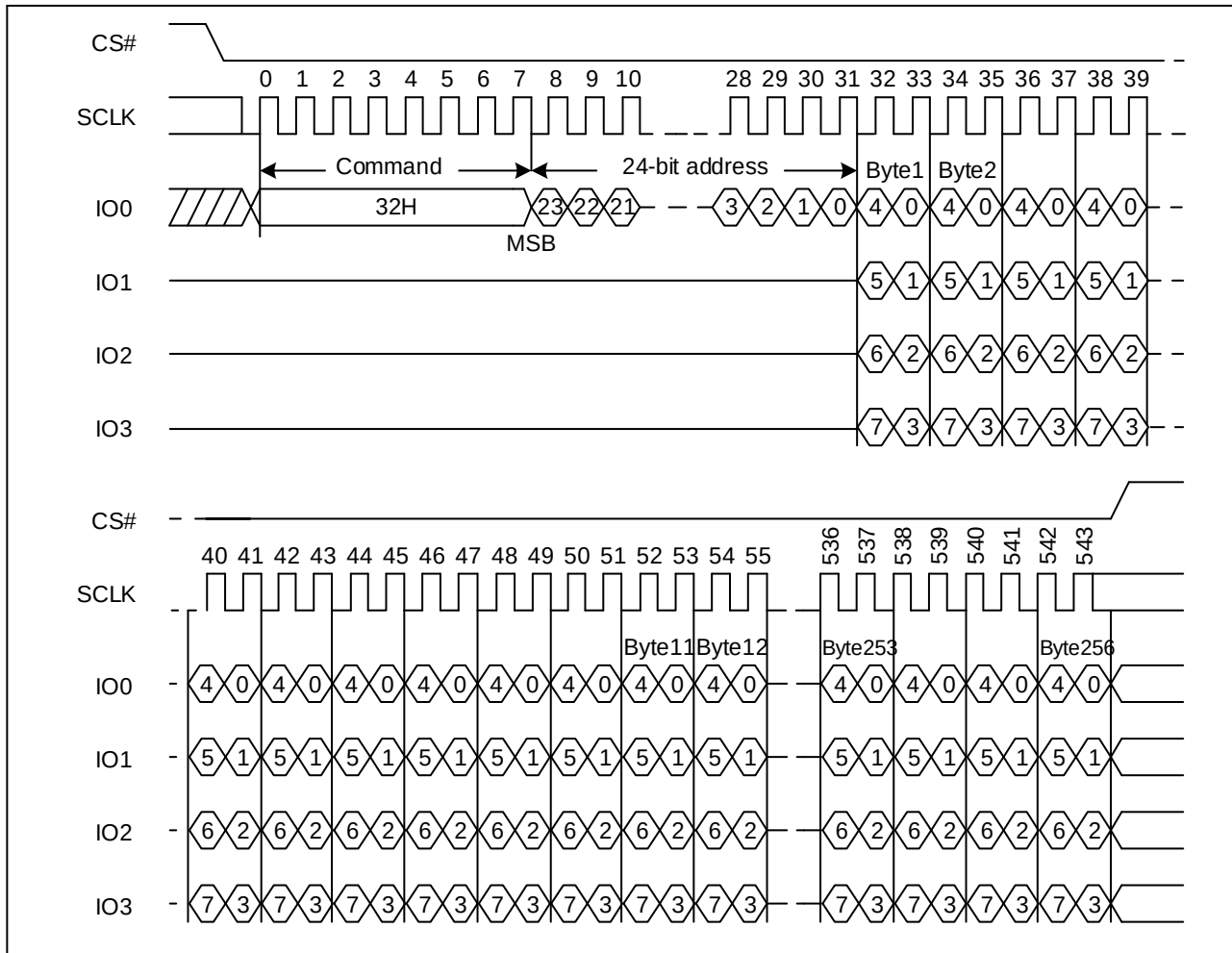
If more than 256 Bytes are sent to the device, previously latched data are discarded and the last 256 data Bytes are guaranteed to be programmed correctly within the same page. If less than 256 data Bytes are sent to device, they are correctly programmed at the requested addresses without having any effects on the other Bytes of the same page. CS# must be driven high after the eighth bit of the last data Byte has been latched in; otherwise the Quad Page Program (PP) command is not executed.

As soon as CS# is driven high, the self-timed Quad Page Program cycle (whose duration is t_{PP}) is initiated. While the Quad Page Program cycle is in progress, the Status Register may be read to check the value of the Write In Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Quad Page Program cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset.

A Quad Page Program command applied to a page which is protected by the Block Protect (BP4, BP3, BP2, BP1, and BP0) is not executed.



Figure 60 Quad Page Program Sequence Diagram (SPI)

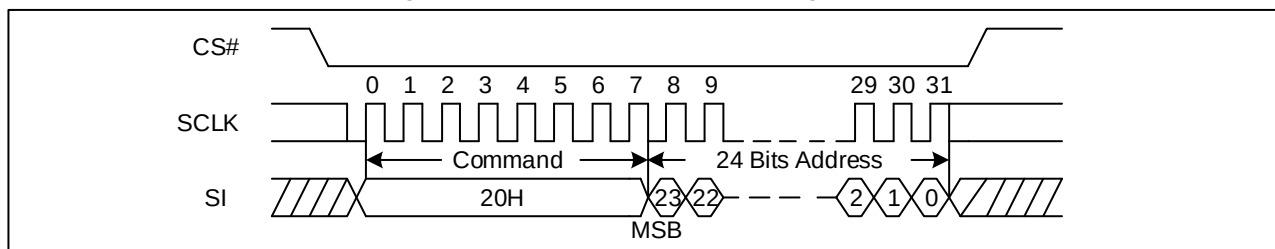


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.26 Sector Erase (SE) (20h/21h)

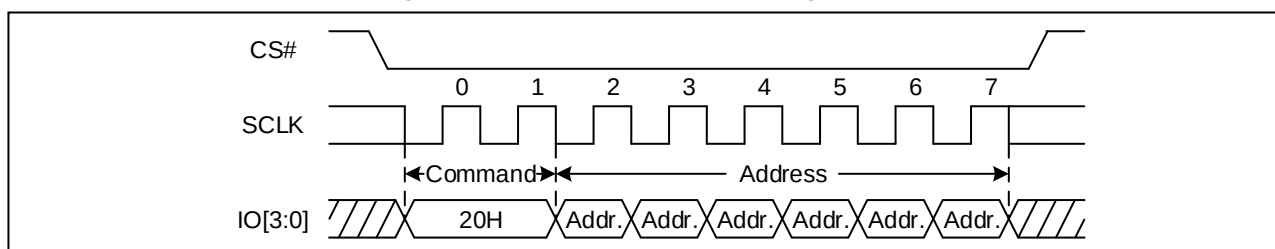
The Sector Erase (SE) command is used to erase all data of the chosen sector. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit. The Sector Erase (SE) command is entered by driving CS# low, followed by the command code, and 3-Byte address or 4-Byte address on SI. Any address inside the sector is a valid address for the Sector Erase (SE) command. CS# must be driven low for the entire duration of the sequence. The Sector Erase command sequence: CS# goes low → sending Sector Erase command → 3-Byte address or 4-Byte address on SI → CS# goes high. CS# must be driven high after the eighth bit of the last address Byte has been latched in; otherwise the Sector Erase (SE) command is not executed. As soon as CS# is driven high, the self-timed Sector Erase cycle (whose duration is t_{SE}) is initiated. While the Sector Erase cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Sector Erase cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset. A Sector Erase (SE) command applied to a sector which is protected by the Block Protect (BP4, BP3, BP2, BP1, and BP0) bits is not executed.

Figure 61 Sector Erase Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 62 Sector Erase Sequence Diagram (QPI)

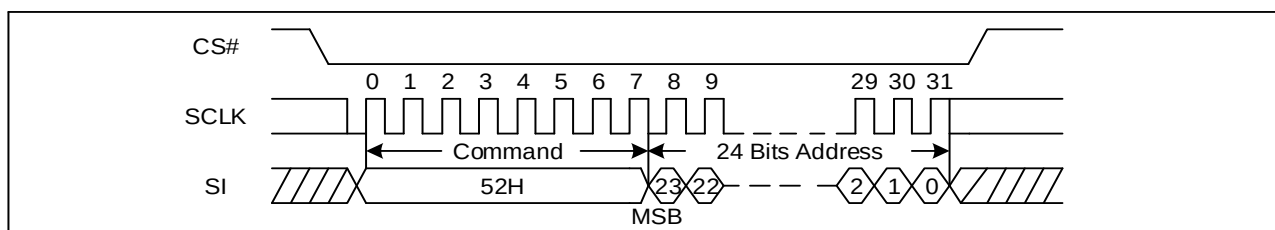


Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.27 32KB Block Erase (BE32) (52h/5Ch)

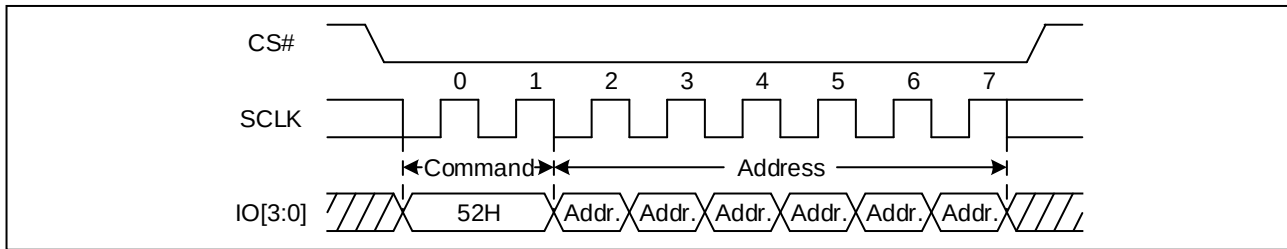
The 32KB Block Erase command is erased the all data of the chosen block. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit. The 32KB Block Erase (BE) command is entered by driving CS# low, followed by the command code, and 3-Byte address or 4-Byte address on SI. Any address inside the block is a valid address for the 32KB Block Erase command. CS# must be driven low for the entire duration of the sequence. The 32KB Block Erase command sequence: CS# goes low → sending 32KB Block Erase command → 3-Byte address or 4-Byte address on SI → CS# goes high. CS# must be driven high after the eighth bit of the last address Byte has been latched in; otherwise the 32KB Block Erase command is not executed. As soon as CS# is driven high, the self-timed Block Erase cycle (whose duration is t_{BE1}) is initiated. While the Block Erase cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Block Erase cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset. A 32KB Block Erase command applied to a block which is protected by the Block Protect (BP4, BP3, BP2, BP1, and BP0) bits is not executed.

Figure 63 32KB Block Erase Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 64 32KB Block Erase Sequence Diagram (QPI)



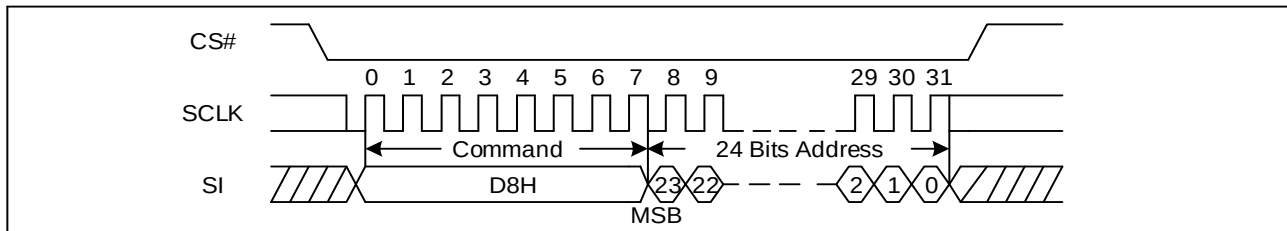
Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.28 64KB Block Erase (BE64) (D8h/DCh)

The 64KB Block Erase command is erased the all data of the chosen block. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit. The 64KB Block Erase command is entered by driving CS# low, followed by the command code, and 3-Byte address or 4-Byte address on SI. Any address inside the block is a valid address for the 64KB Block Erase command. CS# must be driven low for the entire duration of the sequence.

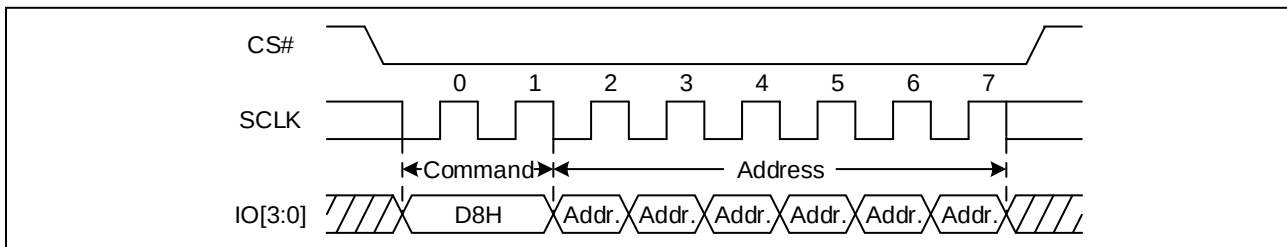
The 64KB Block Erase command sequence: CS# goes low → sending 64KB Block Erase command → 3-Byte address or 4-Byte address on SI → CS# goes high. CS# must be driven high after the eighth bit of the last address Byte has been latched in; otherwise the 64KB Block Erase command is not executed. As soon as CS# is driven high, the self-timed Block Erase cycle (whose duration is t_{BE2}) is initiated. While the Block Erase cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Block Erase cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset. A 64KB Block Erase command applied to a block which is protected by the Block Protect (BP4, BP3, BP2, BP1, and BP0) bits is not executed.

Figure 65 64KB Block Erase Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

Figure 66 64KB Block Erase Sequence Diagram (QPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.29 Enable 4-Byte Mode (B7h)

The Enable 4-Byte Mode command enables accessing the address length of 32-bit for the memory area of the higher density (larger than 128Mb). After sending the Enable 4-Byte Mode command, the ADS bit (S8) will be set to 1 to indicate

the 4-Byte address mode has been enabled. Once the 4-Byte address mode is enabled, the address length becomes 32-bit.

Figure 67 Enable 4-Byte Mode Sequence Diagram (SPI)

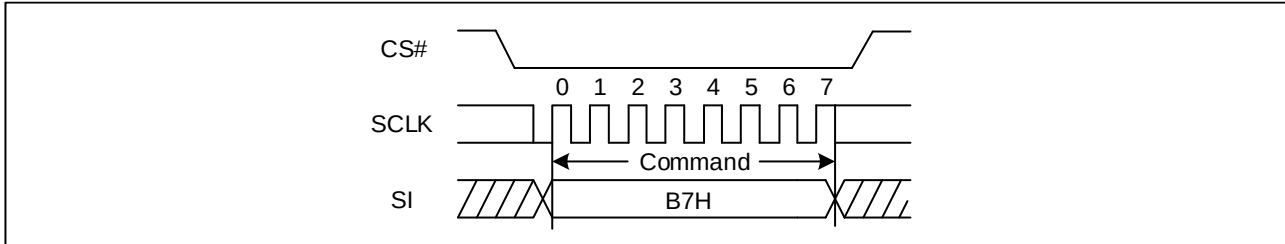
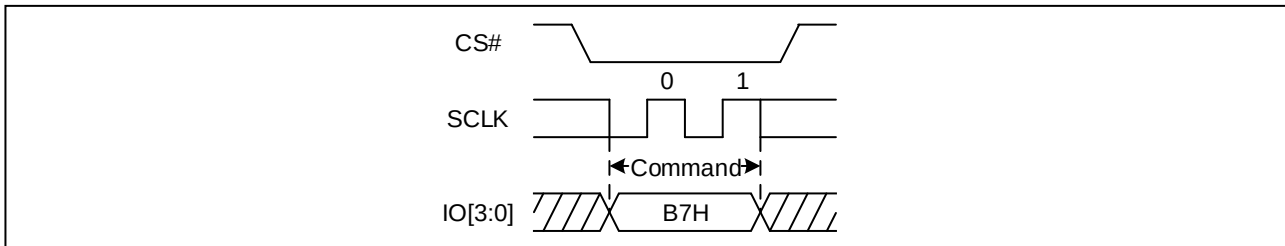


Figure 68 Enable 4-Byte Mode Sequence Diagram (QPI)



9.30 Disable 4-Byte Mode (E9h)

The Disable 4-Byte Mode command is executed to exit the 4-Byte address mode and enter the 3-Byte address mode. After sending the Disable 4-Byte Mode command, the ADS bit (S8) will be clear to be 0 to indicate the 4-Byte address mode has been disabled, and then the address length will return to 24-bit.

Figure 69 Disable 4-Byte Mode Sequence Diagram (SPI)

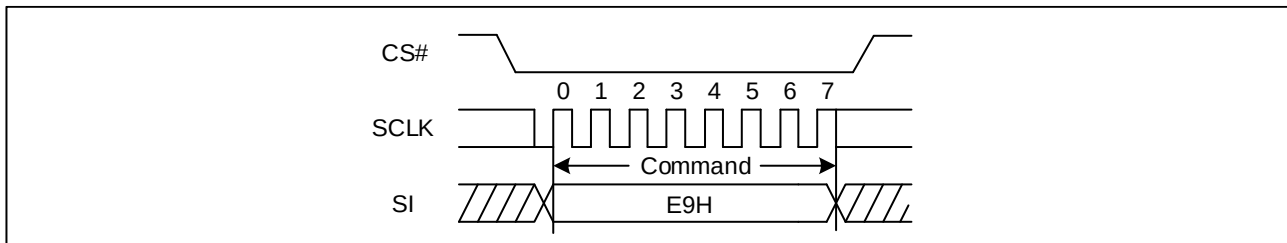
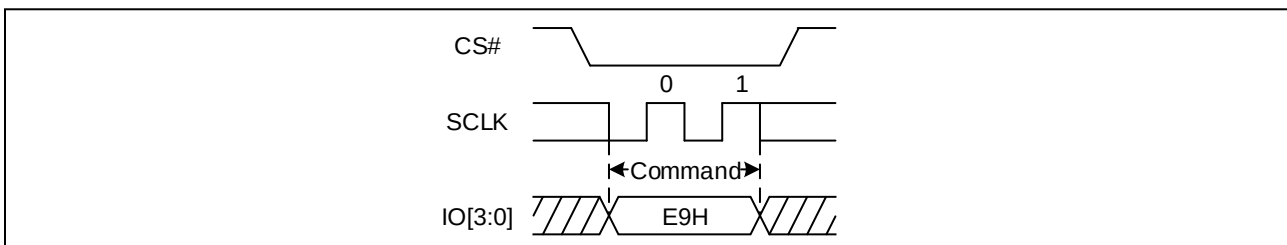


Figure 70 Disable 4-Byte Mode Sequence Diagram (QPI)



9.31 Chip Erase (CE) (60h/C7h)

The Chip Erase (CE) command is used to erase all data of the chip. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit. The Chip Erase (CE) command is entered by driving CS# Low, followed by the command code on Serial Data Input (SI). CS# must be driven Low for the entire duration of the sequence.

The Chip Erase command sequence: CS# goes low → sending Chip Erase command → CS# goes high. CS# must be driven high after the eighth bit of the command code has been latched in; otherwise the Chip Erase command is not executed. As soon as CS# is driven high, the self-timed Chip Erase cycle (whose duration is t_{CE}) is initiated. While the Chip Erase cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Chip Erase cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset. The Chip Erase (CE) command is executed if the Block Protect (BP2, BP1, and BP0) bits are 0. The Chip Erase (CE) command is ignored if one or more sectors are protected.

Figure 71 Chip Erase Sequence Diagram (SPI)

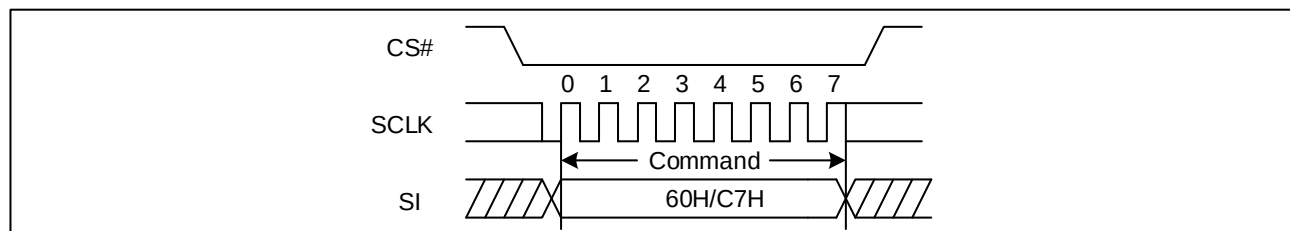
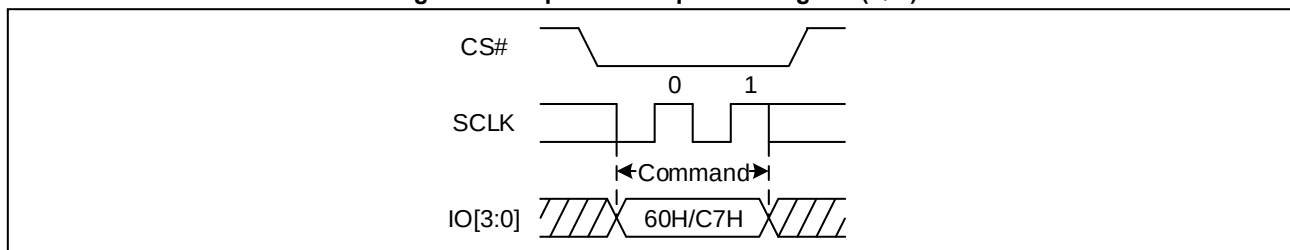


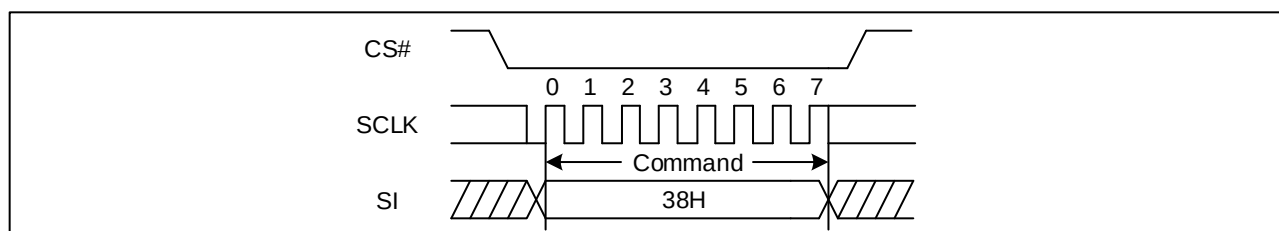
Figure 72 Chip Erase Sequence Diagram (QPI)



9.32 Enable QPI (38h)

The device supports both Standard/Quad SPI and QPI mode. The “Enable QPI (38h)” command can switch the device from SPI mode to QPI mode. In order to switch the device to QPI mode, “Enable QPI (38h)” command must be issued. When the device is switched from SPI mode to QPI mode, the existing Write Enable Latch and Program/Erase Suspend status, and the Wrap Length setting will remain unchanged.

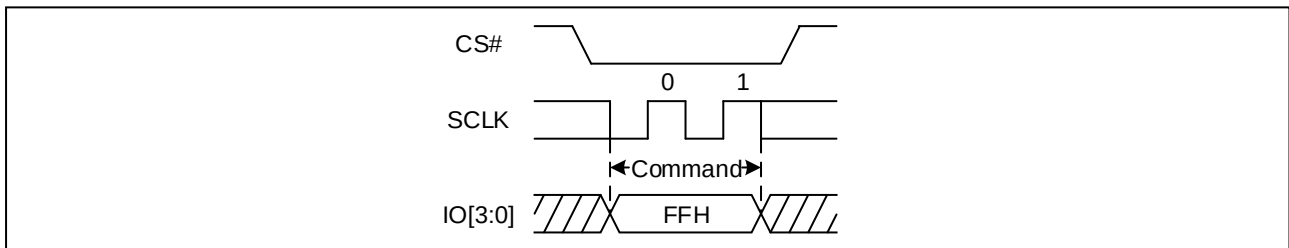
Figure 73 Enable QPI mode command Sequence Diagram



9.33 Disable QPI (FFh)

To exit the QPI mode and return to Standard/Quad SPI mode, the “Disable QPI (FFh)” command must be issued. When the device is switched from QPI mode to SPI mode, the existing Write Enable Latch and Program/Erase Suspend status, and the Wrap Length setting will remain unchanged.

Figure 74 Disable QPI mode command Sequence Diagram



9.34 Deep Power-Down (DP) (B9h)

Executing the Deep Power-Down (DP) command is the only way to put the device in the lowest consumption mode (the Deep Power-Down Mode). It can also be used as an extra software protection mechanism, while the device is not in active use, since in this mode, the device ignores all Write, Program and Erase commands. Driving CS# high deselects the device, and puts the device in the Standby Mode (if there is no internal cycle currently in progress). But this mode is not the Deep Power-Down Mode. The Deep Power-Down Mode can only be entered by executing the Deep Power-Down (DP) command. Once the device has entered the Deep Power-Down Mode, all commands are ignored except the Release from Deep Power-Down (ABh) or Enable Reset (66h) and Reset (99h) commands. These commands can release the device from this mode. The Release from Deep Power-Down command releases the device from deep power down mode.

The Deep Power-Down Mode automatically stops at Power-Down, and the device is in the Standby Mode after Power-Up. The Deep Power-Down command sequence: CS# goes low → sending Deep Power-Down command → CS# goes high. CS# must be driven high after the eighth bit of the command code has been latched in; otherwise the Deep Power-Down (DP) command is not executed. As soon as CS# is driven high, it requires a delay of t_{DP} before the supply current is reduced to I_{CC2} and the Deep Power-Down Mode is entered. Any Deep Power-Down (DP) command, while an Erase, Program or Write cycle is in progress, is rejected without having any effects on the cycle that is in progress.

Figure 75 Deep Power-Down Sequence Diagram (SPI)

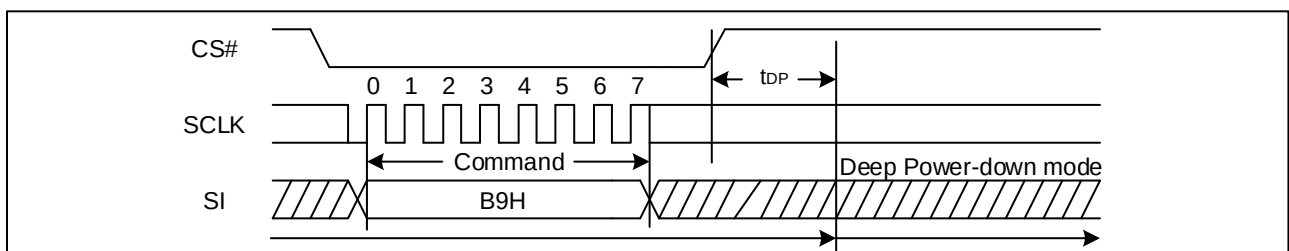
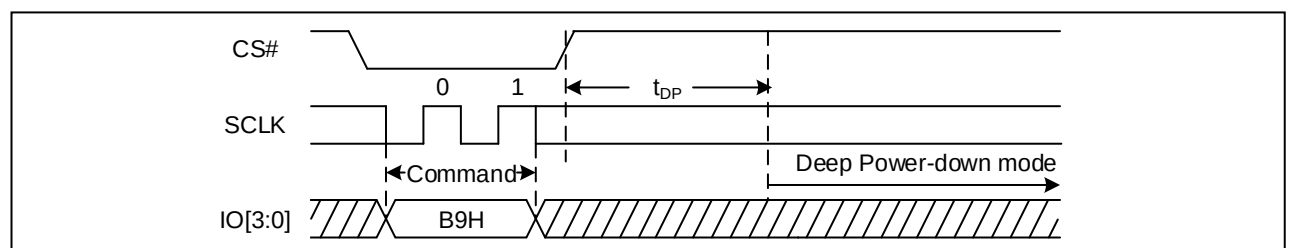


Figure 76 Deep Power-Down Sequence Diagram (QPI)



9.35 Release from Deep Power-Down (ABh)

To release the device from the Power-Down state, the command is issued by driving the CS# pin low, shifting the instruction code "ABh" and driving CS# high. Release from Power-Down will take the time duration of t_{RES1} (See AC



Characteristics) before the device will resume normal operation and other command are accepted. The CS# pin must remain high during the t_{RES1} time duration.

When used to release the device from the Power-Down state, the command is the same as previously described, After this time duration the device will resume normal operation and other command will be accepted. If the Release from Power-Down command is issued while an Erase, Program or Write cycle is in process (when WIP equal 1) the command is ignored and will not have any effects on the current cycle.

Figure 77 Release Power-Down Sequence Diagram (SPI)

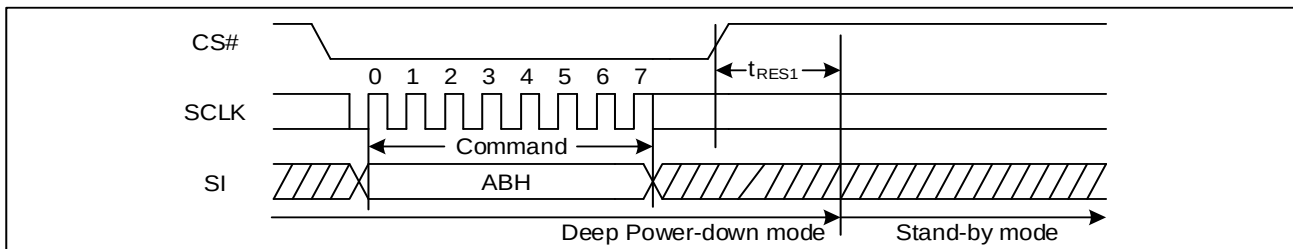


Figure 78 Release Power-Down Sequence Diagram (QPI)

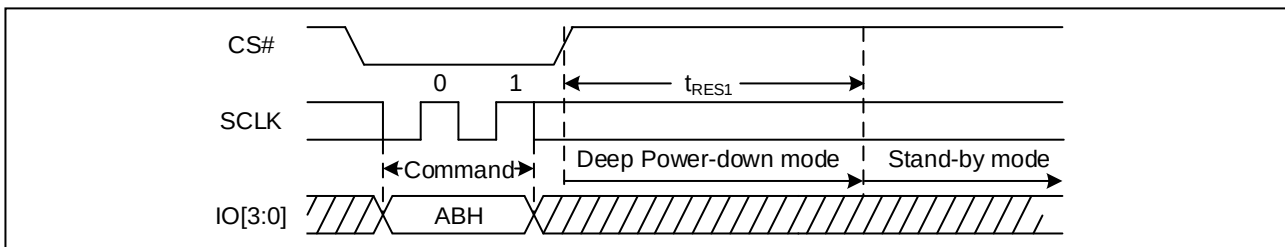


Figure 79. Release Power-Down/Read Device ID Sequence Diagram (SPI)

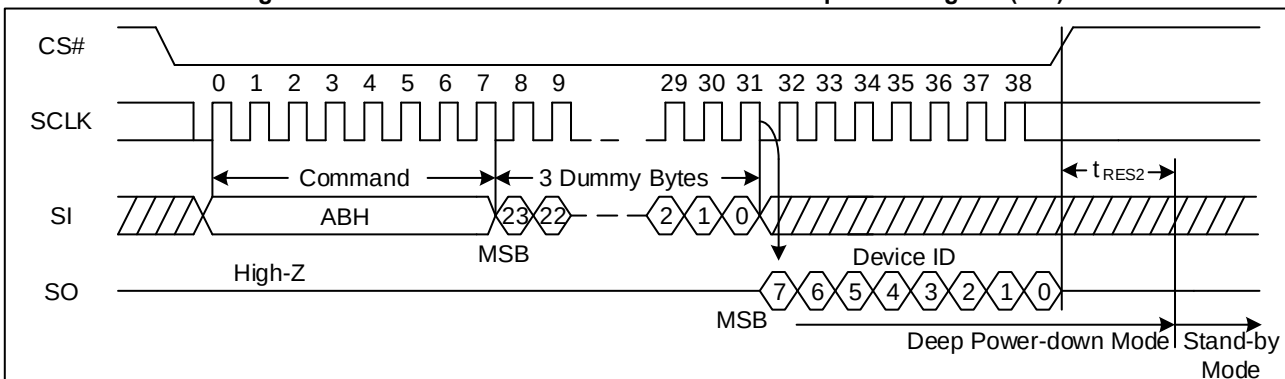
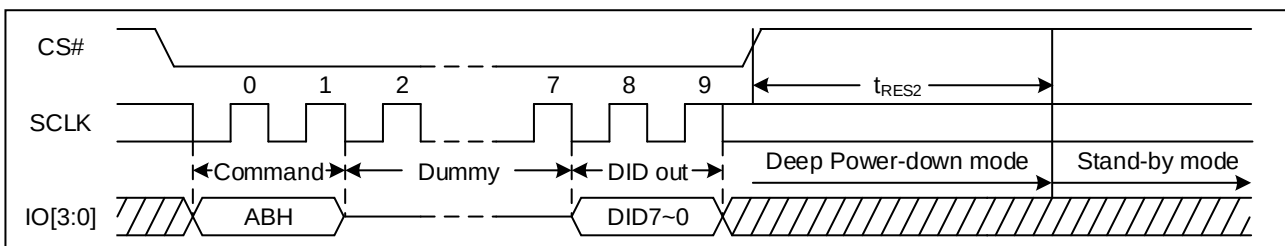


Figure 80. Release Power-Down/Read Device ID Sequence Diagram (QPI)



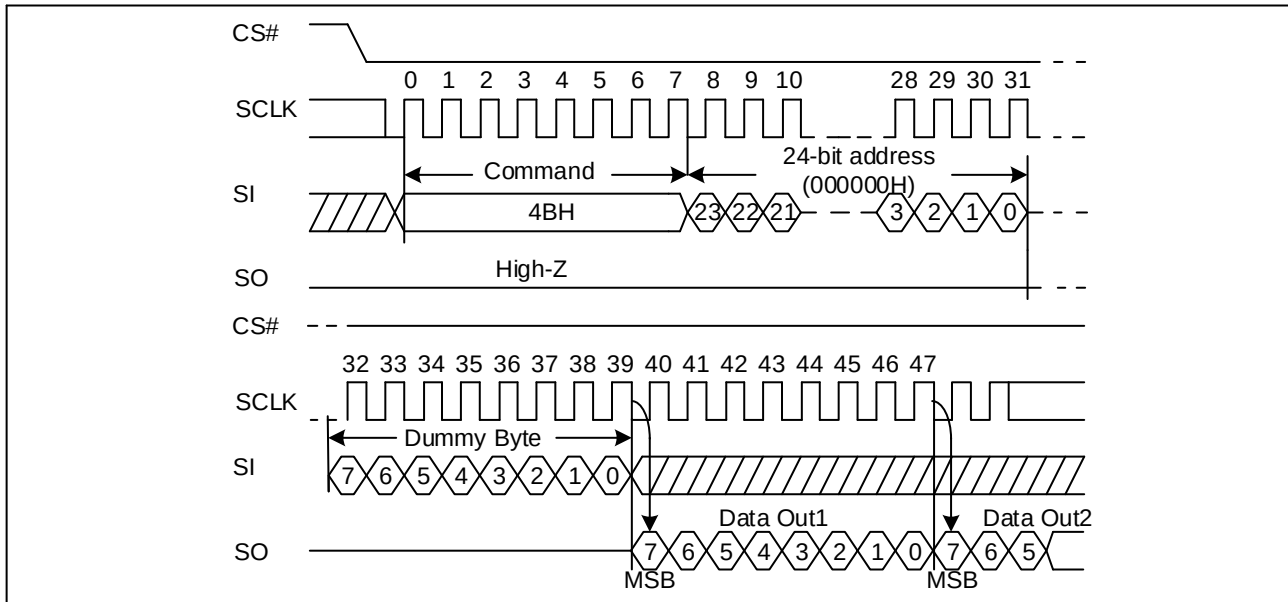
9.36 Read Unique ID (4Bh)

The Read Unique ID command accesses a factory-set read-only 128bit number that is unique to each device. The Unique ID can be used in conjunction with user software methods to help prevent copying or cloning of a system.



The Read Unique ID command sequence: CS# goes low → sending Read Unique ID command → 3-Byte (000000h) or 4-Byte (00000000h) Address → 1 Byte Dummy → 128bit Unique ID Out → CS# goes high.

Figure 81 Read Unique ID Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.37 Read Manufacture ID/ Device ID (REMS) (90h)

The Read Manufacturer/Device ID command is an alternative to the Release from Power-Down / Device ID command that provides both the JEDEC assigned Manufacturer ID and the specific Device ID.

The command is initiated by driving the CS# pin low and shifting the command code “90h” followed by a 24-bit address (A23-A0) of 000000h. After which, the Manufacturer ID and the Device ID are shifted out on the falling edge of SCLK with most significant bit (MSB) first.

Figure 82. Read Manufacture ID/ Device ID Sequence Diagram (SPI)

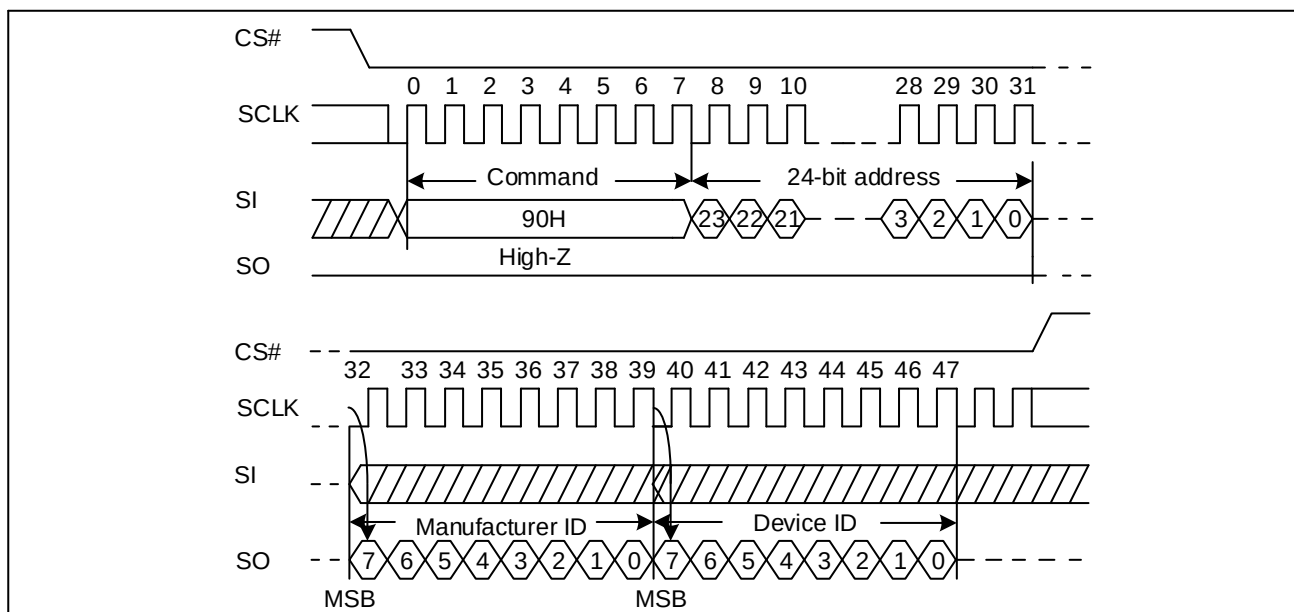
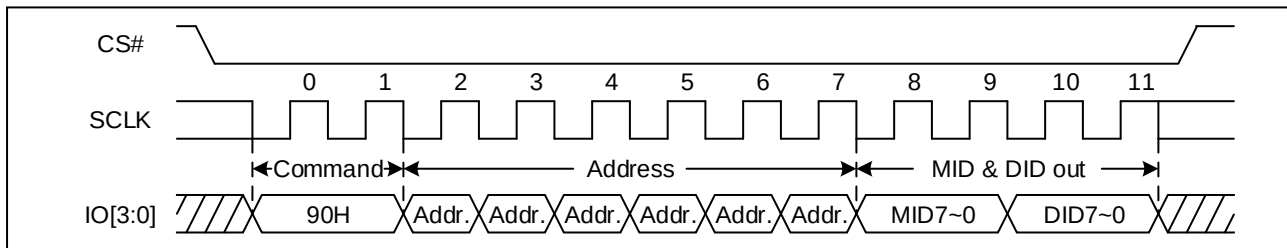


Figure 83. Read Manufacture ID/ Device ID Sequence Diagram (QPI)



9.38 Read Identification (RDID) (9Fh)

The Read Identification (RDID) command allows the 8-bit manufacturer identification to be read, followed by three Bytes of device identification. The device identification indicates the memory type in the first Byte, and the memory capacity of the device in the second Byte. The Read Identification (RDID) command while an Erase or Program cycle is in progress, is not decoded, and has no effect on the cycle that is in progress. The Read Identification (RDID) command should not be issued while the device is in Deep Power-Down Mode.

The device is first selected by driving CS# low. Then, the 8-bit command code for the command is shifted in. This is followed by the 24-bit device identification, stored in the memory. Each bit is shifted out on the falling edge of Serial Clock. The Read Identification (RDID) command is terminated by driving CS# high at any time during data output. When CS# is driven high, the device is in the Standby Mode. Once in the Standby Mode, the device waits to be selected, so that it can receive, decode and execute commands.

Figure 84 Read Identification ID Sequence Diagram (SPI)

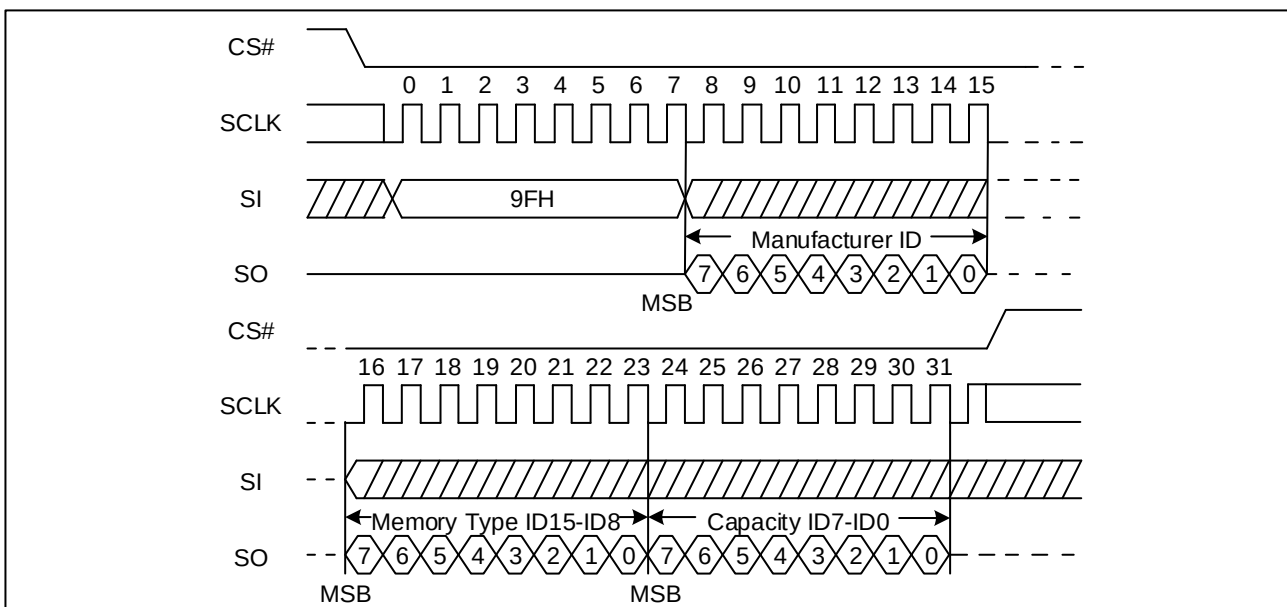
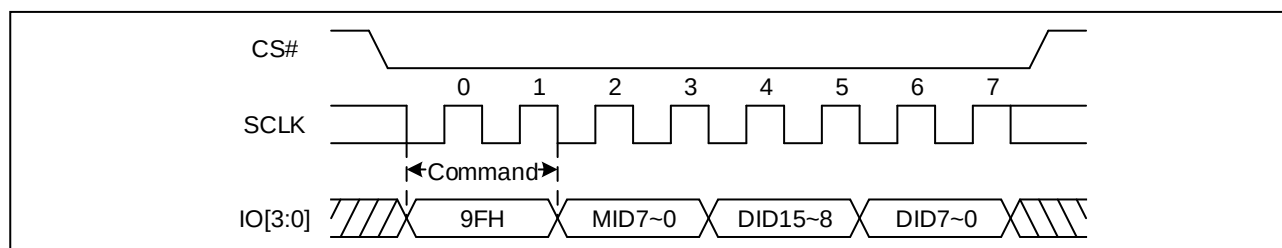


Figure 85 Read Identification ID Sequence Diagram (QPI)



9.39 Program/Erase/Data Integrity Check Suspend (PES) (75h)

The Program/Erase/Data Integrity Check Suspend command “75h”, allows the system to interrupt a page program or sector/block erase operation or Data Integrity Check and then read data from any other sector or block. The Write Status Register command (01h,11h,B1h,AAh,55h) and Erase/Program Security Registers command (44h, 42h) and Erase commands (20h/21h, 52h/5Ch, D8h/DCh, 60h/C7h) and Page Program command (02h/12h, 32h/34h) and Data Integrity Check (5Bh) and 50h, 27h, 28h, 29h, E1h, E3h, E4h, 7Eh, 98h are not allowed during Program/Data Integrity Check suspend. The Write Status Register command (01h,11h,B1h,AAh,55h) and Erase Security Registers command (44h) and Erase commands (20h/21h, 52h/5Ch, D8h/DCh, 60h/C7h) and Data Integrity Check (5Bh) and 50h, 27h, 28h, 29h, E1h, E3h, E4h, 7Eh, 98h are not allowed during Erase suspend. Program/Erase/Data Integrity Check Suspend is valid only during the page program or sector/block erase or Data Integrity Check operation. A maximum of time of “tsus” (See AC Characteristics) is required to suspend the program/erase/Data Integrity Check operation. The Program/Erase/Data Integrity Check Suspend command will be accepted by the device only if the SUS3/SUS2/SUS1 bit in the Status Register equal to 0 and WIP bit equal to 1 while a Page Program or a Sector or Block Erase or Data Integrity Check operation is on-going. If the SUS3/SUS2/SUS1 bit equal to 1 or WIP bit equal to 0, the Suspend command will be ignored by the device. The WIP bit will be cleared from 1 to 0 within “tsus” and the SUS3/SUS2/SUS1 bit will be set from 0 to 1 immediately after Program/Erase/Data Integrity Check Suspend. A power-off during the suspend period will reset the device and release the suspend state.

Figure 86 Program/Erase/Data Integrity Check Suspend Sequence Diagram (SPI)

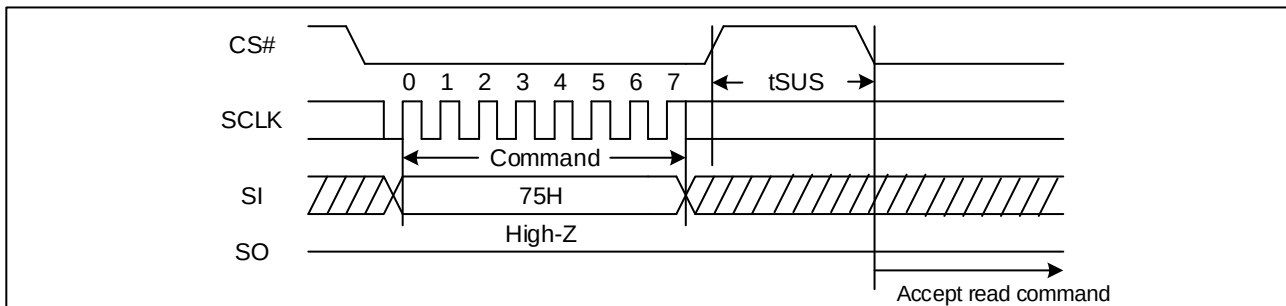
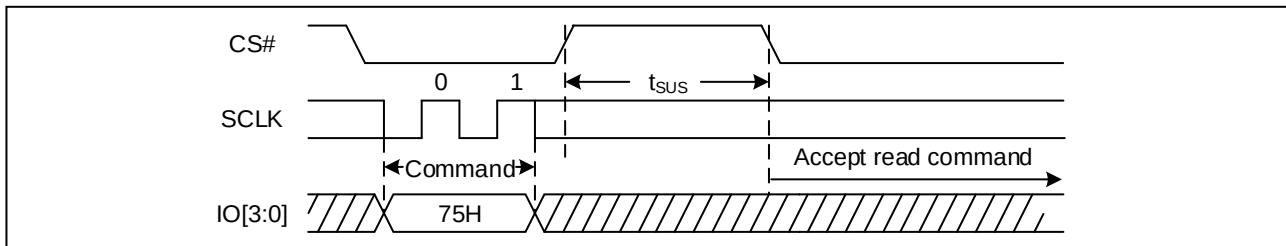


Figure 87 Program/Erase/Data Integrity Check Suspend Sequence Diagram (QPI)



9.40 Program/Erase/Data Integrity Check Resume (PER) (7Ah)

The Program/Erase/Data Integrity Check Resume command must be written to resume the program or sector/block erase or Data Integrity Check operation after a Program/Erase/Data Integrity Check Suspend command. The Program/Erase/Data Integrity Check Resume command will be accepted by the device only if the SUS3/SUS2/SUS1 bit equal to 1 and the WIP bit equal to 0. After issued the SUS3/SUS2/SUS1 bit in the status register will be cleared from 1 to 0 immediately, the WIP bit will be set from 0 to 1 within 200ns and the Sector or Block will complete the erase operation or the page will complete the program operation or the Data Integrity Check will complete the Data Integrity Check operation. The Program/Erase/Data Integrity Check Resume command will be ignored unless a Program/Erase/Data Integrity Check Suspend is active.

Figure 88 Program/Erase/Data Integrity Check Resume Sequence Diagram

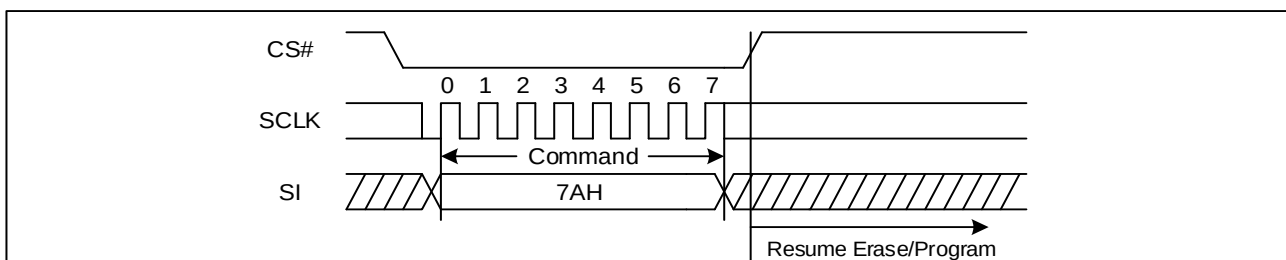
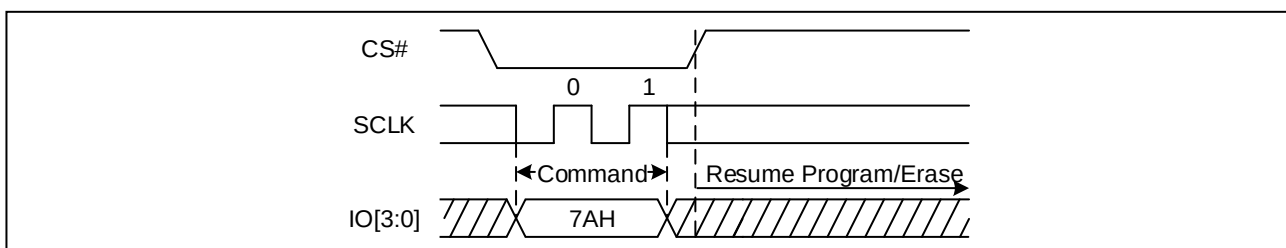


Figure 89 Program/Erase/Data Integrity Check Resume Sequence Diagram (QPI)



9.41 Erase Security Registers (44h)

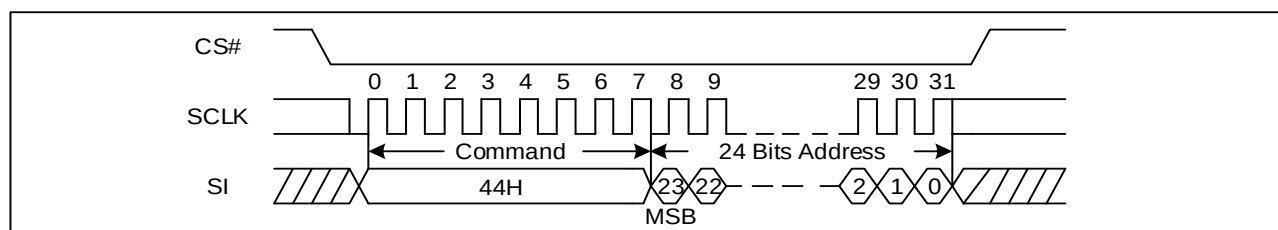
The GD25LF256F provides 4K-Byte Security Registers which can be erased and programmed individually. These registers may be used by the system manufacturers to store security and other important information separately from the main memory array.

The Erase Security Registers command is similar to Sector/Block Erase command. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit.

The Erase Security Registers command sequence: CS# goes low → sending Erase Security Registers command → CS# goes high. CS# must be driven high after the eighth bit of the last address Byte has been latched in; otherwise the Erase Security Registers command is not executed. As soon as CS# is driven high, the self-timed Erase Security Registers cycle (whose duration is t_{SE}) is initiated. While the Erase Security Registers cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Erase Security Registers cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset. The Security Registers Lock Bit in the Status Register can be used to OTP protect the security registers. Once the bit is set to 1, the Security Registers will be permanently locked; the Erase Security Registers command will be ignored.

Address	A23-16	A15-12	A11-0
Security Register #1	00H	0 0 0 1	Don't care
Security Register #2	00H	0 0 1 0	Don't care
Security Register #3	00H	0 0 1 1	Don't care

Figure 90 Erase Security Registers command Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.42 Program Security Registers (42h)

The Program Security Registers command is similar to the Page Program command. The security register contains 16 pages content. A Write Enable (WREN) command must previously have been executed to set the Write Enable Latch (WEL) bit before sending the Program Security Registers command. The Program Security Registers command is entered by driving CS# Low, followed by the command code (42h), three address Bytes and at least one data Byte on SI. As soon as CS# is driven high, the self-timed Program Security Registers cycle (whose duration is t_{PP}) is initiated. While the Program Security Registers cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Program Security Registers cycle, and is 0 when it is completed. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset.

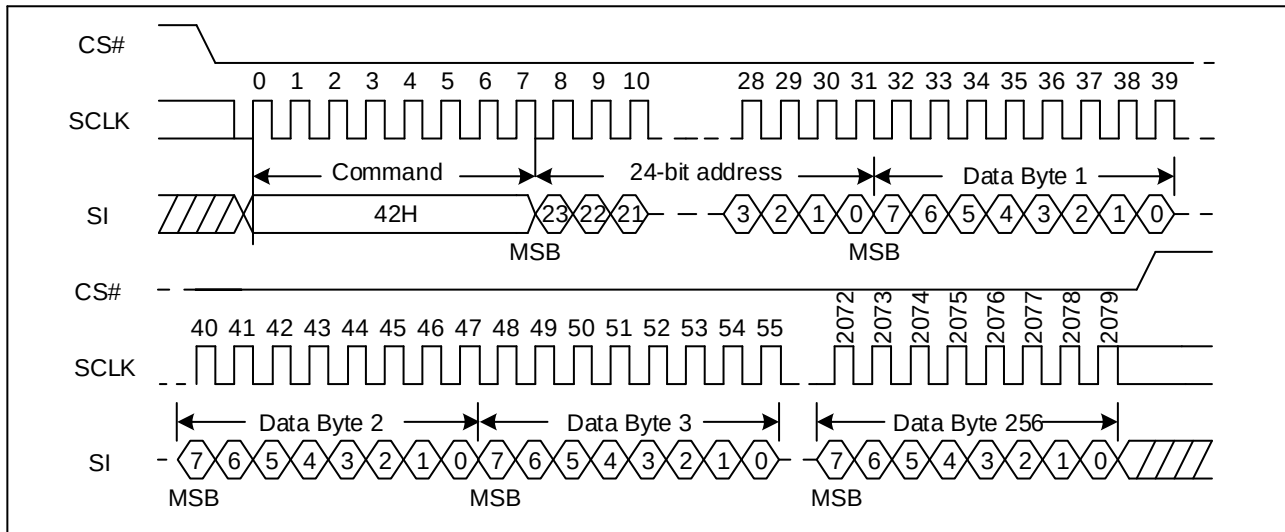
If the Security Registers Lock Bit is set to 1, the Security Registers will be permanently locked. Program Security Registers command will be ignored.

Address	A23-16	A15-12	A11-8	A7-0
Security Register #1	00H	0 0 0 1	Page Address	Byte Address
Security Register #2	00H	0 0 1 0	Page Address	Byte Address



Security Register #3	00H	0 0 1 1	Page Address	Byte Address
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Figure 91 Program Security Registers command Sequence Diagram (SPI)



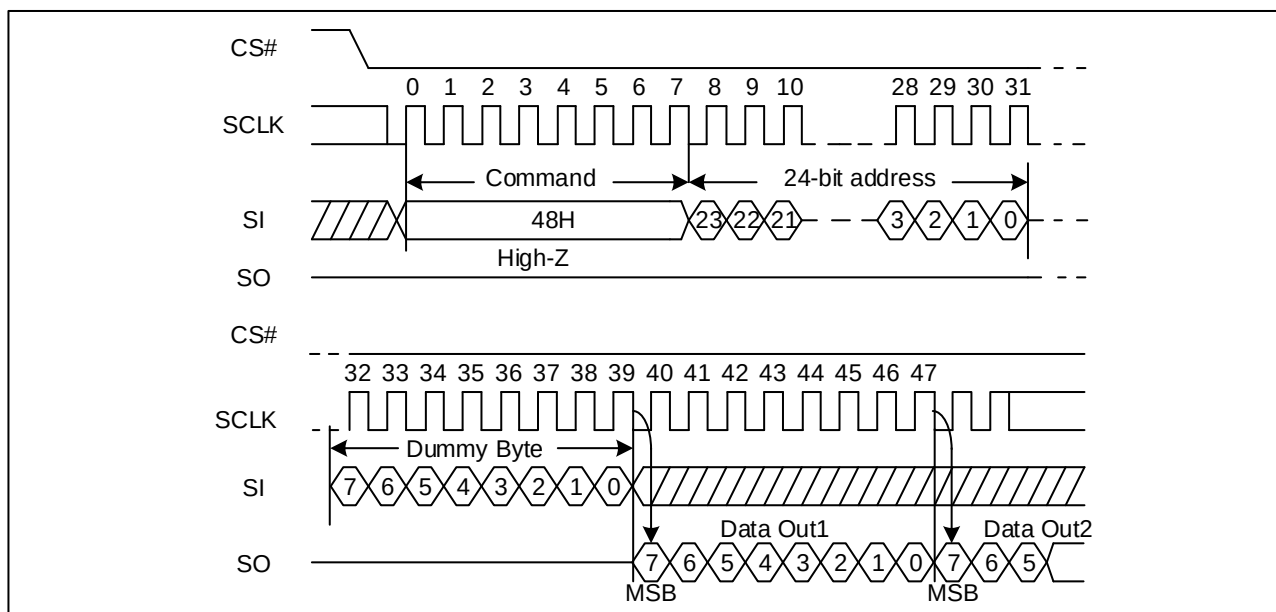
Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.43 Read Security Registers (48h)

The Read Security Registers command is similar to Fast Read command. The command is followed by a 3-Byte or 4-Byte address (A23-A0 or A31-A0) and a dummy Byte, and each bit is latched-in on the rising edge of SCLK. Then the memory content, at that address, is shifted out on SO, and each bit is shifted out, at a Max frequency f_c , on the falling edge of SCLK. The first Byte addressed can be at any location. The address is automatically incremented to the next higher address after each Byte of data is shifted out. Once the A11-A0 address reaches the last Byte of the register (Byte FFFH), it will reset to 000H, the command is completed by driving CS# high.

Address	A23-16	A15-12	A11-8	A7-0
Security Register #1	00H	0 0 0 1	Page Address	Byte Address
Security Register #2	00H	0 0 1 0	Page Address	Byte Address
Security Register #3	00H	0 0 1 1	Page Address	Byte Address

Figure 92 Read Security Registers command Sequence Diagram (SPI)



Note: The device default is in 24-bit address mode. For 4-Byte mode, the address length becomes 32-bit.

9.44 Global Block/Sector Lock (7Eh) or Unlock (98h)

All Block/Sector Lock bits can be set to 1 by the Global Block/Sector Lock command, or can set to 0 by the Global Block/Sector Unlock command.

The Global Block/Sector Lock command (7Eh) sequence: CS# goes low → SI: Sending Global Block/Sector Lock command → CS# goes high.

The Global Block/Sector Unlock command (98h) sequence: CS# goes low → SI: Sending Global Block/Sector Unlock command → CS# goes high.

Figure 93 Global Block/Sector Lock Sequence Diagram (SPI)

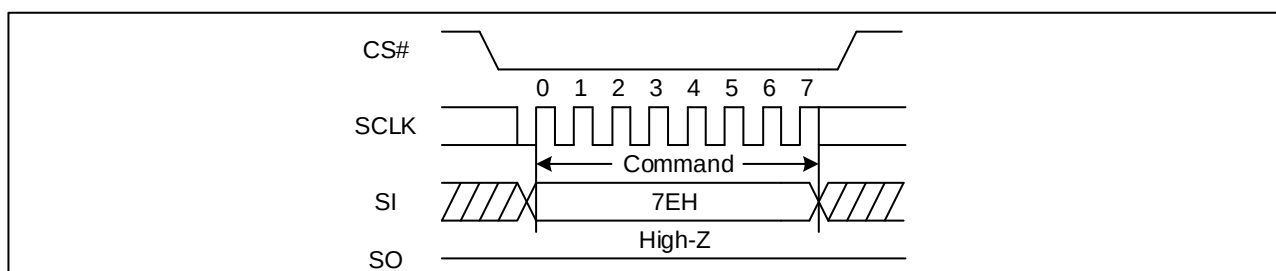


Figure 94 Global Block/Sector Lock Sequence Diagram (QPI)

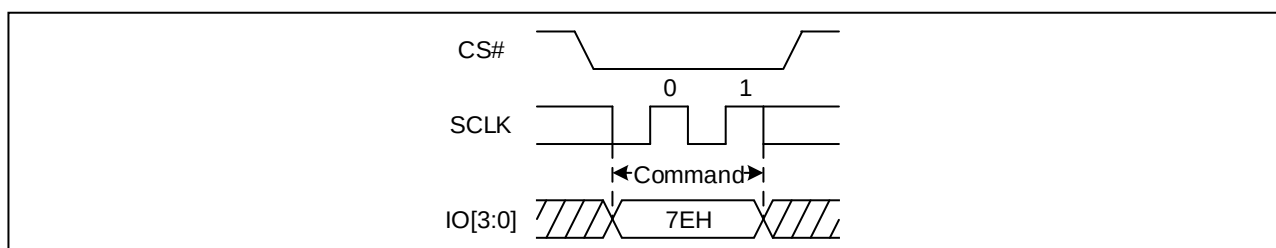


Figure 95 Global Block/Sector Unlock Sequence Diagram (SPI)

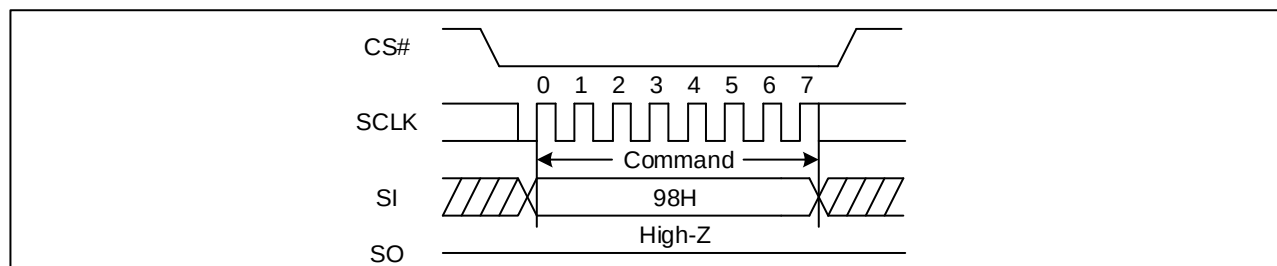
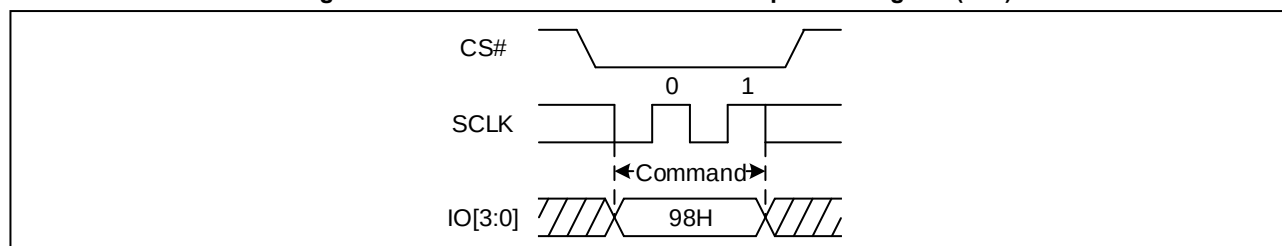


Figure 96 Global Block/Sector Unlock Sequence Diagram (QPI)



9.45 Read Password Register (27h)

The Read Password Register command is accepted and executed only when the device is in normal protection mode (PWD bit=0). When the device is in Advanced Password Protection mode (PWD bit=1), the Read Password Register is ignored; and the Password Register is lock and not read accessible (permanently stored and hidden). The Password Register is all “1’s” when shipped from the factory. This instruction is usually executed after a Program Password Register instruction to verify the pre-program password data.

The Read Password Register command sequence is initiated by driving CS# low followed by the Read Password Register command opcode 27h and the 64-bit Password (Data Byte [1:8]) data output.

Figure 97 Read Password Register Sequence Diagram (SPI)

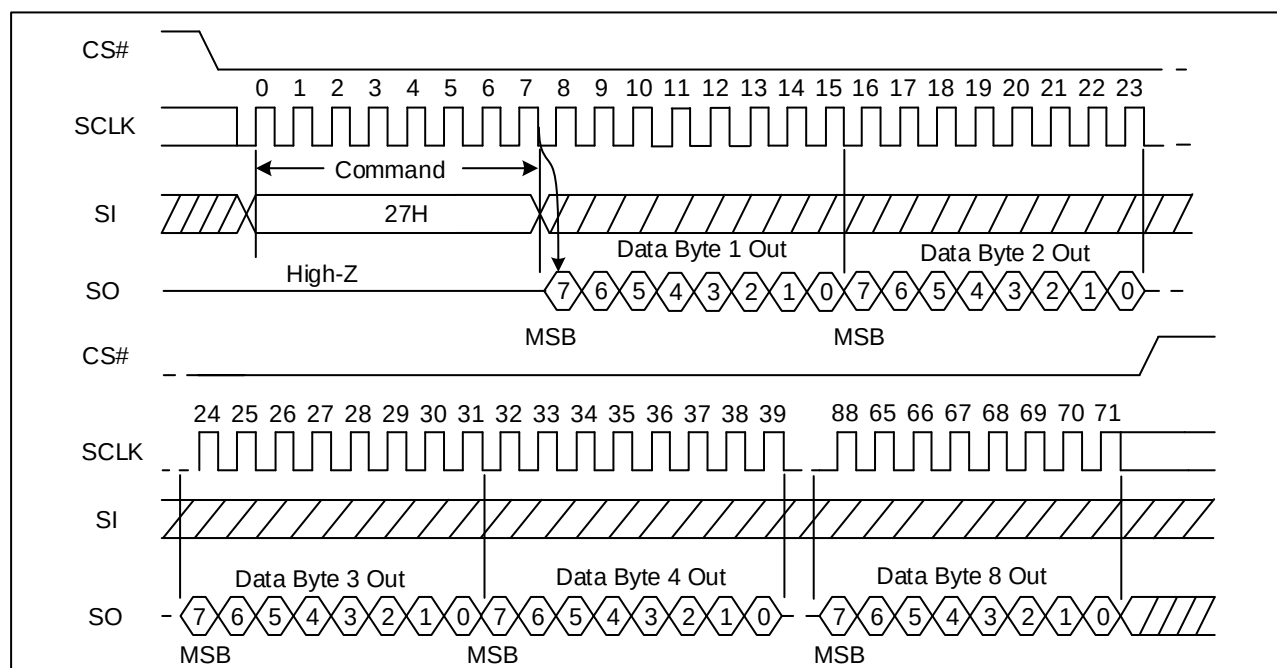
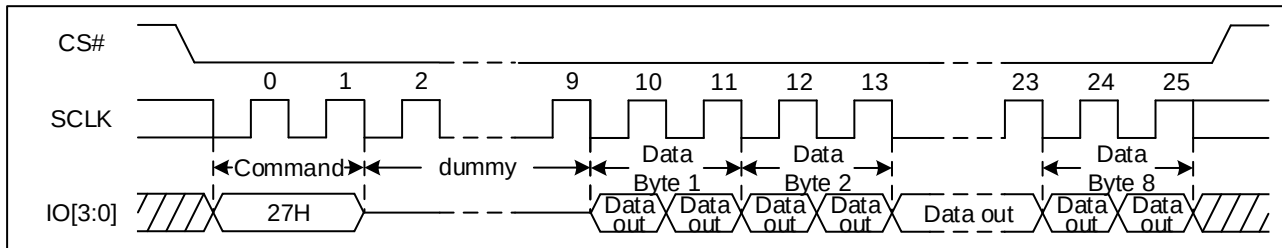


Figure 98 Read Password Register Sequence Diagram (QPI)



9.46 Program Password Register (28h)

The Program Password Register (28H) command is accepted and executed only when the device is in normal protection mode (PWD bit=0). When the device entered the Advanced Password Protection mode (PWD bit=1), the Program Password Register is ignored since the Password Register is lock, not programmable, and permanently stored/hidden. The Password Register is all “1’s” when shipped from the factory. When programmable, the Password Register supports only program operation (program 1 to 0). Once the bits are programmed to 0s, they can no longer be erased to 1s.

A Write Enable (WREN) command must be executed to set the Write Enable Latch (WEL) bit before sending the Program Password Register command. The Program Password Register command is entered by driving CS# Low, followed by the command code 28h and the 64-bit Password (Data Byte[1:8]) data input. CS# must be driven low for the entire duration of the sequence. As soon as CS# is driven high, the self-timed Program Password Register cycle (whose duration is tPPWD) is initiated. While the Program Password cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Program Password cycle, and transition to 0 after program completion. At some unspecified time before the cycle is completed, the Write Enable Latch (WEL) bit is reset. The exact number of bytes (64-bit or 8 bytes) of data must be sent to the device to execute the Program Password instruction, otherwise, the instruction is not executed.

Figure 99 Program Password Register Sequence Diagram (SPI)

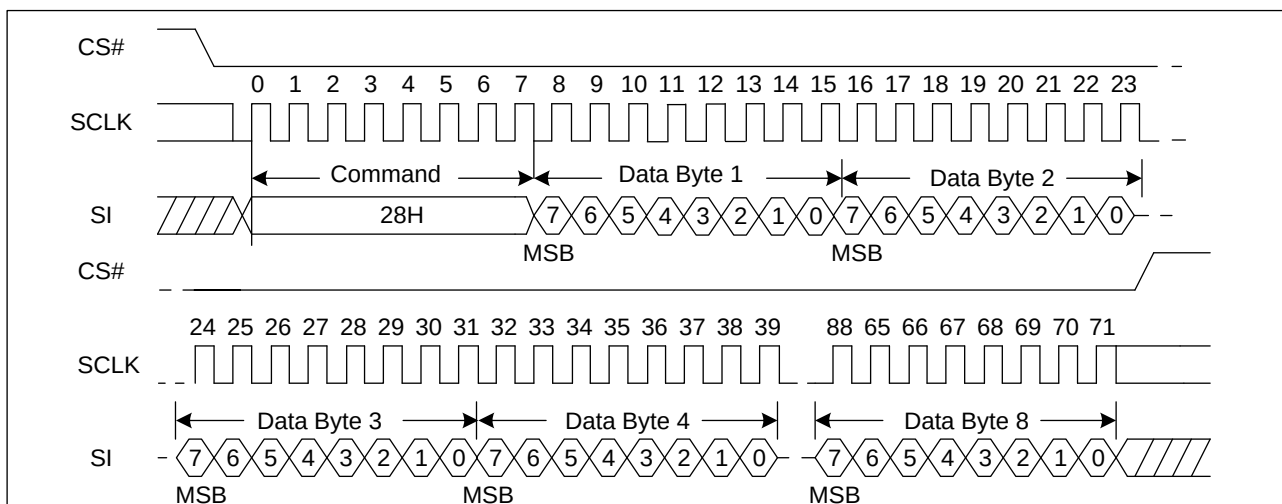
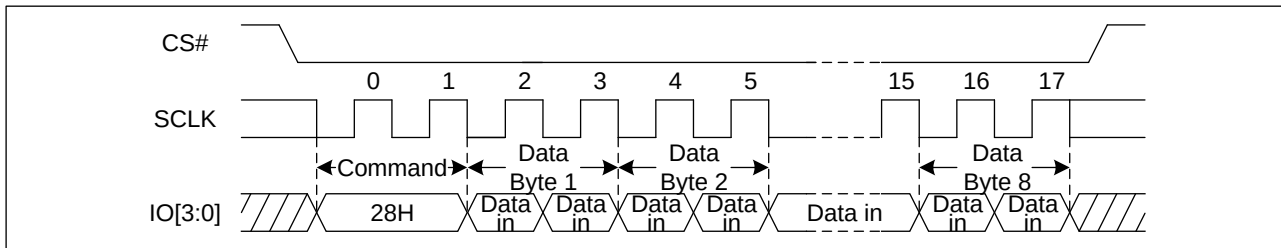


Figure 100 Program Password Register Sequence Diagram (QPI)



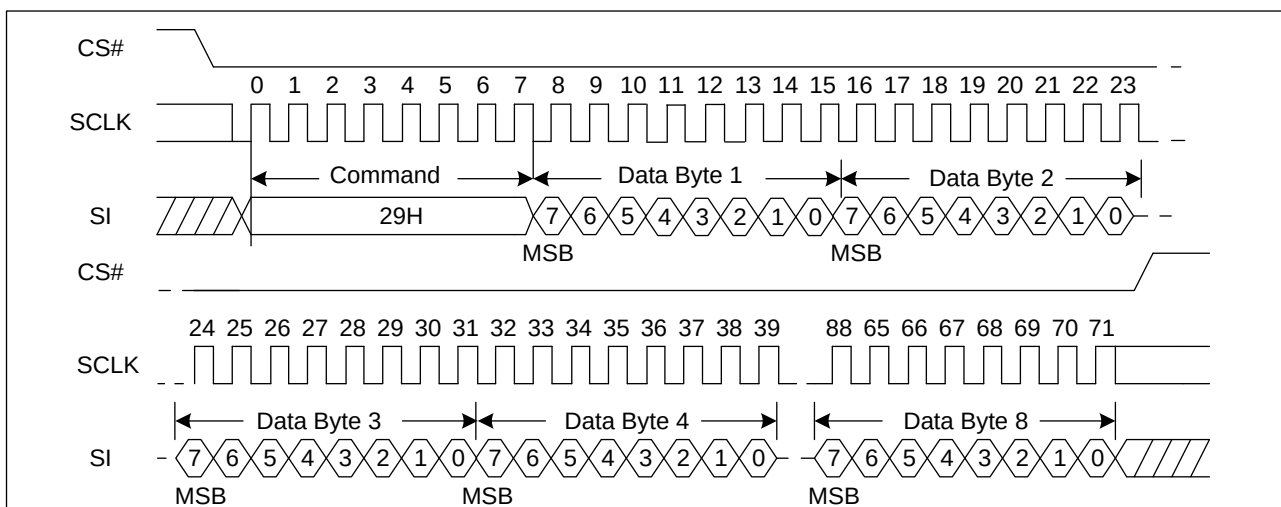
9.47 Password Unlock/Lock (29h)

Once the Advanced Password Protection Mode is entered (after setting PWD bit to '1'), a reset, power cycle, or Password Lock command is required to clear the PWDVfy bit to '0'. This will password protect the Block Protect bits (BPx, CMP, SRP1, SRP0) and NL Registers.

To clear the PWDVfy bit to '0', the Password Lock command is initiated by driving CS# Low, followed by the command code 29h and the mismatch 64-bit Password (Data Byte[1:8]) data input compare to the hidden/stored password. CS# must be driven high after the 64-bit of data, otherwise the Password Lock command is not executed. As soon as CS# is driven high, the self-timed Password Verify operation is initiated. While the password verification operation is in progress, the Status Register may be read to check the Write-In Progress (WIP) bit, the progress time takes at least 3us (this restriction makes it impossible to crack 64-bit password). The Write-In Progress (WIP) bit is a '1' during the self-timed password verify cycle and is a '0' when it is completed. The PWDVfy bit is cleared to '0' after a successful Password Lock sequence.

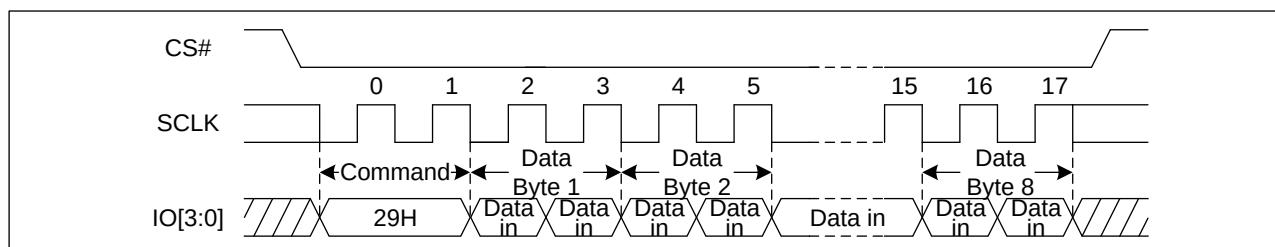
The Password Unlock steps follows the same command sequence as Password Lock sequence except that the matching 64-bit Password (Data Byte[1:8]) data is required to match/verify the hidden/stored password. This will set the PWDVfy bit to '1' and enable writes to the Block Protect (BPx, CMP, SRP1, SRP0) bits and NL Registers. As soon as Password Unlock command sequence is initiated after CS# is driven high, the self-timed Password Verify operation (whose duration is tPWDVfy) is initiated. While the password verification operation is in progress, the Status Register may be read to check the Write-In Progress (WIP) bit, the progress time takes at least 3us (this restriction makes it impossible to crack 64-bit password). The Write-In Progress (WIP) bit is a '1' during the self-timed password verify cycle and is a '0' when it is completed. The PWDVfy bit is set to '1' after a successful Password Lock sequence.

Figure 101 Password Lock/Unlock Sequence Diagram (SPI)



Note: To unlock (unprotect) the Block Protect (BPx, CMP, SRP1, SRP0) Bits and NL Registers, matching 64-bit password input to the stored/hidden password is required. To Lock (protect) the Block Protect (BPx, CMP, SRP1, SRP0) Bits and NL Registers, mismatching 64-bit password input is required.

Figure 102 Password Lock/Unlock Sequence Diagram (QPI)



Note: To unlock (unprotect) the Block Protect (BPx, CMP, SRP1, SRP0) and NL Registers, matching 64-bit password input to the stored/hidden password is required. To Lock (protect) the Block Protect (BPx, CMP, SRP1, SRP0) Bits and NL Registers, mismatching 64-bit password input is required.

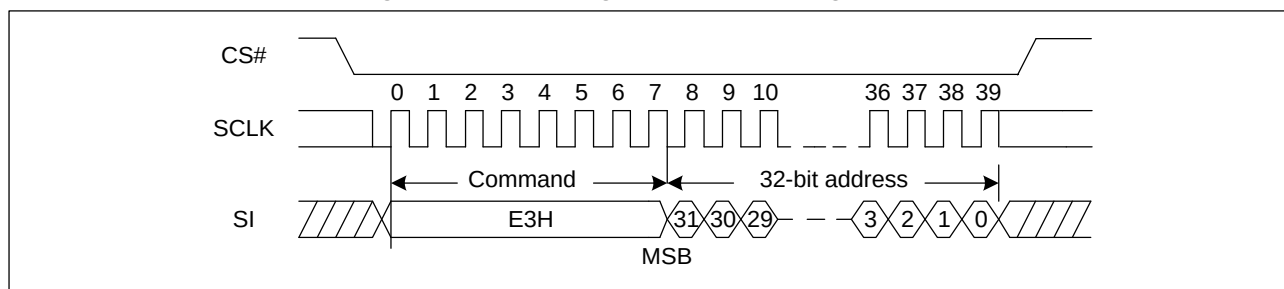
9.48 Set Nonvolatile Lock Register (E3h)

Before the Set NL Register commands are accepted besides the WEL bit= 1 requirement, the device must be either in normal protection mode (PWD bit = 0) or in Advanced Password Protection Mode (PWD bit = 1) and Password Verify enabled with PWDVfy bit set to '1'. The Advanced Password Protection Mode is setup right after PWD is set to '1' or after Password Unlock is executed successfully. On both cases, the device is Password Verify enabled with PWDVfy bit set from '0' to '1'. When Password Verify is disabled (PWD bit =1 & PWDVfy bit=0), executing Set NL Register instruction sequences are ignored.

The Set NL Register command sets the non-volatile NL Register block protection to 'FFH' of the target address within the range of either a sector or a block. When NL Register is '00H' (default), the corresponding sector or block is unprotected or unlocked state; and it can be set to 'FFH' to lock (protect) so the corresponding memory sector or block are protected against program or erase. Please check the NL Registers section for more information on the NL Registers mapping across the full memory.

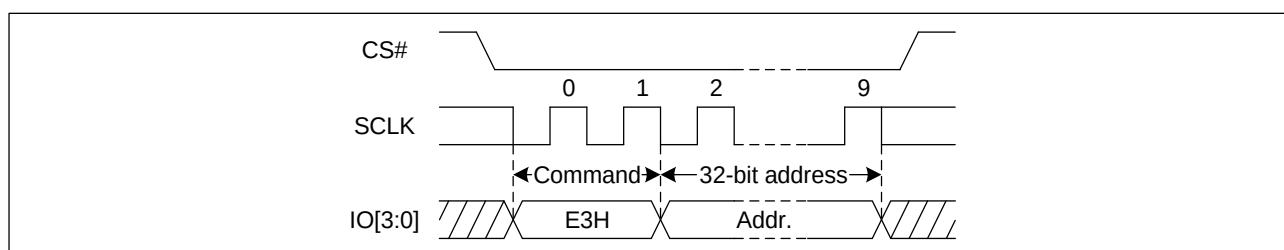
A Write Enable (WREN) command must be executed prior to initiating the Set NL Register command. The Set NL Register command is initiated by driving CS# low followed by latching in command opcode E3h and the 32-Bit Address on every rising edge of SCLK. For SPI sequence, CS# must be driven high after the least significant address bit input's SCLK (least significant address nibble input's SCLK for QPI), otherwise the Set NL Register command is not executed. As soon as CS# is driven high, the self-timed NL Register setting cycle (whose duration is tNLP) is initiated. The Write In Progress (WIP) bit is 1 during the self-timed NL Register setting cycle, and transition to 0 upon completion. When the cycle is completed, the Write Enable Latch (WEL) is reset. The NL Register corresponding to the input address used in the Set NL Register command sequence is also set to 'FFH' (protected).

Figure 103 Set NL Register Sequence Diagram (SPI)



Note: The Program NL Register command uses only 32-bit address input in either 3-Byte or 4-Byte Address mode.

Figure 104 Set NL Register Sequence Diagram (QPI)



Note: The Set NL Register command uses only 32-bit address input in either 3-Byte or 4-Byte Address mode.

9.49 Clear All Nonvolatile Lock Registers (E4h)

Before the Clear All NL Registers commands are accepted besides the WEL bit= 1 requirement, the device must be either in normal protection mode (PWD bit = 0) or in Advanced Password Protection Mode (PWD bit = 1) and Password Verify enabled with PWDVfy bit set to '1'. The Advanced Password Protection Mode is setup right after PWD is set to '1' or after Password Unlock is executed successfully. On both cases, the device is Password Verify enabled with PWDVfy bit set from '0' to '1'. When Password Verify is disabled (PWD bit=1 & PWDVfy bit=0), executing Clear All NL Registers instruction sequences are ignored.

The Clear All NL Registers command clears the whole non-volatile NL Register to '00H' that unlocks all the sector/block protection. When a NL Register is 'FFH', the corresponding sector or block is protected or locked state; Each NL Register is set to 'FFH' individually, however they can only be cleared as a group (all NL Registers) that can unlock (unprotect) the full memory opening the memory for either program or erase operation.

A Write Enable (WREN) command must be executed prior to initiating the Clear All NL Registers command. The Clear All NL Registers command is initiated by driving CS# low followed by latching in command opcode E4h on every rising edge of SCLK. For SPI, CS# must be driven high after the 8th SCLK (after 2nd CLK for QPI), otherwise the Clear All NL Registers command is not executed. As soon as CS# is driven high, the self-timed NL Registers clearing cycle (whose duration is tCNLR) is initiated. The Write In Progress (WIP) bit is 1 during the self-timed NL Registers clearing cycle, and transition to 0 upon completion. When the cycle is completed, the Write Enable Latch (WEL) is reset. All the NL Registers are cleared to '00H' (unprotected) after Clear All NL Registers completion.

Figure 105 Clear All NL Registers Sequence Diagram (SPI)

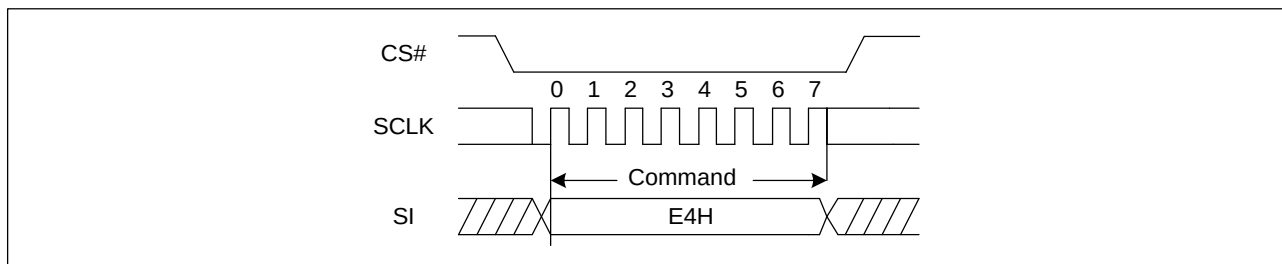
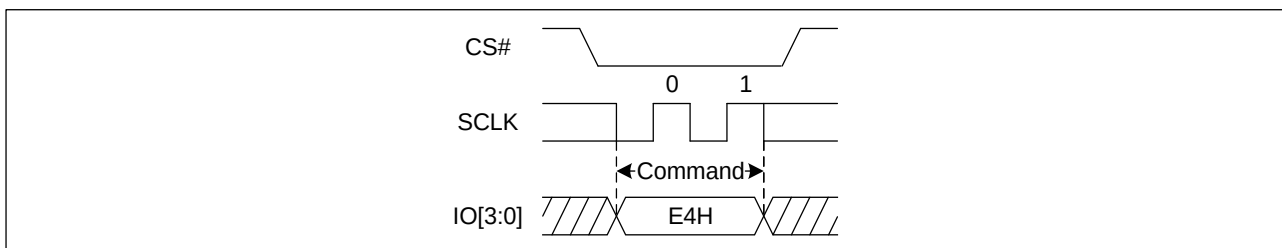


Figure 106 Clear All NL Registers Sequence Diagram (QPI)

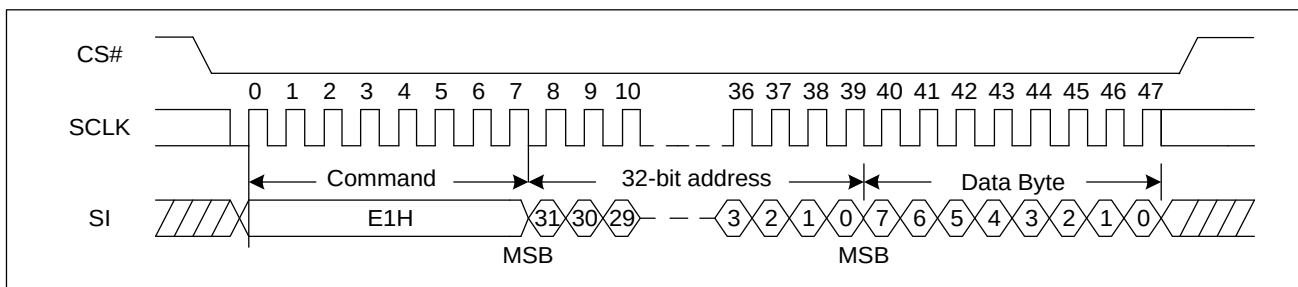


9.50 Write Volatile Lock Register (E1h)

The Write VL Register command writes to the VL Register either a 'FFh' data to protect or a '00h' data to unprotect a corresponding individual sector or block address. A byte size data input is used to do the write. If any other data other than 00H or FFH (ie. 01h, 02h... FEh) is used in the Write VL Register instruction data input sequence, the instruction is ignored. When a sector or block is protected, the corresponding VL Register is 'FFh' and when a sector or block is unprotected, the corresponding VL Register is '00h'. Please check the VL Register section for more information on the VL Registers mapping across the full memory.

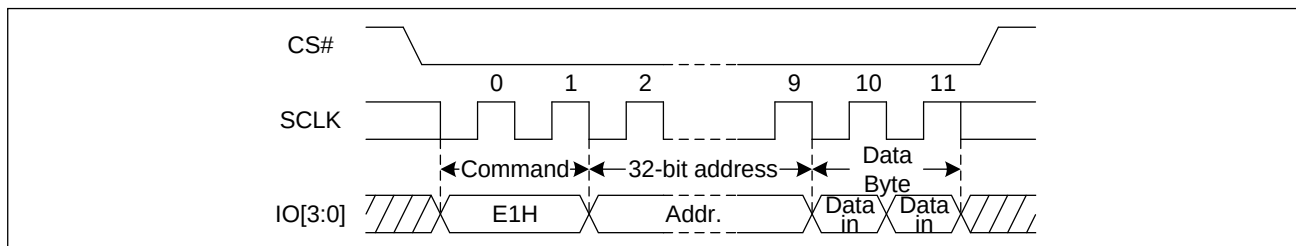
A Write Enable (WREN) command must be first executed before sending the Write VL Register command. Once properly setup, Write VL Register instruction is initiated by driving CS# low followed by sending the command opcode E1h, the 32-Bit Address targeting the sector or block to be protected or unprotected, and the byte of data to be written (either 00H or FFh will only be accepted). The command, address, and data input are latched in on the rising edge of the SCLK. As soon as CS# is driven high, the Write to VL Register cycle is initiated instantaneously (no wait time). When the cycle is completed, the Write Enable Latch (WEL) is reset.

Figure 107 Write VL Register Sequence Diagram (SPI)



Note: The Write VL Register command uses only 32-bit address input in either 3-Byte or 4-Byte Address mode.

Figure 108 Write VL Register Sequence Diagram (QPI)

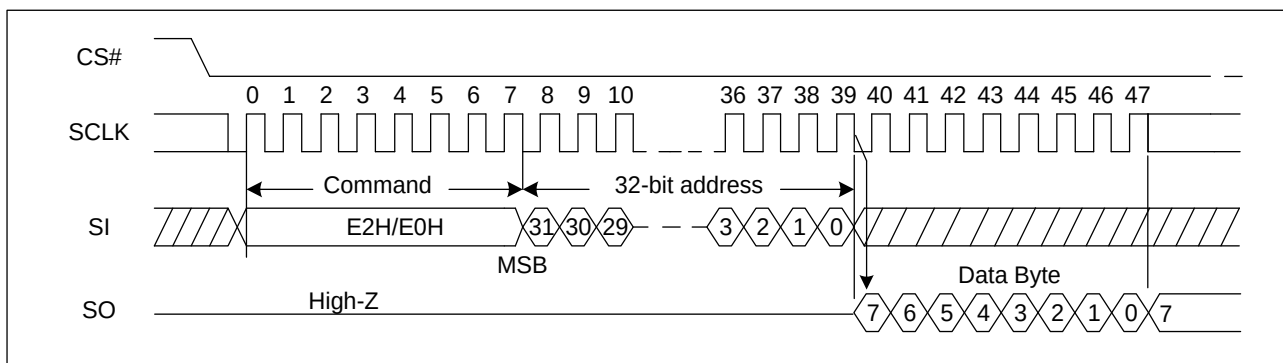


Note: The Write VL Register command uses only 32-bit address input in either 3-Byte or 4-Byte Address mode.

9.51 Read Nonvolatile Lock Register (E2h) / Read Volatile Lock Register(E0h)

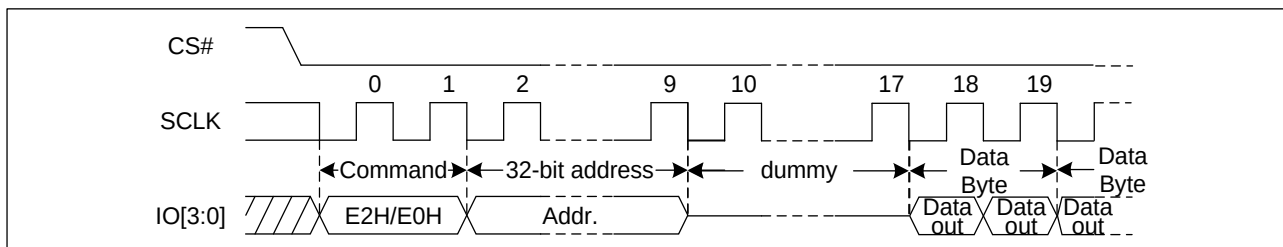
The Read NL Register (E2h)/Read VL Register (E0h) command sequence reads the NL/VL Register data of the corresponding sector or block used in the input address. The input command sequence is as follows: command E2h/E0h opcode and followed by a 4-Byte address (A31-A0). The input sequence uses the rising edge of SCLK to latch-in data. After the last input address SCLK falling edge, the NL/VL Register byte data follows as data output. Read NL Register/VL Register is ignored if executed in the middle of an Erase, Program or Write cycle.

Figure 109 Read NL Register/VL Register Sequence Diagram (SPI)



Note: The Read NL Register (E2h) / VL Register(E0h) command uses only 32-bit address input in either 3-Byte or 4-Byte Address mode.

Figure 110 Read NL Register/VL Register Sequence (QPI)



Note: The Read NL Register(E2h) / VL Register (E0h) command uses only 32-bit address input in either 3-Byte or 4-Byte Address mode.

9.52 Read Interface CRC Register (64h)

The Read Interface CRC Register (64h) command sequence reads the Interface CRC Register data. The input command sequence is as follows: command 64h opcode and followed by a 4-Byte address (A31-A0). The 4-Byte address (A31-A0) can be seen as dummy cycle, and can be all 0's. The input sequence uses the rising edge of SCLK to latch-in data. After the last input address SCLK falling edge, the Interface CRC Register byte data follows as data output.

If the host device upon receiving the slave's CRC checksum finds a mismatch with its own calculated CRC checksum, it can re-issue the Program / Erase command to the Flash. For Flash, multiple Program / Erase to the same location due to CRC checksum errors will affect data endurance.

Figure 111 Read Interface CRC Register Sequence Diagram (SPI)

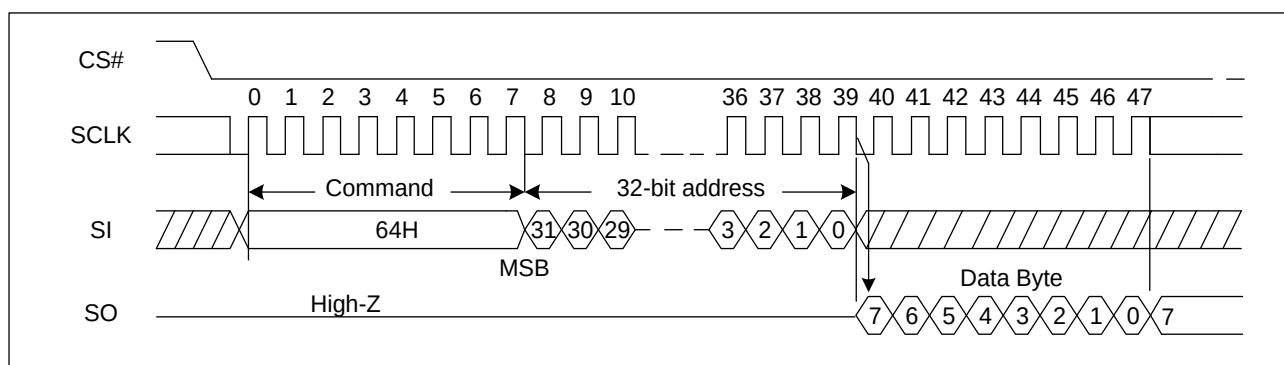
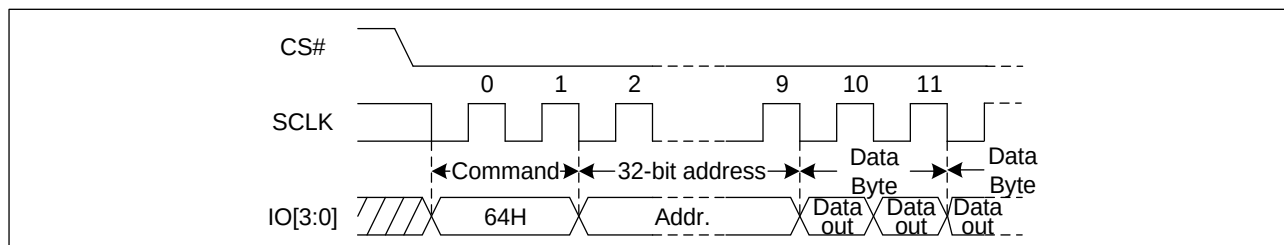


Figure 112 Read Interface CRC Register Sequence (QPI)



9.53 Data Integrity Check (5Bh)

The Data Integrity Check (5Bh) command is a hardware accelerated CRC calculation over a user defined address range in the memory array. The Data Integrity Check command is entered by driving CS# low, followed by the command code, and 4-Byte address of beginning, and 4-Byte address of ending on SI. CS# must be driven low for the entire duration of the sequence.

The Data Integrity Check command sequence: CS# goes low → sending Data Integrity Check command → 4-Byte address of beginning on SI → 4-Byte address of ending on SI → CS# goes high. CS# must be driven high after the eighth bit of the last address Byte has been latched in, otherwise the Data Integrity Check command is not executed. As soon as CS# is driven high, the self-timed Data Integrity Check cycle is initiated. While the Data Integrity Check cycle is in progress, the Status Register may be read to check the value of the Write in Progress (WIP) bit. The Write in Progress (WIP) bit is 1 during the self-timed Data Integrity Check cycle, and is 0 when it is completed.

Figure 113 Data Integrity Check Sequence Diagram (SPI)

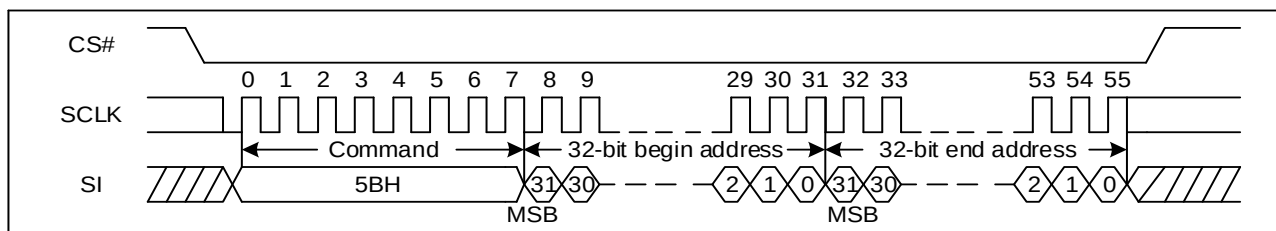
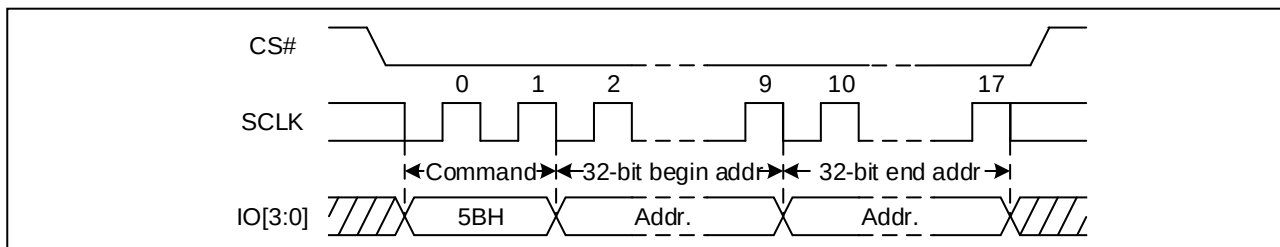


Figure 114 Data Integrity Check Sequence (QPI)



9.54 Enable Reset (66h) and Reset (99h)

If the Reset command is accepted, any on-going internal operation will be terminated and the device will return to its default power-on state and lose all the current volatile settings, such as Volatile Status Register bits, Write Enable Latch status (WEL), Program/Erase Suspend status, Read Parameter setting (P7-P0), Deep Power Down Mode, Continuous Read Mode bit setting (M7-M0) and Wrap Bit Setting (W6-W4).

The “Enable Reset (66h)” and the “Reset (99h)” commands can be issued in either SPI or QPI mode. The “Enable Reset (66h)” and “Reset (99h)” command sequence as follow: CS# goes low → Sending Enable Reset command → CS# goes high → CS# goes low → Sending Reset command → CS# goes high. Once the Reset command is accepted by the device, the device will take approximately t_{RST}/t_{RST_E} to reset. During this period, no command will be accepted. Data corruption may happen if there is an on-going or suspended internal Erase or Program operation when Reset command sequence is accepted by the device. It is recommended to check the WIP bit and the SUS1/SUS2 bit in Status Register before issuing the Reset command sequence.

Figure 115 Enable Reset and Reset command Sequence Diagram (SPI)

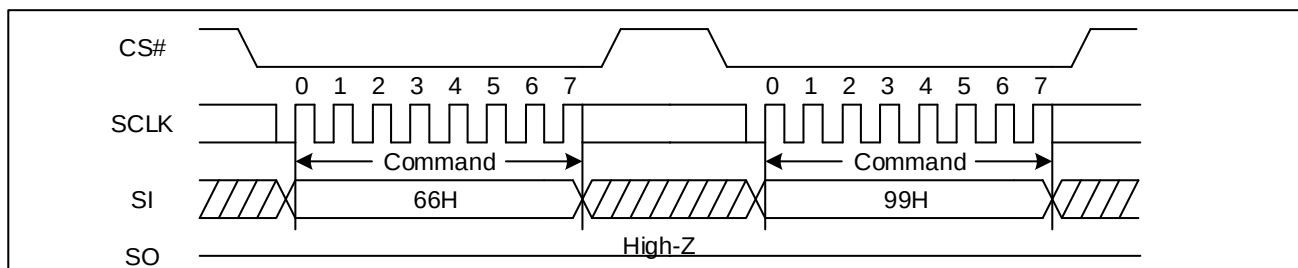
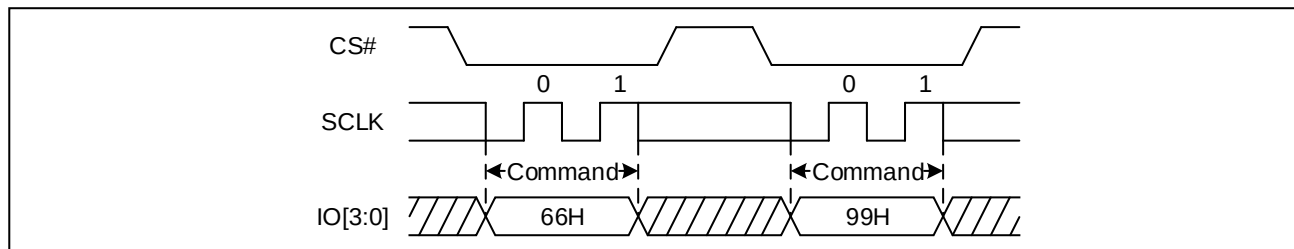


Figure 116 Enable Reset and Reset command Sequence Diagram (QPI)



9.55 Read Serial Flash Discoverable Parameter (5Ah)

The Serial Flash Discoverable Parameter (SFDP) standard provides a consistent method of describing the functional and feature capabilities of serial flash devices in a standard set of internal parameter tables. These parameter tables can be interrogated by host system software to enable adjustments needed to accommodate divergent features from multiple vendors. The concept is similar to the one found in the Introduction of JEDEC Standard, JESD68 on CFI. SFDP is a standard of JEDEC Standard No.216C.

Figure 117 Read Serial Flash Discoverable Parameter command Sequence Diagram (SPI)

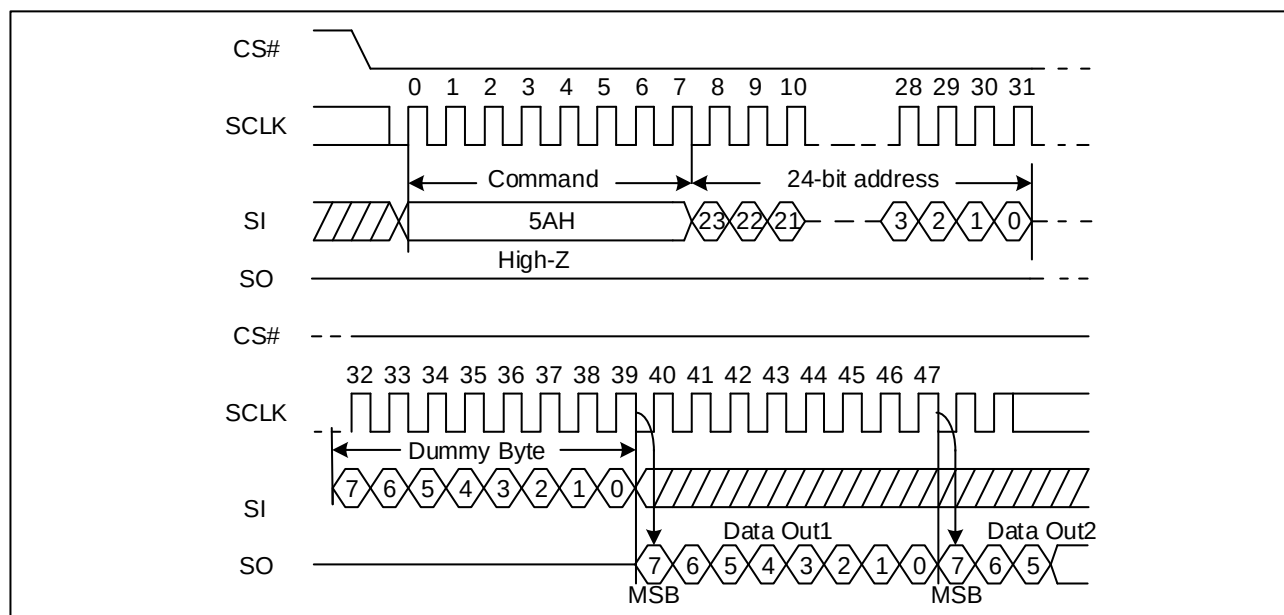


Figure 118 Read Serial Flash Discoverable Parameter command Sequence Diagram (QPI)

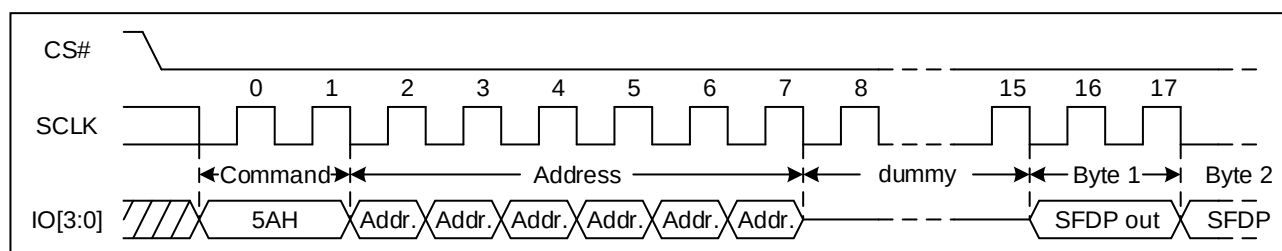


Table 20 Signature and Parameter Identification Data Values (Please contact GigaDevice for details)

10 ELECTRICAL CHARACTERISTICS

10.1 Power-On Timing

Figure 119. Power-On Timing Sequence Diagram

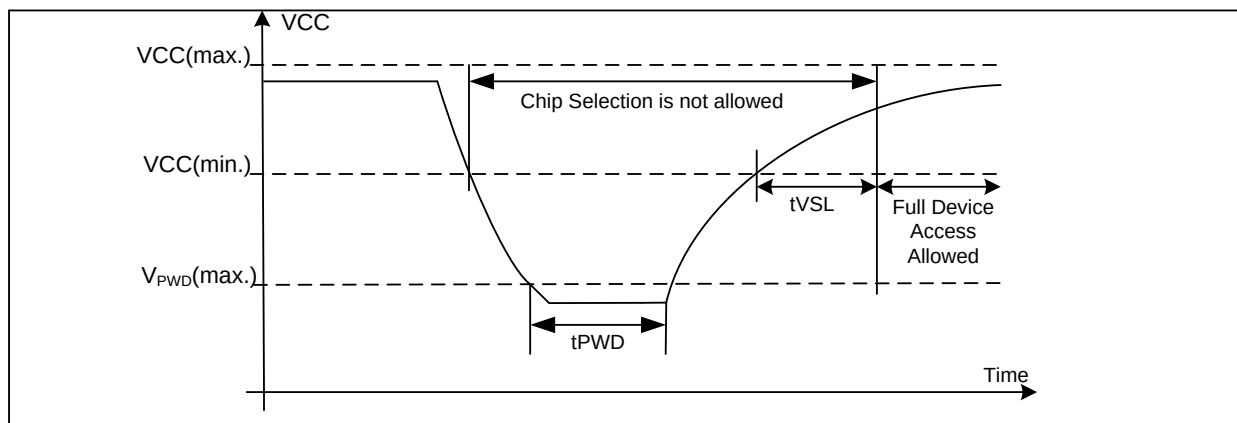


Table 21 Power-Up Timing and Write Inhibit Threshold

Symbol	Parameter	Min.	Max.	Unit
tVSL	VCC (min.) to device operation	1.5		ms
VWI	Write Inhibit Voltage	1	1.5	V
VPWD	VCC voltage needed to below VPWD for ensuring initialization will occur		0.5	V
tPWD	The minimum duration for ensuring initialization will occur	300		μs

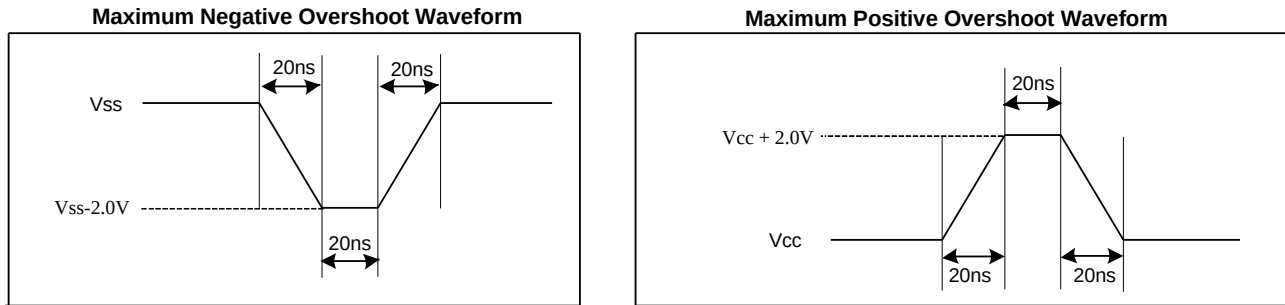
10.2 Initial Delivery State

The device is delivered with the memory array erased: all bits are set to 1 (each Byte contains FFH). The Status Register contains 00H, except that QE bit (S9) are set to 1.

10.3 Absolute Maximum Ratings

Parameter	Value	Unit
Ambient Operating Temperature (T_A)	-40 to 85	°C
	-40 to 105	
	-40 to 125	
Storage Temperature	-65 to 150	°C
Transient Input/Output Voltage (note: overshoot)	-2.0 to VCC+2.0	V
Applied Input/Output Voltage	-0.6 to VCC+0.4	V
VCC	-0.6 to 2.5	V

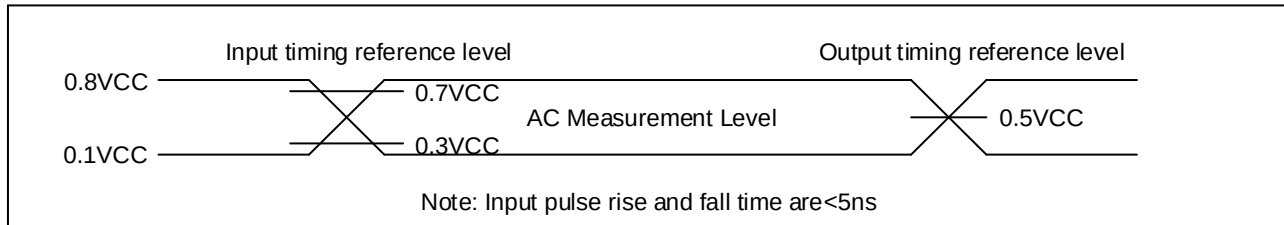
Figure 120. Input Test Waveform and Measurement Level



10.4 Capacitance Measurement Conditions

Symbol	Parameter	Min	Typ.	Max	Unit	Conditions
C _{IN}	Input Capacitance			6	pF	V _{IN} =0V
C _{OUT}	Output Capacitance			8	pF	V _{OUT} =0V
C _L	Load Capacitance	30			pF	
	Input Rise And Fall time			5	ns	
	Input Pulse Voltage	0.1V _{CC} to 0.8V _{CC}			V	
	Input Timing Reference Voltage	0.3V _{CC} to 0.7V _{CC}			V	
	Output Timing Reference Voltage	0.5V _{CC}			V	

Figure 121. Absolute Maximum Ratings Diagram





10.5 DC Characteristics

(T_A = -40°C~85°C, VCC=1.65~2.0V)

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit.
I _{LI}	Input Leakage Current				±2	μA
I _{LO}	Output Leakage Current				±2	μA
I _{CC1}	Standby Current	CS#=VCC, VIN=VCC or VSS		15	80	μA
I _{CC2}	Deep Power-Down Current	CS#=VCC, VIN=VCC or VSS		2	30	μA
I _{CC3}	Operating Current (Read)	CLK=0.1VCC / 0.9VCC at 166MHz, Q=Open(x4 I/O)		15	25	mA
		CLK=0.1VCC / 0.9VCC at 80MHz, Q=Open(x4 I/O)		9	15	mA
		CLK=0.1VCC / 0.9VCC at 104MHz DTR, Q=Open(x4 I/O)		15	25	mA
I _{CC4}	Operating Current (PP)	CS#=VCC		12	20	mA
I _{CC5}	Operating Current (WRSR)	CS#=VCC		12	20	mA
I _{CC6}	Operating Current (SE)	CS#=VCC		12	20	mA
I _{CC7}	Operating Current (BE)	CS#=VCC		12	20	mA
I _{CC8}	Operating Current (CE)	CS#=VCC		12	20	mA
V _{IL}	Input Low Voltage		-0.5		0.3VCC	V
V _{IH}	Input High Voltage		0.7VCC		VCC+0.4	V
V _{OL}	Output Low Voltage	I _{OL} = 100μA			0.2	V
V _{OH}	Output High Voltage	I _{OH} = -100μA	VCC-0.2			V

Notes:

1. Typical value at T_A = 25°C, VCC = 1.8V.
2. Value guaranteed by design and/or characterization, not 100% tested in production.



($T_A = -40^{\circ}\text{C} \sim 105^{\circ}\text{C}$, $V_{CC} = 1.65 \sim 2.0\text{V}$)

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit.
I_{LI}	Input Leakage Current				± 2	μA
I_{LO}	Output Leakage Current				± 2	μA
I_{CC1}	Standby Current	$CS\# = V_{CC}$, $V_{IN} = V_{CC}$ or V_{SS}		15	170	μA
I_{CC2}	Deep Power-Down Current	$CS\# = V_{CC}$, $V_{IN} = V_{CC}$ or V_{SS}		2	60	μA
I_{CC3}	Operating Current (Read)	$CLK = 0.1V_{CC} / 0.9V_{CC}$ at 166MHz, $Q = \text{Open}(x4 \text{ I/O})$		15	30	mA
		$CLK = 0.1V_{CC} / 0.9V_{CC}$ at 80MHz, $Q = \text{Open}(x4 \text{ I/O})$		9	20	mA
		$CLK = 0.1V_{CC} / 0.9V_{CC}$ at 104MHz DTR, $Q = \text{Open}(x4 \text{ I/O})$		15	30	mA
I_{CC4}	Operating Current (PP)	$CS\# = V_{CC}$		12	25	mA
I_{CC5}	Operating Current (WRSR)	$CS\# = V_{CC}$		12	25	mA
I_{CC6}	Operating Current (SE)	$CS\# = V_{CC}$		12	25	mA
I_{CC7}	Operating Current (BE)	$CS\# = V_{CC}$		12	25	mA
I_{CC8}	Operating Current (CE)	$CS\# = V_{CC}$		12	25	mA
V_{IL}	Input Low Voltage		-0.5		$0.3V_{CC}$	V
V_{IH}	Input High Voltage		$0.7V_{CC}$		$V_{CC} + 0.4$	V
V_{OL}	Output Low Voltage	$I_{OL} = 100\mu\text{A}$			0.2	V
V_{OH}	Output High Voltage	$I_{OH} = -100\mu\text{A}$	$V_{CC} - 0.2$			V

Notes:

1. Typical value at $T_A = 25^{\circ}\text{C}$, $V_{CC} = 1.8\text{V}$.
2. Value guaranteed by design and/or characterization, not 100% tested in production.



(T_A = -40°C~125°C, VCC=1.65~2.0V)

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit.
I _{LI}	Input Leakage Current				±2	μA
I _{LO}	Output Leakage Current				±2	μA
I _{CC1}	Standby Current	CS#=VCC, VIN=VCC or VSS		15	280	μA
I _{CC2}	Deep Power-Down Current	CS#=VCC, VIN=VCC or VSS		2	110	μA
I _{CC3}	Operating Current (Read)	CLK=0.1VCC / 0.9VCC at 166MHz, Q=Open(x4 I/O)		15	30	mA
		CLK=0.1VCC / 0.9VCC at 80MHz, Q=Open(x4 I/O)		9	20	mA
		CLK=0.1VCC / 0.9VCC at 104MHz DTR, Q=Open(x4 I/O)		15	30	mA
I _{CC4}	Operating Current (PP)	CS#=VCC		12	25	mA
I _{CC5}	Operating Current (WRSR)	CS#=VCC		12	25	mA
I _{CC6}	Operating Current (SE)	CS#=VCC		12	25	mA
I _{CC7}	Operating Current (BE)	CS#=VCC		12	25	mA
I _{CC8}	Operating Current (CE)	CS#=VCC		12	25	mA
V _{IL}	Input Low Voltage		-0.5		0.3VCC	V
V _{IH}	Input High Voltage		0.7VCC		VCC+0.4	V
V _{OL}	Output Low Voltage	I _{OL} = 100μA			0.2	V
V _{OH}	Output High Voltage	I _{OH} = -100μA	VCC-0.2			V

Notes:

1. Typical value at T_A = 25°C, VCC = 1.8V.
2. Value guaranteed by design and/or characterization, not 100% tested in production.



10.6 AC Characteristics

(T_A = -40°C~85°C, VCC=1.65~2.0V)

Symbol	Parameter	Min.	Typ.	Max.	Unit.
f _{C1}	Serial Clock Frequency for all instructions except 03h, 13h, BDh, BEh, EDh, EEh			166	MHz
f _{C2}	Serial Clock Frequency For: DTR Dual I/O (BDh, BEh), DTR Quad I/O (EDh, EEh)			104	MHz
f _R	Serial Clock Frequency For: Read (03h, 13h)			60	MHz
t _{CLH}	Serial Clock High Time	45% (1/f _{CMax})			ns
t _{CLL}	Serial Clock Low Time	45% (1/f _{CMax})			ns
t _{CLCH} t _{CHCL}	Serial Clock Rise/Fall Time (Slew Rate)	0.1			V/ns
t _{SLCH}	CS# Active Setup Time	5			ns
t _{CHSH}	CS# Active Hold Time	5			ns
t _{CLSH}	CS# Active Hold Time (DTR)	5			ns
t _{SHCH}	CS# Not Active Setup Time	5			ns
t _{CHSL}	CS# Not Active Hold Time	5			ns
t _{SHSL}	CS# High Time (Read)	20			ns
	CS# High Time (Write)	40			ns
t _{SHQZ}	Output Disable Time			8	ns
t _{CLQX} t _{CHQX}	Output Hold Time	1.2			ns
t _{DVCH} t _{DVCL}	Data In Setup Time	2			ns
t _{CHDX} t _{CLDX}	Data In Hold Time	2			ns
t _{QSV}	Clock transient to DQS valid time	Align to 30pF t _{CLQV}			ns
t _{DQSQ}	SIO Valid Skew Related to DQS (TFBGA, SOP,15pF)			0.8	ns
t _{QHS}	SIO Hold Skew Factor (TFBGA, SOP,15pF)			1	ns
t _{ECSV}	ECS# Setup Time			10	ns
t _{CLQV} t _{CHQV}	Clock Low To Output Valid (VCC=1.65~2.0V, loading=30pF) ⁽⁵⁾			7	ns
	Clock Low To Output Valid (VCC=1.65~2.0V, loading=15pF) ⁽⁵⁾			6	ns
t _{WHS�}	Write Protect Setup Time Before CS# Low	20			ns
t _{SHWL}	Write Protect Hold Time After CS# High	100			ns
t _{DP}	CS# High To Deep Power-Down Mode			3	μs
t _{RES1}	CS# High To Standby Mode Without Electronic Signature Read			30	μs



t _{RES2}	CS# High To Standby Mode With Electronic Signature Read			30	μs
t _{SUS}	CS# High To Next Command After Suspend			20	μs
t _{RS} ⁽⁶⁾	Latency Between Resume And Next Suspend	100			μs
t _{RST}	CS# High To Next Command After Reset (Except From Erase or Safe boot fail)			30	μs
t _{RST_E}	CS# High To Next Command After Reset (From Erase)			25	ms
t _{RST_S}	CS# High To Next Command After Reset (From Safe boot fail)			1.5	ms
t _{CRCS}	CRC Calculation Setup Time		17		μs
t _{CRCR}	CRC Calculation Rate (Calculation rate over a large (>1024-byte) block of data)		65		MBps
t _{PS}	Program/Erase Command CS# Invalid to Program Suspend / Erase Suspend Command CS# Invalid (Interface CRC)			10	μs
t _W	Write Status/Non-Volatile Configuration Register Cycle Time		5	20	ms
t _{BP}	Byte Program Time		30	150	μs
t _{PP}	Page Programming Time		0.25	1.2	ms
t _{PASSP}	Program password		0.4	0.7	ms
t _{PWDVfy}	Password Verification Time		3		μs
t _{NLP}	Program Nonvolatile Lock bit time		50		μs
t _{NLE}	Erase Nonvolatile Lock array		30	300	ms
t _{SE}	Sector Erase Time		30	300	ms
t _{BE1}	Block Erase Time (32K Bytes)		0.12	0.8	s
t _{BE2}	Block Erase Time (64K Bytes)		0.15	1.2	s
t _{CE}	Chip Erase Time (GD25LF256F)		75	180	s

Notes:

1. Typical value at T_A = 25°C.
2. Value guaranteed by design and/or characterization, not 100% tested in production.
3. Time of CS# High To Next Command After Reset from 01h/11h/B1h command would be t_W + t_{RST}
4. Time of CS# High To Next Command After Reset from 28h command would be t_{PASSP} + t_{RST}
5. If DQS function is used in STR Read mode, DQS Pin loading must be less than 15pF.
6. Minimum timing must be observed before issuing the next suspend command, and a period equal to or longer than the minimum timing is required in order for the program or erase operation to make progress, but the operation time may exceed the maximum value.



(T_A = -40°C~105°C, VCC=1.65~2.0V)

Symbol	Parameter	Min.	Typ.	Max.	Unit.
f _{C1}	Serial Clock Frequency for all instructions except 03H, 13H, BDh, BEh, EDh, EEh			166	MHz
f _{C2}	Serial Clock Frequency For: DTR Dual I/O (BDh, BEh), DTR Quad I/O (EDh, EEh)			104	MHz
f _R	Serial Clock Frequency For: Read (03h, 13h)			60	MHz
t _{CLH}	Serial Clock High Time	45% (1/f _{CMax})			ns
t _{CLL}	Serial Clock Low Time	45% (1/f _{CMax})			ns
t _{CLCH} t _{CHCL}	Serial Clock Rise/Fall Time (Slew Rate)	0.1			V/ns
t _{SLCH}	CS# Active Setup Time	5			ns
t _{CHSH}	CS# Active Hold Time	5			ns
t _{CLSH}	CS# Active Hold Time (DTR)	5			ns
t _{SHCH}	CS# Not Active Setup Time	5			ns
t _{CHSL}	CS# Not Active Hold Time	5			ns
t _{SHSL}	CS# High Time (Read)	20			ns
	CS# High Time (Write)	40			ns
t _{SHQZ}	Output Disable Time			8	ns
t _{CLQX} t _{CHQX}	Output Hold Time	1.2			ns
t _{DVCH} t _{DVCL}	Data In Setup Time	2			ns
t _{CHDX} t _{CLDX}	Data In Hold Time	2			ns
t _{QSV}	Clock transient to DQS valid time	Align to 30pF t _{CLQV}			ns
t _{DQSQ}	SIO Valid Skew Related to DQS (TFBGA, SOP,15pF)			0.8	ns
t _{QHS}	SIO Hold Skew Factor (TFBGA, SOP,15pF)			1	ns
t _{ECSV}	ECS# Setup Time			10	ns
t _{CLQV} t _{CHQV}	Clock Low To Output Valid (VCC=1.65~2.0V, loading=30pF) ⁽⁵⁾			7	ns
	Clock Low To Output Valid (VCC=1.65~2.0V, loading=15pF) ⁽⁵⁾			6	ns
t _{WHSL}	Write Protect Setup Time Before CS# Low	20			ns
t _{SHWL}	Write Protect Hold Time After CS# High	100			ns
t _{DP}	CS# High To Deep Power-Down Mode			3	μs
t _{RES1}	CS# High To Standby Mode Without Electronic Signature Read			30	μs
t _{RES2}	CS# High To Standby Mode With Electronic Signature			30	μs



	Read				
t _{SUS}	CS# High To Next Command After Suspend			20	μs
t _{RS} ⁽⁶⁾	Latency Between Resume And Next Suspend	100			μs
t _{RST}	CS# High To Next Command After Reset (Except From Erase or Safe boot fail)			30	μs
t _{RST_E}	CS# High To Next Command After Reset (From Erase)			25	ms
t _{RST_S}	CS# High To Next Command After Reset (From Safe boot fail)			1.5	ms
t _{CRCS}	CRC Calculation Setup Time		17		μs
t _{CRCR}	CRC Calculation Rate (Calculation rate over a large (>1024-byte) block of data)		65		MBps
t _{PS}	Program/Erase Command CS# Invalid to Program Suspend / Erase Suspend Command CS# Invalid (Interface CRC)			10	μs
t _W	Write Status/Non-Volatile Configuration Register Cycle Time		5	20	ms
t _{BP}	Byte Program Time		30	160	μs
t _{PP}	Page Programming Time		0.25	1.5	ms
t _{PASSP}	Program password		0.4	0.7	ms
t _{PWDFY}	Password Verification Time		3		μs
t _{NLP}	Program Nonvolatile Lock bit time		50		μs
t _{NLE}	Erase Nonvolatile Lock array		30	400	ms
t _{SE}	Sector Erase Time		30	400	ms
t _{BE1}	Block Erase Time (32K Bytes)		0.12	1	s
t _{BE2}	Block Erase Time (64K Bytes)		0.15	1.5	s
t _{CE}	Chip Erase Time (GD25LF256F)		75	250	s

Notes:

1. Typical value at T_A = 25°C.
2. Value guaranteed by design and/or characterization, not 100% tested in production.
3. Time of CS# High To Next Command After Reset from 01h/11h/B1h command would be t_W + t_{RST}
4. Time of CS# High To Next Command After Reset from 28h command would be t_{PASSP} + t_{RST}
5. If DQS function is used in STR Read mode, DQS Pin loading must be less than 15pF.
6. Minimum timing must be observed before issuing the next suspend command, and a period equal to or longer than the minimum timing is required in order for the program or erase operation to make progress, but the operation time may exceed the maximum value.



(T_A = -40°C~125°C, VCC=1.65~2.0V)

Symbol	Parameter	Min.	Typ.	Max.	Unit.
f _{C1}	Serial Clock Frequency for all instructions except 03h, 13h, BDh, BEh, EDh, EEh			166	MHz
f _{C2}	Serial Clock Frequency For: DTR Dual I/O (BDh, BEh), DTR Quad I/O (EDh, EEh)			104	MHz
f _R	Serial Clock Frequency For: Read (03h, 13h)			60	MHz
t _{CLH}	Serial Clock High Time	45% (1/f _{CMax})			ns
t _{CLL}	Serial Clock Low Time	45% (1/f _{CMax})			ns
t _{CLCH} t _{CHCL}	Serial Clock Rise/Fall Time (Slew Rate)	0.1			V/ns
t _{SLCH}	CS# Active Setup Time	5			ns
t _{CHSH}	CS# Active Hold Time	5			ns
t _{CLSH}	CS# Active Hold Time (DTR)	5			ns
t _{SHCH}	CS# Not Active Setup Time	5			ns
t _{CHSL}	CS# Not Active Hold Time	5			ns
t _{SHSL}	CS# High Time (Read)	20			ns
	CS# High Time (Write)	40			ns
t _{SHQZ}	Output Disable Time			8	ns
t _{CLQX} t _{CHQX}	Output Hold Time	1.2			ns
t _{DVCH} t _{DVCL}	Data In Setup Time	2			ns
t _{CHDX} t _{CLDX}	Data In Hold Time	2			ns
t _{QSV}	Clock transient to DQS valid time	Align to 30pF t _{CLQV}			ns
t _{DQSQ}	SIO Valid Skew Related to DQS (TFBGA, SOP, 15pF)			0.8	ns
t _{QHS}	SIO Hold Skew Factor (TFBGA, SOP, 15pF)			1	ns
t _{ECSV}	ECS# Setup Time			10	ns
t _{CLQV} t _{CHQV}	Clock Low To Output Valid (VCC=1.65~2.0V, loading=30pF) ⁽⁵⁾			7	ns
	Clock Low To Output Valid (VCC=1.65~2.0V, loading=15pF) ⁽⁵⁾			6	ns
t _{WHSL}	Write Protect Setup Time Before CS# Low	20			ns
t _{SHWL}	Write Protect Hold Time After CS# High	100			ns
t _{DP}	CS# High To Deep Power-Down Mode			3	μs
t _{RES1}	CS# High To Standby Mode Without Electronic Signature Read			30	μs
t _{RES2}	CS# High To Standby Mode With Electronic Signature			30	μs



	Read				
t _{SUS}	CS# High To Next Command After Suspend			20	μs
t _{RS} ⁽⁶⁾	Latency Between Resume And Next Suspend	100			μs
t _{RST}	CS# High To Next Command After Reset (Except From Erase or Safe boot fail)			30	μs
t _{RST_E}	CS# High To Next Command After Reset (From Erase)			25	ms
t _{RST_S}	CS# High To Next Command After Reset (From Safe boot fail)			1.5	ms
t _{CRCS}	CRC Calculation Setup Time		17		μs
t _{CRCR}	CRC Calculation Rate (Calculation rate over a large (>1024-byte) block of data)		65		MBps
t _{PS}	Program/Erase Command CS# Invalid to Program Suspend / Erase Suspend Command CS# Invalid (Interface CRC)			10	μs
t _W	Write Status/Non-Volatile Configuration Register Cycle Time		5	25	ms
t _{BP}	Byte Program Time		30	180	μs
t _{PP}	Page Programming Time		0.25	1.8	ms
t _{PASSP}	Program password		0.4	0.8	ms
t _{PWDVFY}	Password Verification Time		3		μs
t _{NLP}	Program Nonvolatile Lock bit time		50		μs
t _{NLE}	Erase Nonvolatile Lock array		30	500	ms
t _{SE}	Sector Erase Time		30	500	ms
t _{BE1}	Block Erase Time (32K Bytes)		0.12	1.2	s
t _{BE2}	Block Erase Time (64K Bytes)		0.15	1.5	s
t _{CE}	Chip Erase Time (GD25LF256F)		75	250	s

Notes:

1. Typical value at T_A = 25°C.
2. Value guaranteed by design and/or characterization, not 100% tested in production.
3. Time of CS# High To Next Command After Reset from 01h/11h/B1h command would be t_W + t_{RST}
4. Time of CS# High To Next Command After Reset from 28h command would be t_{PASSP} + t_{RST}
5. If DQS function is used in STR Read mode, DQS Pin loading must be less than 15pF.
6. Minimum timing must be observed before issuing the next suspend command, and a period equal to or longer than the minimum timing is required in order for the program or erase operation to make progress, but the operation time may exceed the maximum value.

Figure 122. Serial Input Timing

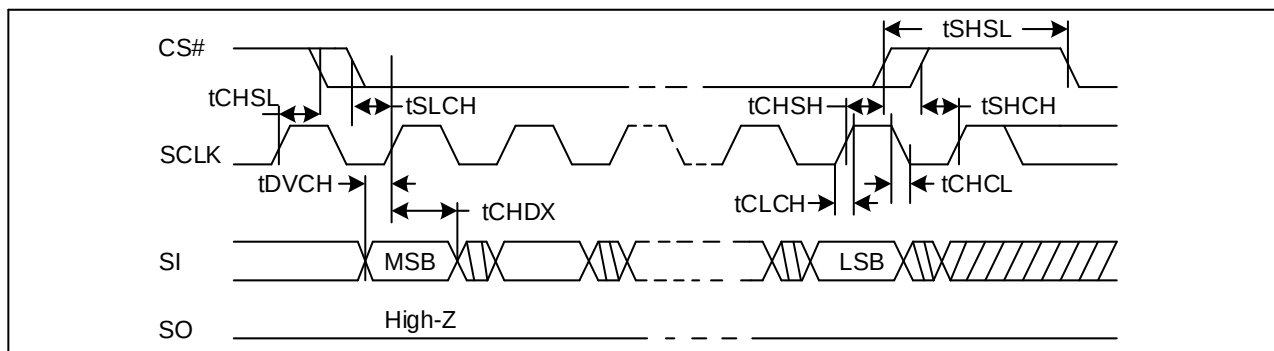


Figure 123. Output Timing

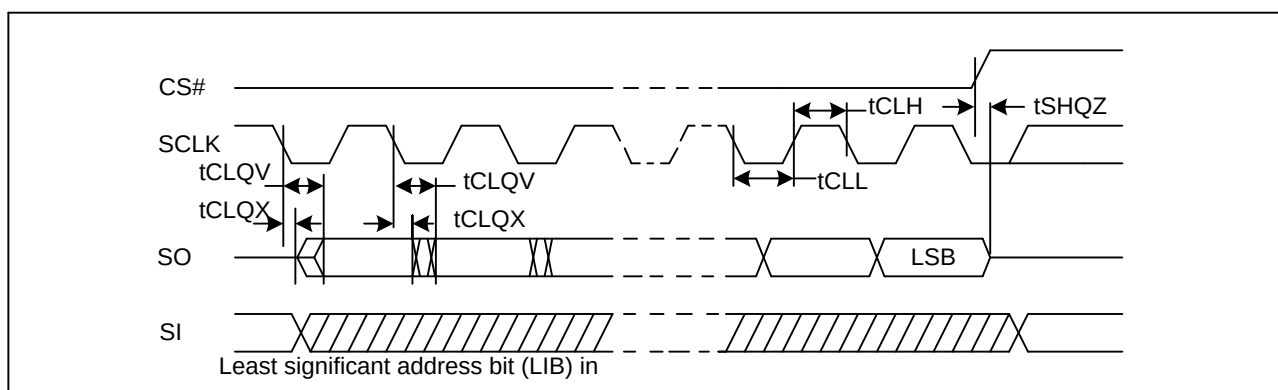


Figure 124. Serial Input Timing (DTR)

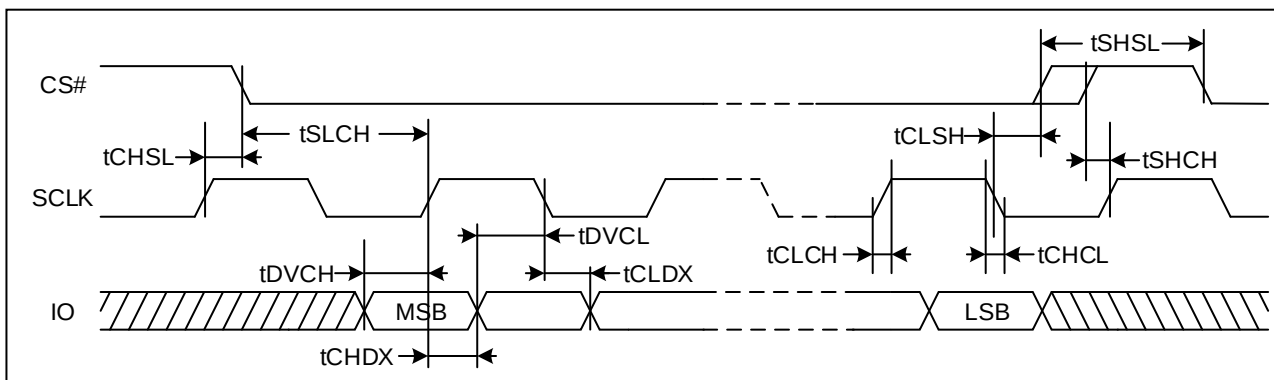


Figure 125. Serial Output Timing (DTR)

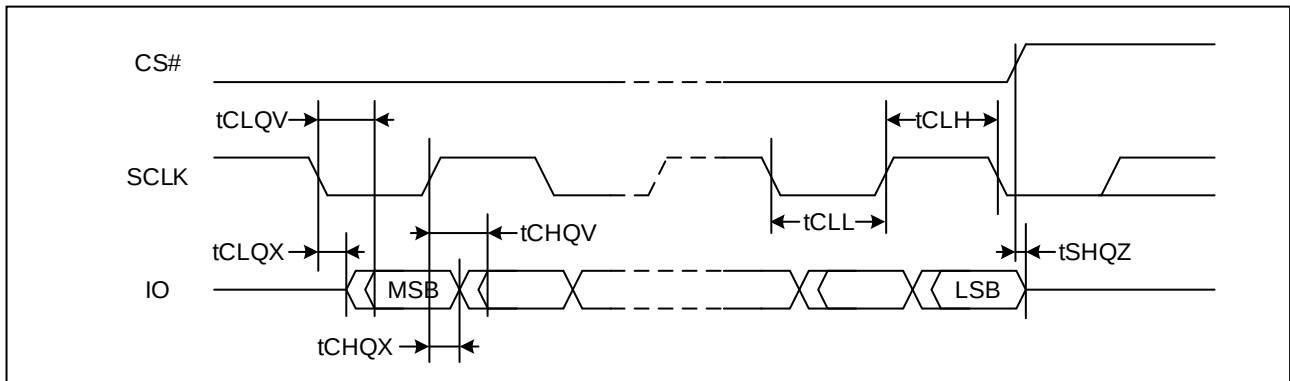


Figure 126. DQS Output Timing (STR)

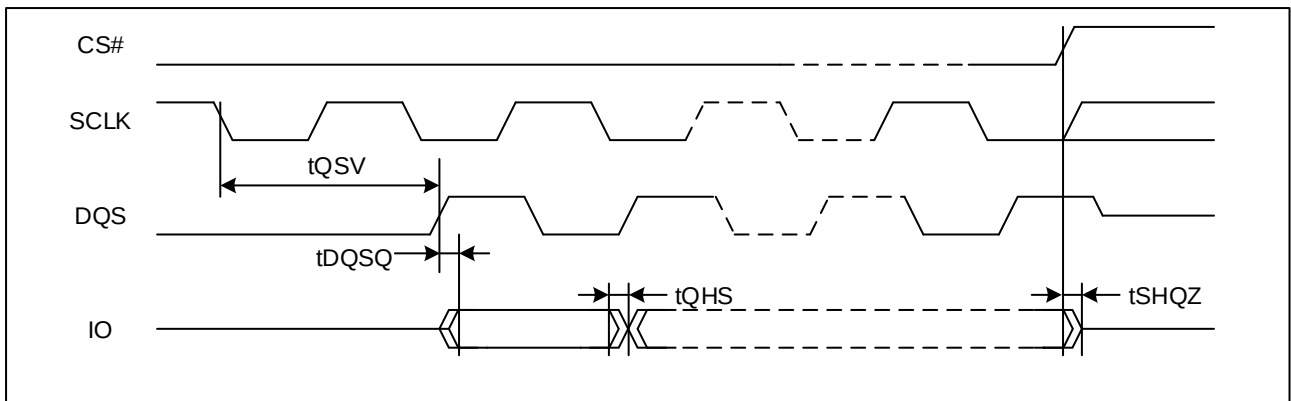


Figure 127. DQS Output Timing (DTR)

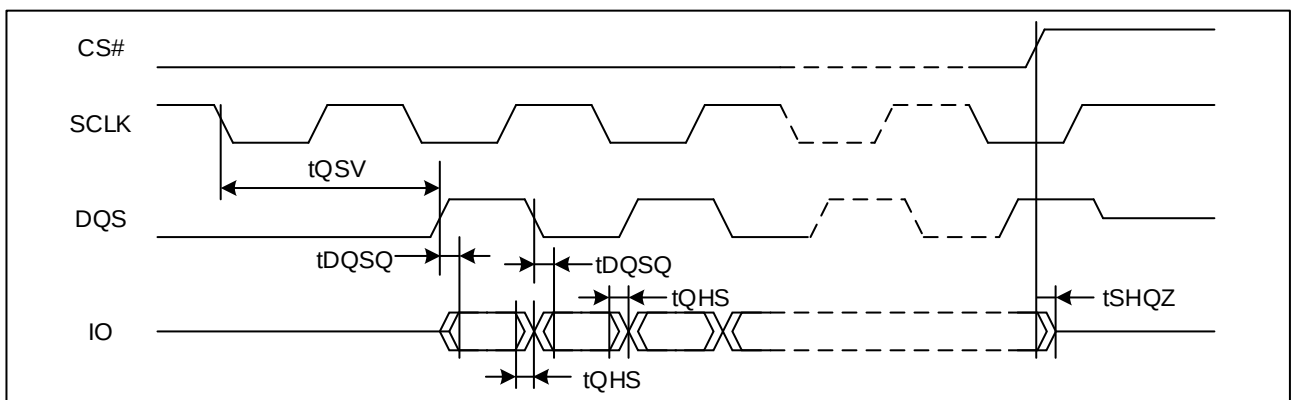


Figure 128. Resume to Suspend Timing Diagram

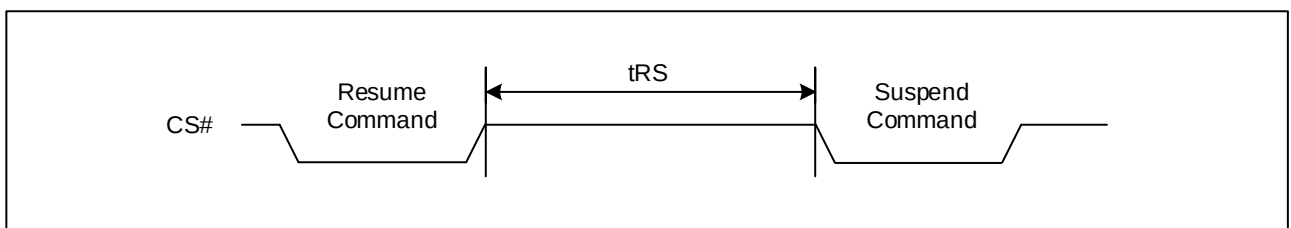


Figure 129. WP# Timing

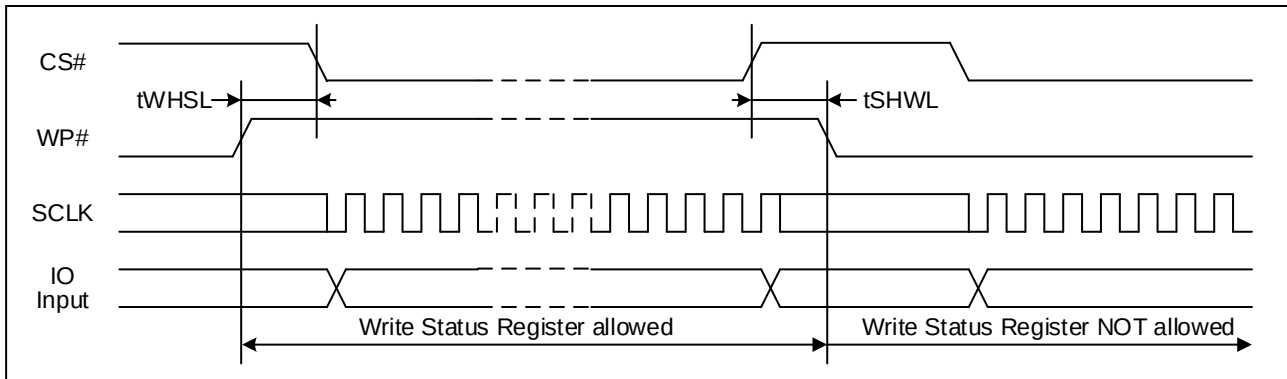


Figure 130. RESET Timing

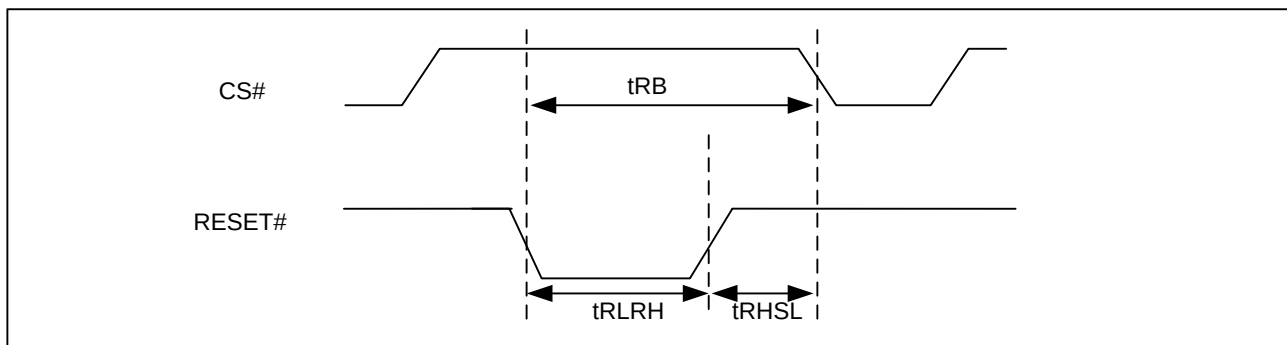


Table 22. Reset Timing

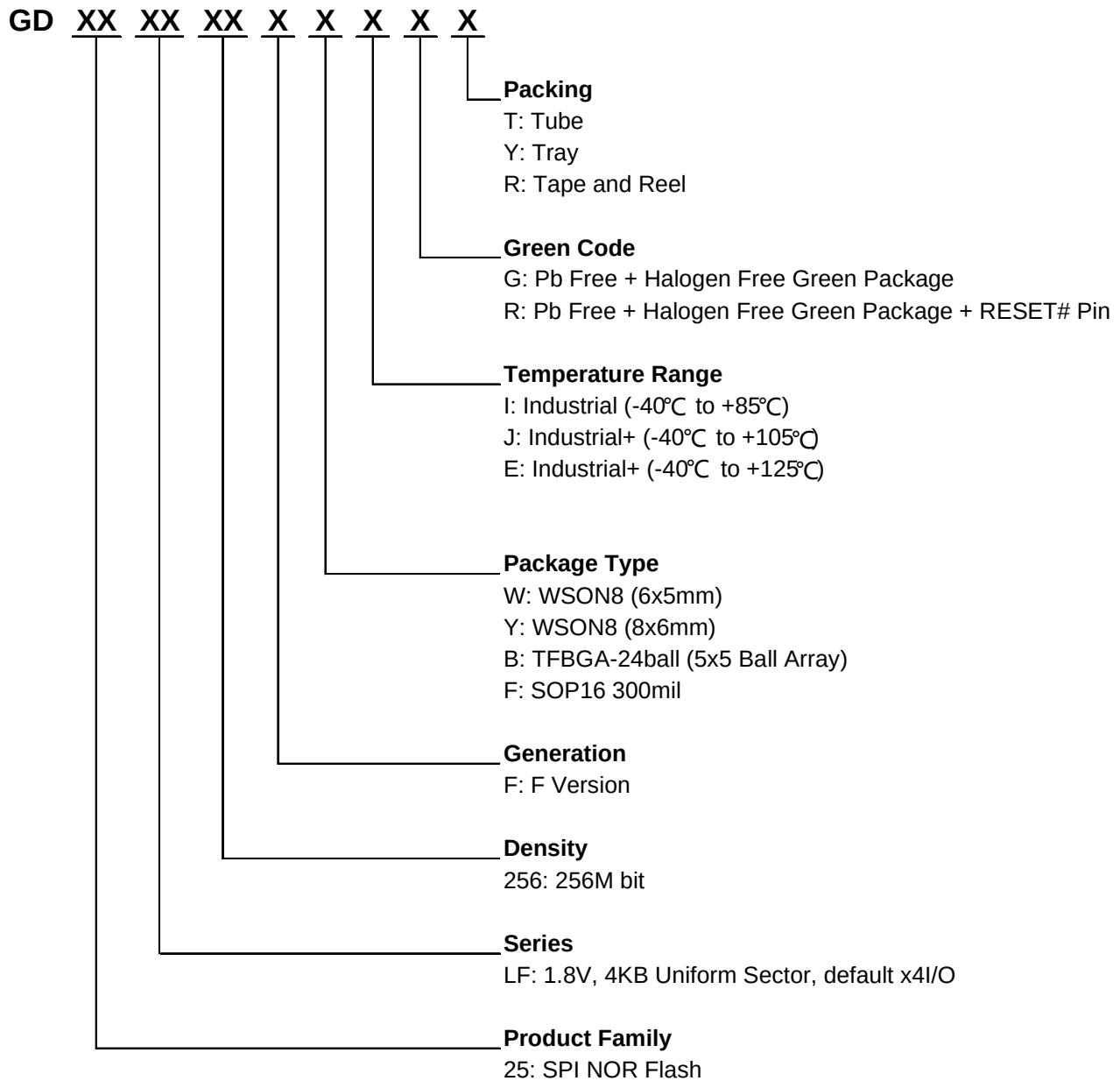
Symbol	Parameter	Min.	Typ.	Max.	Unit.
t_{RLRH}	Reset Pulse Width	1			μs
t_{RHSL}	Reset Hold time before next Operation	50			ns
t_{RB}	Reset Recovery Time (Except From Erase or Safe boot fail)			40	μs
	Reset Recovery Time (From Erase)			25	ms
	Reset Recovery Time (From Safe boot fail)			1.5	ms

Notes:

1. Time of Reset Recovery Time from 01h/B1h command would be $t_W + t_{RB}$
2. The device need t_{RB} (max) at most to get ready for all commands after RESET# low.



11 ORDERING INFORMATION





11.1 Valid Part Numbers

Please contact GigaDevice regional sales for the latest product selection and available form factors.

Temperature Range I: Industrial (-40°C to +85°C)

Product Number	Density	Package Type	Packing Options
GD25LF256FWIG	256Mbit	WSO8 (6x5mm)	Y/R
GD25LF256FYIG	256Mbit	WSO8 (8x6mm)	Y/R
GD25LF256FBIR	256Mbit	TFBGA-24ball (5x5 Ball Array)	Y/R
GD25LF256FFIR	256Mbit	SOP16 300mil	T/Y/R

Temperature Range J: Industrial (-40°C to +105°C)

Product Number	Density	Package Type	Packing Options
GD25LF256FWJG	256Mbit	WSO8 (6x5mm)	Y/R
GD25LF256FYJG	256Mbit	WSO8 (8x6mm)	Y/R
GD25LF256FBJR	256Mbit	TFBGA-24ball (5x5 Ball Array)	Y/R
GD25LF256FFJR	256Mbit	SOP16 300mil	T/Y/R

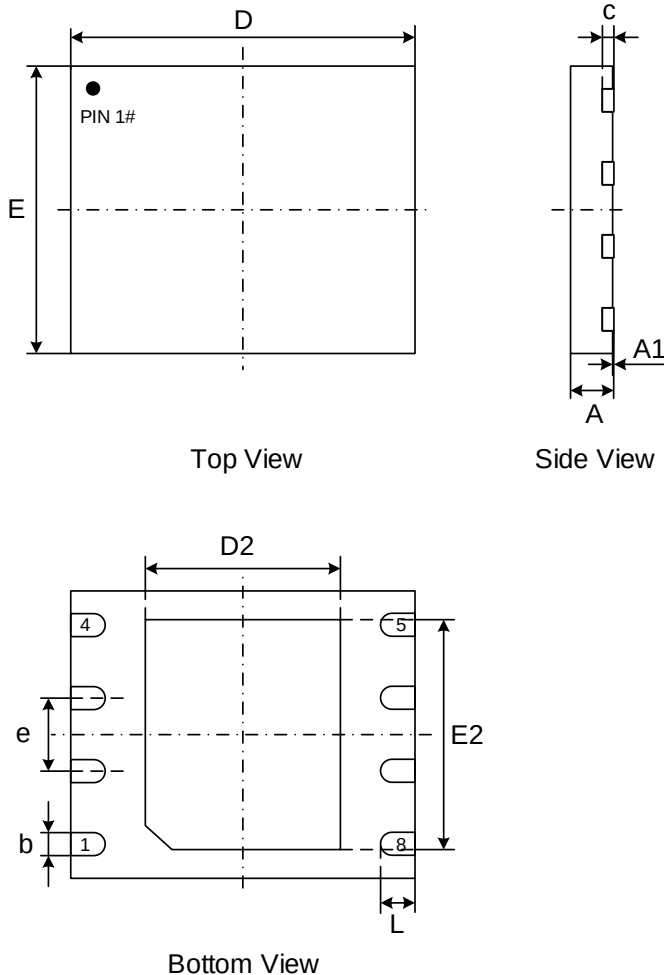
Temperature Range E: Industrial (-40°C to +125°C)

Product Number	Density	Package Type	Packing Options
GD25LF256FWEG	256Mbit	WSO8 (6x5mm)	Y/R
GD25LF256FYEG	256Mbit	WSO8 (8x6mm)	Y/R
GD25LF256FBER	256Mbit	TFBGA-24ball (5x5 Ball Array)	Y/R
GD25LF256FFER	256Mbit	SOP16 300mil	T/Y/R



12 PACKAGE INFORMATION

12.1. Package WSON8 (6x5mm)



Dimensions

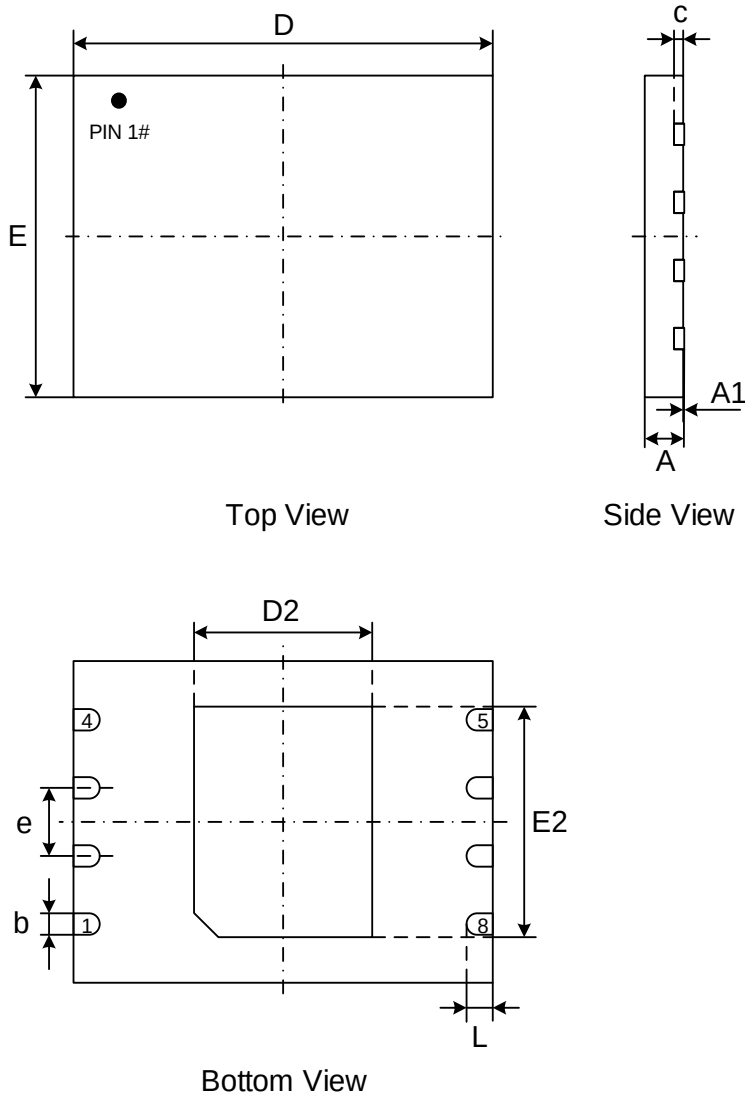
Symbol		A	A1	c	b	D	D2	E	E2	e	L
Unit											
mm	Min	0.70	0.00	0.180	0.35	5.90	3.30	4.90	3.90	1.27	0.50
	Nom	0.75	0.02	0.203	0.40	6.00	3.40	5.00	4.00		0.60
	Max	0.80	0.05	0.250	0.50	6.10	3.50	5.10	4.10		0.75

Note:

1. The exposed metal pad area on the bottom of the package is not connected to any internal signal. It is OK to connect it to the system ground (GND) or leave it floating.
2. Coplanarity $\leq 0.08\text{mm}$. Package edge tolerance $\leq 0.10\text{mm}$.
3. The lead shape may be of little difference according to different package lead frames. These lead shapes are compatible with each other.



12.2. Package WSON8 (8x6mm)



Dimensions

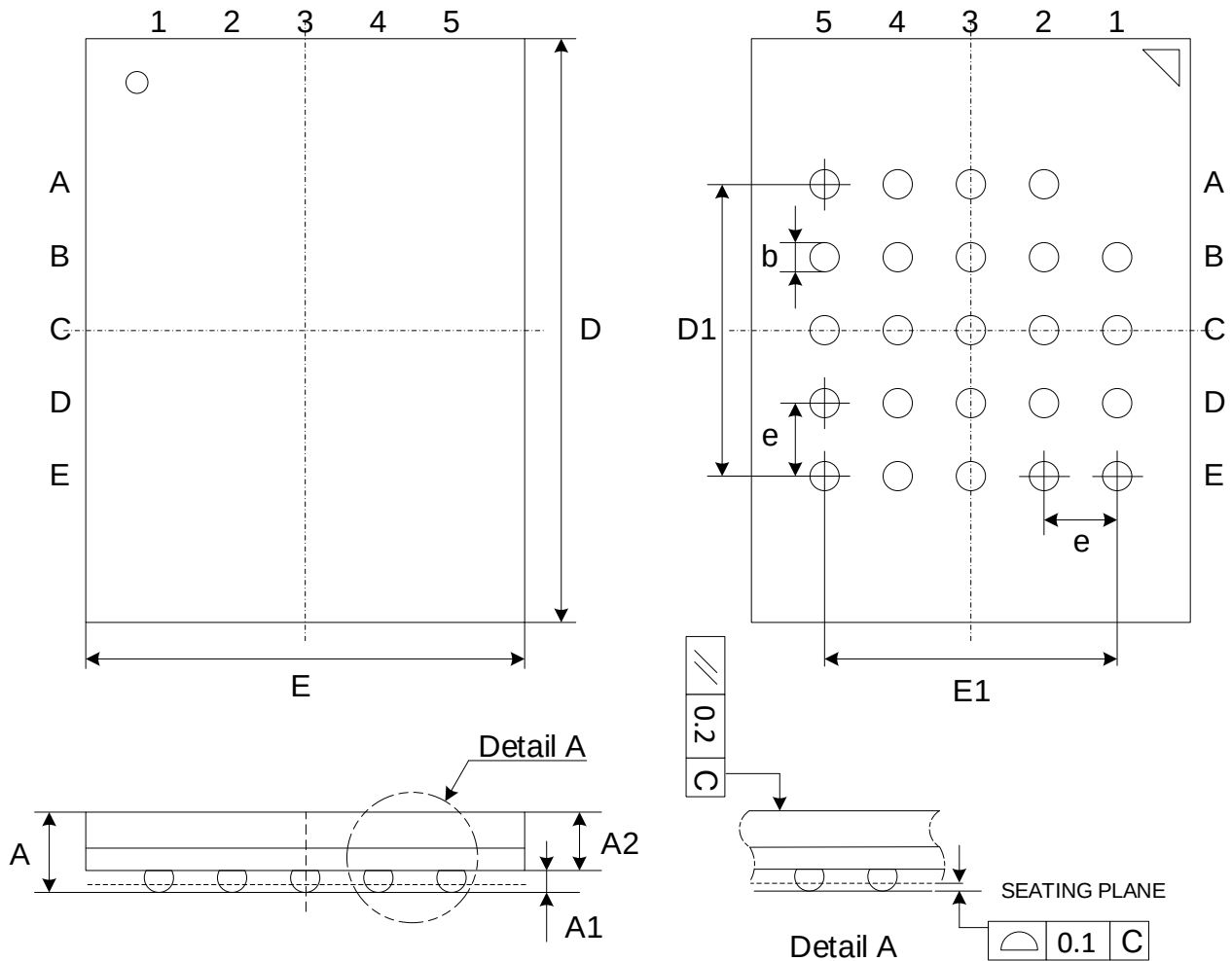
Symbol		A	A1	c	b	D	D2	E	E2	e	L
Unit											
mm	Min	0.70	0.00	0.180	0.35	7.90	3.30	5.90	4.20	1.27	0.45
	Nom	0.75	0.02	0.203	0.40	8.00	3.40	6.00	4.30		0.50
	Max	0.80	0.05	0.250	0.45	8.10	3.50	6.10	4.40		0.55

Note:

1. The exposed metal pad area on the bottom of the package is not connected to any internal signal. It is OK to connect it to the system ground (GND) or leave it floating.
2. Coplanarity $\leq 0.08\text{mm}$. Package edge tolerance $\leq 0.10\text{mm}$.
3. The lead shape may be of little difference according to different package lead frames. These lead shapes are compatible with each other.



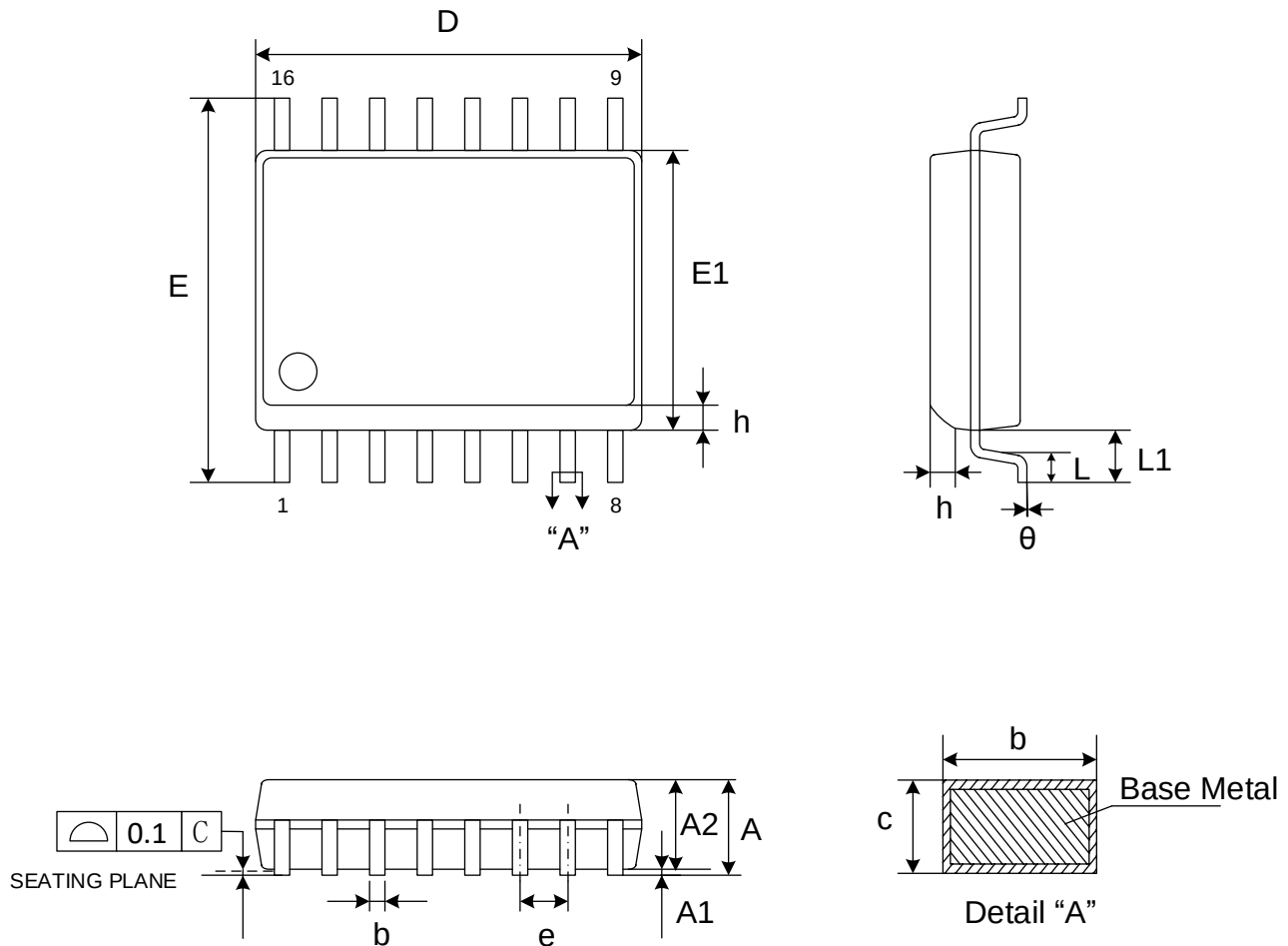
12.3. Package TFBGA-24BALL (5x5 ball array)



Dimensions

Symbol		A	A1	A2	b	E	E1	D	D1	e
Unit										
mm	Min	-	0.25	-	0.35	5.90	4.00	7.90	4.00	1.00
	Nom	-	0.30	0.80	0.40	6.00		8.00		
	Max	1.20	0.35	-	0.45	6.10		8.10		

12.4. Package SOP16 300MIL



Dimensions

Symbol		A	A1	A2	b	c	D	E	E1	e	L	L1	h	θ
Unit														
mm	Min	-	0.10	2.05	0.31	0.10	10.20	10.10	7.40	1.27	0.40	1.40	0.25	0
	Nom	-	0.20	-	0.41	0.25	10.30	10.30	7.50		-		-	
	Max	2.65	0.30	2.55	0.51	0.33	10.40	10.50	7.60		1.27		0.75	8

Note:

1. Dimension D does not include mold flash, protrusions or gate burrs. Mold flash, protrusions or gate burrs shall not exceed 0.15mm per end.
2. Dimension E1 does not include interlead flash or protrusion. Interlead flash or protrusion shall not exceed 0.25mm per end.



13 REVISION HISTORY

Version No	Description	Page	Date
1.0	Initial Release	All	2023-10-20
1.1	Modify ECC description Modify Volatile Configuration Register Note Modify Figure 56 DTR Burst Read with Wrap Sequence Diagram Modify SFDP from 216B to 216C Modify ICC1 max.value from 260uA to 280uA@-40℃~125℃ Modify t _{CLCH} /t _{CHCL} from 0.2V/ns to 0.1V/ns Modify t _{NLP} from 60us to 50us@-40℃~125℃ Modify t _{BP} max.value from 110/120/120us to 150/160/180us Modify t _{PS} from 12us to 10us Modify Typo	P23 P42 P70 P97 P103 P104,106,108 P105,107,109 P105,107,109 P105,107,109 ---	2024-2-20
1.2	Modify Typo Update “SRP1=1/SRP0=X” description and Note Modify Power-Up/Down Timing Modify Package TFBGA-24BALL (5x5 ball array) Modify Package SOP16 300mil Note	-- P31 P100 P118 P119	2024-8-12
1.3	Modify ECCFA description Modify Ordering Information	P42,44 P114-115	2025-4-3

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